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Quasi-Steady System-Level Modeling of an Expander-Cycle Rocket Engine: Nominal Validation and Off-Design Analysis

Tesi di Laurea Magistrale in
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Abstract

Expander-cycle engines play a crucial role in upper-stage propulsion due to their high efficiency, but the strong thermo-fluid coupling inherent to this architecture makes the system modeling particularly challenging. Most system-level approaches focus on fully dynamic engine models, often implemented within proprietary computational environments. Alternatively, quasi-steady approaches are typically applied to design point analyses and are rarely followed by a critical assessment of their validity under off-nominal conditions. This work develops and critically assesses a fully coupled quasi-steady system-level model of a cryogenic expander-cycle engine, treating the quasi-steady assumption as a structural physical hypothesis rather than a numerical simplification. The modeling framework is based on dedicated and modular representations of the main subsystems, including turbomachinery, regenerative cooling jacket, combustion chamber, and fluid network, whose strong thermo-fluid coupling is resolved through iterative closure of global mass, energy, and shaft power balances.

The RL10A-3-3A engine is adopted as the reference configuration. At nominal operating condition, the model predicts the reference chamber pressure and total mass flow rate with relative deviations below 1%, validating not only the individual subsystem models, but also the overall coupling strategy and quasi-steady formulation. The framework is then extended to off-design regimes via valve modulation, enabling identification of physically admissible equilibrium states down to 90% of nominal thrust. Throughout the analysis, the resulting operating points are evaluated through embedded diagnostic checks, allowing the definition of feasibility frontiers and the identification of critical regimes associated with choking, pressure ratio limitations, and solution breakdown.

The proposed framework establishes a validated quasi-steady modeling platform and explicitly defines its domain of applicability and structural limitations. It further identifies the modeling elements that can be reused in alternative engine architectures and outlines the structural transition required for time-dependent formulations.

Keywords: Expander-cycle rocket engine, Quasi-steady system modeling, Off-design equilibrium analysis, Critical operating regimes

Abstract in lingua italiana

I motori a ciclo *expander* svolgono un ruolo cruciale nella propulsione grazie alla loro elevata efficienza, ma il loro forte accoppiamento termo-fluidodinamico rende la modellazione a livello di sistema particolarmente complessa. La maggior parte degli approcci si concentra su modelli dinamici completi, spesso implementati in *software* proprietari, mentre gli approcci quasi-stazionari sono generalmente limitati ad analisi al punto di progetto e raramente accompagnati da una valutazione critica della loro validità in condizioni *off-design*.

Nel presente lavoro è stato sviluppato e analizzato in modo critico un modello quasi-stazionario completamente accoppiato a livello di sistema di un motore criogenico a ciclo *expander*, trattando la quasi-stazionarietà come ipotesi fisica strutturale e non solamente come semplificazione numerica. Il *framework* adotta una decomposizione modulare dei principali sottosistemi, quali turbomacchine, *cooling jacket* rigenerativo, camera di combustione e rete idraulica, il cui accoppiamento è risolto mediante la chiusura iterativa dei bilanci globali di massa, energia e potenza d'albero.

Il motore RL10A-3-3A viene preso come caso di riferimento. In condizioni nominali il modello predice la pressione in camera e la portata totale con errori inferiori all'1%, validando sia i singoli sottosistemi sia la strategia di accoppiamento adottata. Il *framework* è stato poi esteso a regimi *off-design* tramite la modulazione delle valvole, consentendo l'identificazione di stati di equilibrio fisicamente ammissibili fino al 90% della spinta nominale. I punti operativi sono valutati mediante controlli diagnostici integrati, che permettono di delineare i limiti di validità e di individuare i regimi critici associati a *choking*, limiti di rapporto di pressione e *breakdown* della soluzione.

Il *framework* costituisce una piattaforma quasi-stazionaria validata, e ne definisce esplicitamente il dominio di applicabilità e le limitazioni strutturali. Inoltre sono individuati gli elementi riutilizzabili in architetture alternative ed è definita una transizione strutturale necessaria per formulazioni dipendenti dal tempo.

Keywords: Motore a ciclo expander, Modellazione di sistema quasi-stazionaria, Analisi off-design dei punti di equilibrio operativi, Regimi operativi critici

Contents

1 | Introduction

System-level modeling represents a fundamental tool in the development and assessment of liquid rocket engines. During the preliminary design stages, it supports effective decisions and trade-offs in engine architecture. In the latter phases it is crucial for reducing program risks, refining the requirements, and finally predicting and improving the system performance in a repeatable and traceable manner with reduced cost and development time [1]. Liquid rocket engines may experience various failure modes during operation, not only because of their complexity, but also due to the extreme conditions they must withstand in terms of pressure and temperature. System-level simulations can further improve their safety and reliability, by enabling a deeper understanding of engine behavior under varying operating conditions [2]. In practice, such models are used to explore engine operating conditions, assess feasibility and performance margins, and identify critical behaviors.

From a modeling standpoint, liquid rocket engines are characterized by strong thermo-fluid coupling among turbomachinery, cooling jacket, combustion processes and distributed hydraulic networks. In particular, expander-cycle engines exhibit a particularly tight energetic coupling. In this architecture, turbine power is generated by the heat absorbed by the propellant along the regenerative cooling path. As a consequence, heat transfer, mass flow distribution and shaft power balance are structurally linked. The overall engine behavior therefore emerges from the simultaneous satisfaction of mechanical, hydraulic and thermodynamic balances, making system-level prediction intrinsically nonlinear and highly interdependent.

While high-fidelity dynamic simulations are widely employed for transient analyses, quasi-steady formulations remain extensively used for system-level evaluations due to their reduced complexity and computational cost. However, these approaches are often limited to design-point calculations and are not accompanied by a systematic assessment of their domain of validity.

The present work addresses this aspect by developing a fully coupled quasi-steady system-level model of a cryogenic expander-cycle engine, explicitly treating the quasi-steady assumption as a physical modeling hypothesis and critically evaluating its applicability. The reference engine selected for this work is the RL10A-3-3A, an expander-cycle engine. The modeling framework is based on dedicated and modular representations of the main subsystems characterizing the reference engine, namely turbomachinery, regenerative cooling jacket, combustion chamber and hydraulic network, interconnected according to the real engine architecture. The solution algorithm is based on nested loops, and the system solves the resulting set of nonlinear equations by iterating between the subsystem blocks. The solution algorithm is applied not only to the evaluation of the nominal engine operating point, but is also extended to off-design regimes for the identification of physically admissible equilibrium states under throttling conditions. Non-feasible solutions are explicitly detected and analyzed in order to delineate the admissible operating domain of the quasi-steady formulation.

1.1. Motivation and Objectives

The present work is motivated by the need for a physically consistent and critically assessed quasi-steady modeling framework for cryogenic expander-cycle engines. In response to the considerations discussed above, the thesis addresses the following research objectives:

- **Formulation of a fully coupled quasi-steady system-level model of a cryogenic expander-cycle engine**

  The first objective is to construct a modular framework in which turbomachinery, cooling jacket, combustion chamber and hydraulic network models are simultaneously coupled through explicit mechanical, hydraulic and thermodynamic closure relations.

- **Extension of the quasi-steady formulation beyond the nominal operating point.**

  The nominal configuration is treated as a validation result of the coupled formulation, verifying the internal consistency of subsystem models and closure strategy. The second objective is to verify if the framework can be extended to off-design and throttling regimes, for the identification of physically admissible equilibrium states under valve modulation.

- **Critical assessment of the quasi-steady hypothesis and identification of its domain of validity.**

The third objective is to analyze the structural limits of the adopted formulation by detecting non-feasible configurations, interpreting critical regimes, and evaluating how system-level parameters influence the admissible operating domain.

- **Investigation of the structural extensibility of the framework.**

  The last objective is to examine the possible extension of the proposed quasi-steady approach toward alternative engine architectures and toward time-driven formulations, analyzing the modifications required in the closure strategy and in the organization of the solution process.

1.2. Organization of the Thesis

The work is organized as follows:

- **Chapter 1** provides a brief introduction of the motivations which led to this work, followed by the main research questions addressed in the thesis.
- **Chapter 2** provides a review of the relevant literature on system-level modeling of liquid rocket engines, with particular emphasis on quasi-steady and dynamic approaches. The main modeling strategies adopted in previous works are analyzed, highlighting their assumptions, limitations and domains of applicability, in order to position the present work within the existing research framework.
- **Chapter 3** presents the formulation of the quasi-steady system-level modeling framework. Dedicated models for turbomachinery, regenerative cooling jacket, combustion chamber, and hydraulic network are described, together with the coupling strategy and the closure equations governing the global equilibrium problem for nominal analysis.
- **Chapter 4** deals with the extension of the framework presented in Chapter 3 to off-nominal analysis. In this chapter, the necessary modifications to the solution algorithm and the overall set of diagnostic checks embedded in the solver for the critical assessment of the analyzed configurations are presented.
- **Chapter 5** provides the results obtained in this analysis. It is divided into three sections: the first focuses on the computation and validation of the nominal operating point against reference data to assess the internal consistency of the proposed framework. The second focuses on the analysis of off-design and throttling regimes. The admissible equilibrium configurations under valve modulation are investigated and critically assessed, and the variation of the main performance parameters is

evaluated. The last section provides an analysis of the possible system lever to increase engine throttle-ability, based on the identified critical regimes.

- **Chapter 6** provides a critical assessment of the quasi-steady formulation. The domain of validity of the adopted hypothesis is analyzed, critical operating regimes are identified, and the structural extensibility of the framework to alternative engine architectures and time-driven formulation is discussed.
- **Chapter 7** summarizes the main contributions of the thesis, delineates the scope and boundaries of validity of the proposed approach, and outlines possible directions for future developments.

2 | Literature Review

2.1. Liquid Rocket Engines and Expander-Cycle Architectures

Rocket Engines are complex systems that convert energy sources into kinetic energy, enabling space travel. Among the many available, the most flexible are Liquid Rocket Engines (LRE), a very efficient class of engines with a wide range of thrust that can even allow for re-ignition.

2.1.1. General Overview of Liquid Rocket Engines

Liquid rocket engines are defined as "rocket engines that use propellants stored as liquids that are fed under pressure from tanks into a thrust chamber" [3].

In general, the main components of a LRE system are the propellant tanks, a feeding system consisting of lines and valves, injectors used to introduce and atomize propellants into the combustion chamber, the combustion chamber itself, where the conversion from chemical to thermal energy occurs, and finally a nozzle where the combustion gases expand, converting their thermal energy into kinetic energy and generating thrust.

The next fundamental classification distinguishes between pressure-fed and turbopump-fed engines. Pressure-fed systems are mostly used for low-thrust propulsion systems, such as attitude control of kick stages, which often include more than one combustion chamber per engine. Their simplicity is their strength, as they only require pressure vessels to pressurize the tanks as the propellants are depleted. On the other hand, turbopump-fed systems are used when larger amounts of propellant or higher thrust are required. They include one or more turbine-pump assemblies, in which an energetic working fluid expands through a turbine, extracting the mechanical power required to drive the pumps and pressurize the upstream flow exiting the tanks. These systems are mechanically more complex, but allow for easier storage of propellants, since they do not need to be constantly pressurized.

Turbopump-fed engines are further classified according to the mechanism by which the propellants driving the turbine are energized. One option involves the use of preburner assemblies, i.e. small miniature combustion chambers that produce hot combustion gases. These configurations are then classified as gas generator cycles if the gases are not reused and therefore dumped and wasted; otherwise, they are identified as staged combustion cycles if the exhaust gases are injected in the main combustion chamber, contributing to overall engine thrust.

The other option is to allow a propellant to acquire thermal energy present in the hot rocket nozzle, passing through channels that act as a heat exchanger in the so-called expander-cycle.

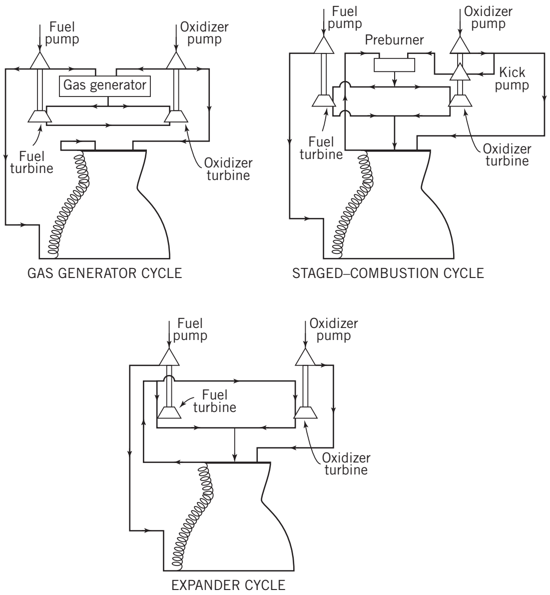


Figure 2.1: Overview of common turbopump cycles [3]

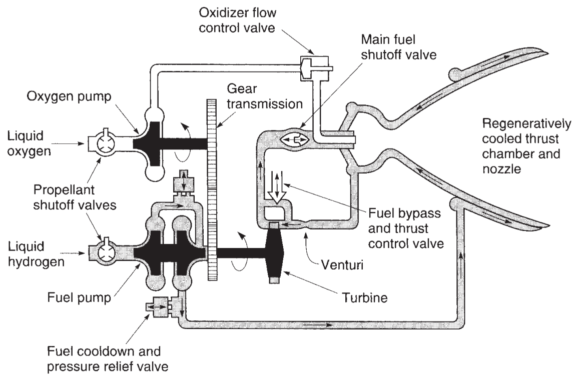


Figure 2.2: Overview of the expander-cycle LRE [3]

2.1.2. Expander-Cycle Engines and Reference Applications

The expander-cycle is a closed-cycle LRE (i.e. all the propellants are eventually fed into the combustion chamber), granting higher performance in terms of specific impulse compared to a gas generator cycle, with a simpler construction and lower mass compared to the staged combustion cycle.

It usually employs cryogenic liquid hydrogen (LH2) as fuel [3], even if nowadays methane is being considered [4]. The fuel is passed through some channels positioned in the nozzle in order to heat it up, while cooling down the thrust chamber. The hot gasified fuel is then expanded in a turbine, which in turn drives both fuel and oxidizer pumps. The thrust control is typically achieved by regulating the bypass of some of the gasified fuel around the turbine, actuating a valve.

Since the only energy pickup method is heat transfer through the nozzle, both heat transfer coefficients and temperature difference must be maximized, which motivates the use of cryogenic propellants. The fuel is the only propellant considered for cooling, as an oxidizer would quickly degrade the surface of the channels and destroy the turbine.

Since the energy harvested by the turbine is entirely obtained by heating the fuel in the cooling jacket rather than by combustion, in theory, it could have a higher specific impulse than a staged combustion cycle LRE. In practice, the ability of the latter to achieve a higher chamber pressure often levels the efficiency gains.

Expander-cycle engines were developed as early as the late 1950's, with first tests in 1960 and achieving flight in 1963. These were the American-built RL10 engines: first developed as light, simple, and efficient engines that reignite in flight, they conquered the upper stage scene, being employed on Atlas, Atlas V, and Delta IV upper stages, and remain to this day the only American-made upper-stage LOX-LH2 engine. One of the most widely used and documented is the RL10A-3-3A, which was employed on the Centaur stage atop Atlas G, Atlas I, Atlas II, and Titan 4 [3].

As these versions lacked deep throttling capability, in 1993 a new version, denoted RL10A-5, was developed. This configuration was capable of throttling down to 30% Rate Thrust Level (RTL), by heavily modifying the feeding lines and widening the turbine bypass valve [3].

2.2. Modeling Approaches for Liquid Rocket Engines

Numerical simulations of LRE have become an essential tool for designers, being able to reduce costs and risks associated with experimental test campaigns, which remain complex and budget heavy [5]. The literature shows a clear development history, from early code approaches in the '60s and '70s, which were rigid, proprietary, and single-use, to modern platforms that are modular and object-oriented [5, 6].

The particular modeling choice depends strictly on the objective of the analysis (nominal design, operative envelope, or safety during transients) and can be divided into three main categories: steady-state, quasi-steady state, and transient.

2.2.1. Steady-State Modeling

Steady-state modeling represents the basis for the preliminary design and performance evaluation at the design point. In this approach, the time derivatives are set to zero, and the problem simplifies to solving a system of nonlinear algebraic equations [7].

The importance of having specific libraries dedicated to nominal regime computations is underlined by Di Matteo [5], who states that employing complex and detailed transient routines/codes in pre-design phases can be detrimental, inefficient, and costly. For these reasons, specific libraries have been developed in many environments, such as EcoSimPro [8], that allow rapid execution of parametric design analyses and cycle balances, essential to define admissible conditions on which more detailed transient studies are then based on. Also, the LPRES package [9] exploits simplified analytical models and user-defined efficiencies to quickly build the cycle topology and obtain reliable results in short time.

2.2.2. Quasi-Steady Modeling

Quasi-steady state modeling lies between steady-state and full transient modeling. The founding principle is the hypothesis that fluid dynamic and thermal transients are sufficiently slower than the propagation of pressure waves, allowing the system to be considered as a continuous succession of equilibrium states.

In a quasi-steady state model, the differential governing equations for the conservation of mass and energy are solved by simplifying or setting to zero the accumulation terms for fluid control volumes, while the momentum equations are often reduced to algebraic pressure balances based on stationary performance maps. This approach is particularly effective in *throttling* phases where thrust modulation is required and for mapping *off-nominal* performance.

An example of this methodology is the use of adimensional performance maps for the turbomachinery. As discussed by Binder et al. [10], pumps and turbines are modeled using head, torque, and efficiency maps, derived from stationary tests and applied instant by instant during the simulations. The literature offers numerous examples of the effectiveness and the limitations of the quasi-steady approach. As an example, Naderi et al. [1] proved the efficacy of this modeling by simulating the RS-25 SSME at various Rated Thrust Levels (RTLs), ranging from 50% to 109% of nominal thrust value. This modular approach, developed in MATLAB/Simulink, utilizes nonlinear algebraic conservation equations, iteratively solved using a Newton-Raphson method to find the equilibrium point of the system at every valve stem variation. Results show an error lower than 5%

compared to flight data for the main parameters, confirming that for controlled regime variations, the quasi-steady state approach is robust.

Moreover, while codebases like ROCETS (ROCket Engine Transients Simulations) were mainly developed for transient analysis, they integrate steady or quasi-steady state libraries, which are fundamental for engine setup. In his work on the RL10A-3-3A engine, Binder [10] explicitly describes the creation of two model versions: one for transient and one for steady-state analysis. The stationary RL10A-3-3A model is used to predict engine performance at various mixture ratios and RTLs, validating the results within 4% of the actual flight data [11]. It is interesting to note the different modeling approaches adopted within the same work: while quasi-steady state modeling used thermal exchange maps based on empirical data, the transient simulation required a finer discretization to capture the thermal inertia of the nozzle metal during start-up. This underlines that the quasi-steady state simulation is the first step to take in order to define the initial conditions that will be fed into the more complex transient simulations.

The quasi-steady hypothesis is adopted even in transient modeling to simplify the computation of components with rapid response. For example, in the code *Liquid Rocket Transient Code* (LRTC) developed by Ruth et al. [12], the combustion chamber model assumes that mass efflux through the nozzle is quasi-stationary. This implies that flux reacts instantaneously to chamber pressure variations, allowing to determine the mass flow exclusively via the throat conditions and the gasdynamic stationary relationships, without the need to solve the pressure wave propagation problem inside the nozzle.

The limitations of these models lie in their inherent hypothesis: no fluidynamic and thermal time-driven transients are considered, with each state being computed as an independent equilibrium point. As a consequence, while it is possible to ensure the feasibility of the system at the retrieved working point, it is not possible to clearly define the control sequence needed to achieve it. That is, controls must be applied slowly, in order to allow for fast transients to dissipate and for the system to reach a new equilibrium state before proceeding. Impulsive variations in control valves are not allowed in this modeling environment.

In conclusion, quasi-steady modeling is essential not only as a standalone tool for the definition of the design point and the stationary control laws, but also as a founding basis for more advanced models, providing the boundary conditions and subsystem characterizations on which dynamic analyses are based on.

2.2.3. Transient and Time-Dependent Approaches

Transient modeling is the most complex and critical frontier in the simulation of liquid propulsion systems. Differently from the other approaches (steady and quasi-steady), that focus on equilibrium points, transient analysis tries to solve the temporal evolution of the system state variables ($\frac{d}{dt} \neq 0$), allowing to predict the behavior of the engine during highly dynamical phases like start-up, shut-down, fast throttling, and failures. The literature highlights that the ability to model these phenomena accurately is essential not only to verify safety margins and valve sequences, but also to determine transient structural loads, temperature peaks, and instabilities that are not evident from stationary analyses [5, 11, 13].

Transient simulation of LREs is an intrinsically *stiff* problem [13], characterized by the coexistence of time constants that span various orders of magnitude: from acoustic or combustion phenomena at high frequency (microseconds), to the dynamics of turbomachinery (milliseconds), and to the thermal inertia of metal masses (seconds). From the physical viewpoint, models must deal with severe discontinuities and nonlinearities. Ruth et al. [12] underline how the traditional finite differences methods can fail in capturing discontinuities in pressure or speed (like water hammer), introducing numerical oscillations known as the Gibbs phenomenon. To avoid it, in their code, they implemented the Method of Characteristics (MOC), capable of tracking the propagation of pressure waves in feeding lines with high fidelity.

Where custom solvers are not preferred, commercially available tools like Simulink are widely used. They comprise built-in solvers that are specifically designed to solve stiff problems, and their block-based modeling environment is specifically designed to model the interaction between complex subsystems, while providing a clear representation of the equations being integrated. Its seamless coexistence with MATLAB makes it an even more powerful tool, since many libraries are available with toolsets and blocksets optimized for spacecraft components [14, 15].

Moreover, during startup phases, cryogenic fluids enter the feeding lines that are still at ambient temperature, generating complex biphase flows, boiling, and chilldown phenomena that can lead to thermal choking or dangerous overpressures. Di Matteo [5] outlines that is crucial to accurately describe these phenomena, especially in expander-cycle engines like the RL10, where the energy used to power the turbine is exclusively harvested this way.

2.2.4. Evolution of Simulation Tools

The literature shows a clear evolution in dedicated software tools. Several countries developed simulation programs mostly for LREs by developing their own Space Launch Vehicles, and as technology and research progressed, there was a clear increase in reliance on simulation data compared to test data [3].

Historical modular codes (ROCETS and derived)

One of the founding bases of the transient simulation is the ROCETS program (ROCket Engine Transient Simulator), developed by Pratt & Whitney and NASA [16]. As described by Binder [10] and Naderi [1], it adopts a modular approach based on *building blocks* (i.e. pumps, turbines, volumes, resistances) developed in FORTRAN language. It solves a stiff differential equations system using implicit integration schemes to ensure numerical stability even with large timesteps [13, 16]. An important aspect underlined by Binder [11] is the need to modify turbopump maps to include low-speed and torque regimes typical of start-up, which are often not included in standard testing data.

Approaches based on custom algorithms

Several researchers proposed ad hoc algorithms to solve the coupling between algebraic and differential equations. Karimi et al. developed an algorithm based on Newton-Raphson loops nested inside an Euler integration routine [7, 17]. This approach physically follows the propellant flow. The pressure is iteratively integrated from tanks, through pumps until the combustion chamber, allowing to solve hydraulic nonlinearities at each timestep.

Acausal and object oriented modeling (EcosimPro/ESPSS and Modelica)

The most recent trend, amply documented, is the transition to acausal modeling [5, 6]. Tools like EcosimPro (with ESA ESPSS library) and Modelica allow to write conservation equations without predetermining the computations direction (input and output), that is handled by the solver. This allows the system to be solved simultaneously as a system of algebraic and differential equations, with the solver itself determining stiffness, the most adequate solver and the timesteps. As a consequence, elements are connected simply by nodes, and new components can be inserted without needing to reassess stability or solving strategy.

Di Matteo [5] used ESPSS to create a complete transient model of the RL10A-3-3A, introducing custom advanced models for the injectors (considering the injector plate thermal inertia) and the cooling channels, overcoming lumped parameter approach limitations, since they could not accurately describe specific thermal and pressure peaks at start-up.

Liu et al. and Borgia et al. [6, 18] demonstrated Modelica flexibility in creating multi-physics library that integrate fluid, mechanical and control dynamics, easing the simulation of complex scenarios. Their models can simulate the entire life-cycle of the propellant, including phenomena like sloshing in tanks and dynamic interaction between structure and propellant. The advantage of Modelica compared to other platforms is its open-source environment, where every components can be modified, in contrast with other more closed platforms like EcosimPro or Simscape.

The symbolic mathematical kernel of these environments extracts the mathematical model from the graphical schematic. The user selects the governing input variables (e.g. valves opening law) and the internal algorithms generate robust mathematical models through symbolic manipulation and equation reordering. The available internal solvers allow dealing with stiff or non-stiff dynamic problems, steady problems, optimization problems, use of constraints, etc. The solvers use dense or sparse versions depending of the size of the problem in order to speed up the simulation [19].

2.3. Quasi-Steady System-Level Modeling of Rocket Engines

While the previous sections introduced the quasi-steady approach as a numerical methodology, this section aims to analyze how it is applied at system level to simulate the interaction between the various engine subsystems (i.e. turbomachinery, feeding lines, combustion chamber). The literature shows that modeling at the system level in a quasi-steady regime does not stop at solving static equations but requires complex algorithms tuned to grant thermodynamic and mechanical coherence in the whole engine.

2.3.1. Iterative Solvers and Cycle Balancing Schemes

The main challenge in modeling at the system level is the solution of cyclical interdependency among components. In a closed-cycle engine (i.e. expander or staged combustion), the output of a component (i.e. turbine) is the input of another one (i.e. pumps, through mechanical coupling), that in turn feeds the first one through another variable (i.e. fuel flow), creating algebraic loops that require robust iterative solvers.

A common methodology found in the literature is the use of iterative nested loops that satisfy the mass and energy balances. They can solve single components for the innermost loops and even system-level balances for the outermost ones. Of course, the innermost loops will be called more often, leading to an increased computational time and the need to render the code more efficiently.

Naderi et al. [1] proposed an algorithm for the quasi-steady simulation of the SSME. Their approach divides the engine into *fuel line* and *oxidizer line*, and uses iterative, nested loops based on the Newton-Raphson Method to solve the two propellant branches sequentially. In this scheme, critical parameters such as pump discharge pressures or preburner mass flows are initially guessed, since many components require them as inputs. Then, by performing mass flow and power balance, all the engine parameters are calculated one by one on each line, and finally, the combustion chamber module is called and simulated, yielding the final results.

Regarding the power balance, while in transient models shaft rotational speed is a state variable retrieved from the time-integration of the angular momentum equation, in the quasi-steady state approach the power balance is an algebraic closure constraint. As highlighted by Binder and Naderi [1, 11], the shaft speed becomes an iteration variable inside a numerical loop, whose value is often iteratively updated until the torque residual is null, removing the link from the physical rotational inertia of the component.

2.3.2. Scope of Applicability and Limitations

The quasi-steady state models applied at the system level have shown to be a valid instrument for specific phases of the engine analysis and development, whilst still maintaining intrinsic criticalities.

The main application is the mapping of the throttling range in off-design conditions. A modular model based on algebraic nonlinear equations can accurately predict engine performance at different throttle levels [1], assuming that the control valves are actuated slowly enough relative to the fluid dynamics.

However, this modeling approach has serious limitations regarding the start-up and shut-down phases. While many models achieve acceptable error levels relative to nominal performance, they fail to capture the thermal and hydraulic peaks associated with fast transient phases. These simulations require models with nodes and thermal and hydraulic capacity, which the quasi-steady approach, based on instantaneous equilibrium, intrinsically ignores. Moreover, more complex phenomena require temporal delay components that, by design, are not present in the closure equations provided.

2.4. Throttling Modeling in Liquid Rocket Engines

The ability to modulate thrust is a key requirement for trajectory optimization and orbital maneuvering. However, throttling modeling presents unique challenges because it pushes components towards the limit of their working ranges, where strong nonlinear effects and, in some cases, dynamic instabilities can arise.

In the literature, throttling in LREs is generally achieved by modulating the propellant mass flow in the combustion chamber. For turbopump-fed engines, this translates to regulating the power extracted by the turbine. Specifically for expander-cycle LREs, this is managed by the Thrust Control Valve (TCV) [3]. This valve acts as a turbine bypass valve, reducing the fraction of gaseous hydrogen that expands in the turbine, lowering its power and, in turn, shaft speed and chamber pressure. At the same time, it is very common to have an Oxidizer Control Valve that regulates the mixture ratio OF, indirectly influencing the global energy balance.

As described before, the most suitable and widely used tool to analyze the off-nominal throttling conditions is the quasi-steady state approach. In this context, the aforementioned valves are treated as system variables, imposing different aperture areas within the feeding lines that allow the identification of new equilibrium points [1, 5, 10].

However, extending the throttling regime to low thrust values introduces criticalities that quasi-steady models can only identify as failures. For example, in an expander-cycle engine the energy that drives the turbine comes from the heat absorbed by the fuel in the cooling jacket. At low regimes, reducing the mass flow rate lowers the heat transfer coefficient, possibly leading to incomplete gasification of the propellant in the cooling jacket or to insufficient power being extracted by the turbine to drive the turbopump assembly [20].

3 | Nominal Quasi-Steady Engine Modeling

The purpose of this chapter is to introduce the modeling framework adopted for the present work. Modeling the behavior of a rocket engine requires not only representing each component in a physically consistent way, but also carefully accounting for how subsystems interact during engine operation. For this reason, each modeled block needs to be assessed not only in terms of its individual performance, but also in terms of its coupling with the other elements within the overall system.

Within this context, the modeling framework developed in this work adopts a quasi-steady, system-level approach. This perspective allows focusing on the development of physically consistent component models and their integration into the overall system. In this chapter, the quasi-steady modeling framework is formulated and applied to determine a nominal operating condition, which serves as the reference configuration for subsequent analyses.

The chapter is organized as follows. First, the main modeling choices and the reference engine configuration are introduced. Subsequently, the modeling approach adopted for each engine subsystem and the overall solution strategy are described. Finally, the validation activities performed at subsystem level are presented, providing the foundation for the analyses developed in the following chapters.

3.1. Problem Definition and Modeling Assumptions

This section introduces the formulation of the nominal quasi-steady state problem addressed in this work. The main objective is the determination of the nominal operating point of a strongly coupled system, such as a rocket engine. To achieve this, the problem is formulated under a set of explicitly stated modeling assumptions, which are required to make the system-level problem well-posed. These assumptions not only facilitate the numerical implementation of the model, but also play a fundamental role in defining the scope of applicability of the present work.

3.1.1. RL10A-3-3A Expander-Cycle Engine

The first modeling choice concerns the selection of the reference engine architecture. Among the various liquid rocket engine cycles, the RL10A-3-3A cryogenic expander-cycle engine has been adopted as the reference configuration for the present study.

The RL10A-3-3A is a bi-propellant cryogenic Liquid Rocket Engine (LRE), powered by Liquid Hydrogen (LH2) as fuel and Liquid Oxygen (LOX) as oxidizer. Developed by Pratt & Whitney since the 1960's, the RL10A-3-3A rocket engine is an important component of the American space infrastructure. Two RL10 engines form the main propulsion system for the Centaur upper stage vehicle, used on both Atlas and Titan launch vehicles [10]. Over the years, the original engine has been improved, incorporating new components, but still remains one of the most reliable engine used by NASA in its space program. It is based on an expander-cycle, in which fuel is used to cool down the combustion chamber through a regenerative cooling system. The thermal energy absorbed by the fuel is then used to drive the turbopump, composed by a turbine and two centrifugal pumps, one for each branch, mounted on the same shaft.

The expander-cycle architecture is characterized by a strong thermal–fluid coupling, mostly related to the regenerative cooling jacket subsystem, in which heat transfer, combustion processes, turbopump, and propellant mass flow rates are directly linked together. Even though presenting all the key features of a liquid rocket engine, expander-cycle engine architecture avoids unnecessary complexity. Indeed, compared to more articulated configurations as staged combustion engines (e.g. the RS25 Space Shuttle Main Engine), in which a larger number of subsystems is involved, expander-cycle engines require a reduced set of subsystems that need to be modeled, while preserving the critical components and the coupling mechanism. In this sense, modeling an expander-cycle allows the construction of a baseline in terms of component models and coupling strategies that can be extended to more complex engine architectures.

Regarding the selection of RL10A-3-3A between all the expander-cycle engines present in literature, including other variants of the RL10 family, the main motivation lies in the extensive treatment reserved to this specific model. In fact, RL10A-3-3A has been adopted extensively as reference case in numerical modeling and simulation studies over the years [5, 10, 11, 21–23]. In addition, multiple technical reports from Pratt & Whitney are available in literature, making this engine an optimal choice as benchmark for the nominal validation of the developed model [24, 25].

Moreover, the use of cryogenic propellants further strengthens the relevance of the selected engine, as such fluids introduce thermodynamic and fluid-dynamic effects typical of the modern rocket engines. Figure 3.1 presents the schematic diagram of RL10A-3-3A engine, in which all the involved subsystems and the different fuel and oxidizer flow paths are highlighted.

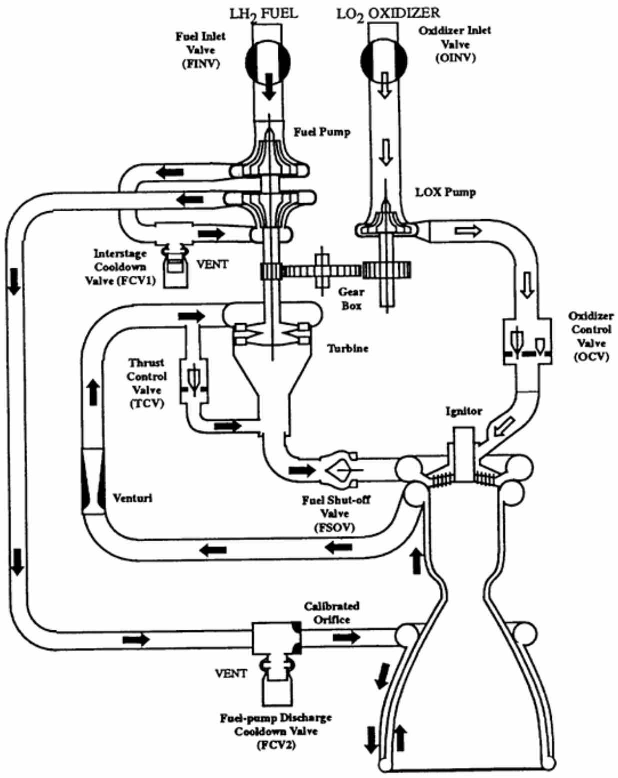


Figure 3.1: RL10A-3-3A engine diagram [10].

In an expander-cycle architecture, the fuel plays a dual role: it serves not only as reactant in the combustion process, but also as the primary energy carrier. After being pressurized by the fuel pump, liquid hydrogen flows through the regenerative cooling jacket, where it absorbs heat from the thrust chamber walls, undergoes phase change, and increases its specific enthalpy. The heated hydrogen is then expanded through the turbine, converting thermal energy into mechanical power. The turbine drives the entire turbopump assembly, as the fuel pump (two-stages centrifugal pump) and the oxidizer pump (single-stage centrifugal pump) are mounted on a common shaft, with a gearbox reducing the angular speed of the LOX pump. In this way, the available turbine power, the shaft rotational speed, and the pressure rise across both pumps are intrinsically linked.

The engine operating points, both under nominal and off-nominal conditions, result from the mutual consistency of the thermal, mechanical and hydraulic subsystems. In particular, the operating condition is determined by the balance between the thermal energy gained by the fuel during regenerative cooling, the mechanical power required by the turbopump assembly, and the mass flow rates established within the fluid network. In the present work, the fluid network is intended as the interconnected hydraulic system comprising feed lines, valves, cooling jacket, turbomachinery and injection elements within each propellant branch.

Within this coupled structure, a set of valves, together with the Venturi, plays a fundamental role in regulating and throttling engine performance. Particularly, two valves directly affect the engine equilibrium point: Thrust Control Valve (TCV) and Oxidizer Control Valve (OCV). The TCV is a normally closed, servo-operated, variable-position bypass valve used to control engine thrust by regulation of turbine power. It allows to control thrust overshoots and to maintain the desired pressure chamber during steady-state operations. To achieve this, when the chamber pressure deviates from the desired value, the action of the control allows the turbine bypass valve to vary the fuel mass flow rate through the turbine, consequently modifying the power extracted by the turbine from the fluid [5, 25]. The OCV is a normally closed, variable position valve. The valve controls oxidizer pump cool-down flow during the engine pre-start cycle and during engine start transients [5, 25]. By modifying the oxidizer mass flow rate and therefore the mixture ratio, it affects combustion temperature and chamber pressure, thereby influencing the thermal energy transferred to the cooling jacket and the overall thermomechanical balance of the cycle.

3.1.2. Global Modeling Assumptions

The quasi-steady-state formulation requires that engine operation be described as a succession of independent equilibrium states, each treated as a quasi-steady operating point. The objective of the model is the determination of these operating equilibria, both under nominal and off-nominal conditions, rather than the description of transient behavior. Consequently, no explicit time dependence is included. Off-nominal conditions are achieved by acting on the valve openings in a user-controlled manner, thereby shifting the system equilibrium toward a different configuration. The nominal operating point is obtained as the solution of the global coupled problem. Only the boundary conditions provided by propellant tanks, represented by the fluid state (p, T) at the inlet valve of each propellant branch, are imposed a priori, along with the engine physical parameters.

Within this quasi-steady framework, no mechanical dynamics is modeled. The turbopump shaft is considered to be rigidly coupled, but its dynamics is not explicitly solved and the effects of its inertia are neglected. Nevertheless, the shaft equilibrium condition enforces mechanical power balance, meaning that the available power produced by the turbine should match the power absorbed by the pumps and the power dissipated by mechanical losses. Within the quasi-steady formulation, each operating equilibrium point must satisfy the shaft balance instantaneously, while the evolution of the shaft rotational speed is not considered, since its time derivative is assumed to be zero. Inertia in this case is not treated as a physical property of the system, but is instead employed as a numerical tool to stabilize and update the shaft rotational speed within the iterative solution procedure.

Similarly, no mass accumulation and no transient compressibility in the fluid network are considered. Each propellant branch of the engine is assumed to be in instantaneous equilibrium. On the contrary, compressibility effects and variations of the fluid thermodynamic state are accounted for within each component model, where their influence is stronger and essential for an accurate representation of individual component operation. Due to the absence of an explicit time scale, the simulation of the start-up and shutdown transients, along with the representation of time-dependent valve actuation, is not suitable within the quasi-steady framework introduced in this work.

The model assumes no two-phase fluid condition throughout the entire system. In particular, the cooling jacket is assumed to operate mainly in supercritical regime, which is reached within the cooling channels and maintained up to the outlet. To ensure the consistency of this assumption, dedicated checks are implemented to verify that the local pressure remains above the saturation pressure. Operating points violating this condition are flagged as unfeasible and excluded from the solution domain.

The hydrogen isomer that is assumed to be used throughout the whole engine is parahydrogen: this is a lower-energy isomer that at ambient temperature composes around 25% of normally found hydrogen. However, at cryogenic temperature, it is the only isomer found, and the speed at which it converts back to ortho-hydrogen is negligible compared to the time it takes the hydrogen to move from the tanks to the combustion chamber [26].

Real-fluid thermodynamic properties are employed throughout the model. Thermophysical quantities such as density and enthalpy are evaluated using CoolProp to ensure consistency under cryogenic and near-critical conditions. This choice is particularly relevant in the cooling jacket, turbomachinery, and combustion chamber, where strong variations in pressure and temperature require accurate enthalpy evaluation for the energy balance.

The propellant feeding lines are represented through a lumped-parameter approach. In these sections, distributed friction and minor losses are condensed into equivalent pressure drops. This choice is motivated by the limited availability of detailed geometric information in the open literature. Introducing a distributed line model would require arbitrary assumptions on lengths, diameters, roughness, and local loss coefficients. The lumped representation therefore provides a physically consistent simplified approach.

The engine valves are assumed to be externally regulated, and no active control law is implemented. The opening positions of the TCV and OCV are specified as problem inputs, along with the thermodynamic states of the propellants in the tanks. Starting from the nominal valve opening configuration, controlled variations of these parameters allow the exploration of off-nominal operating conditions that the engine model can reach without violating its validity.

From a numerical perspective, the problem is treated as a strongly coupled nonlinear system. The solution is obtained via a set of nested, iterative loops that enforce mechanical and thermal balances. Convergence criteria are based on the relative residuals of selected quantities, with tolerances typically in the range of 10^{-3} to 10^{-4}. Convergence failures may indicate either the physical infeasibility of the operating point or limitations in the modeling assumptions. Table 3.1 summarizes the main assumptions discussed in this section for the quasi-steady model implemented.

Table 3.1: Main modeling assumptions of the quasi-steady framework

Category	**Assumption**	**Scope**
System formulation	Operation described as independent quasi-steady equilibrium states	No explicit time dependence
Mechanical modeling	Shaft rotational dynamics neglected ($\dot{\omega} = 0$); power balance enforced	No inertial or transient mechanical effects
Fluid Network	No mass accumulation at system level	No global compressibility dynamics
Phase regime	No two-phase flow allowed; cooling jacket assumed supercritical	Saturation crossing excluded
Thermodynamic modeling	Real-fluid thermodynamic properties (CoolProp)	Accurate cryogenic and near-critical behavior
Feed-line representation	Feed lines modeled with lumped pressure losses	No spatially distributed friction modeling
Valve modeling	Valve openings prescribed; no control law implemented	Off-design explored parametrically
Working fluid assumption	Parahydrogen assumed throughout	Ortho–para conversion neglected
Numerical treatment	Strongly coupled nonlinear system solved iteratively	Convergence required for physical admissibility

The modeling framework introduced in this chapter is intended for the analysis of the engine nominal operating conditions. Limited modifications are required to extend the formulation of the nominal setup to off-nominal conditions, achieved through valves modulation. This extended formulation will be presented in the Chapter 4.

3.2. System Decomposition and Coupling Strategy

The overall engine model is decomposed into a set of individual subsystems, each representing a different component of the propulsion system. The main blocks considered are: turbomachinery, cooling jacket, thrust chamber (intended as combustion chamber and nozzle), feeding lines, Venturi, valves and injectors. These subsystems are then interconnected according to the real engine architecture, forming a strongly coupled system in which mechanical, thermal, hydraulic, and thermodynamic constraints are simultaneously enforced.

The turbomachinery block determines the pressure rise in both propellant branches as a function of the shaft rotational speed and the turbine power. The mechanical equilibrium condition imposes that turbine power matches the power absorbed by the pumps and mechanical losses, thereby determining the shaft rotational speed. The turbine power, in turn, is governed by the thermodynamic state of the fuel at turbine inlet, which is set by the heat absorbed during regenerative cooling. Through this mechanism, the cooling jacket provides the turbine inlet state, the turbine determines the shaft rotational speed, and the shaft speed fixes the pressure levels in the hydraulic network. As a consequence, the shaft rotational speed acts as the primary global coupling variable of the system, linking the thermal behavior of the cooling jacket to the hydraulic pressure levels in both propellant branches.

The hydraulic network distributes the pressure levels set by the turbomachinery across feed lines, Venturi, injectors and valves, thereby determining the mass flow rates delivered to the thrust chamber. The continuity of mass and pressure-loss relations ensures consistency between the pump outlet conditions and the flow rates injected into the combustion chamber.

The thrust chamber provides the downstream thermodynamic closure of the system. For a given total mass flow rate and OF ratio, the chamber pressure is determined by the choked flow condition at the nozzle throat, which relates the mass flow rate to the combustion chamber characteristic velocity and throat area. In this way, chamber pressure is not imposed a priori but emerges from the interaction between the propellant feed system and the choking constraint.

The overall operating condition of the engine is therefore defined by the simultaneous satisfaction of three coupled constraints: mechanical equilibrium at the shaft, hydraulic continuity in the propellant network, and thermodynamic closure at the nozzle throat. The thermal interaction in the cooling jacket connects these constraints by determining the turbine inlet state and, consequently, the shaft rotational speed.

The detailed formulation of each component model is presented in Section 3.3, while the numerical solution strategy is described in Section 3.4.

3.3. Components Modeling

3.3.1. Thermophysical Propellant Properties with CoolProp Library

Since the selected engine uses cryogenic propellants in liquid form, far from their ambient conditions, and undergo significant temperature and pressure changes during their journey through the engine, a fast, precise, and reliable way to estimate their thermophysical properties during the cycle is needed.

The adopted solution is CoolProp, an open-source thermophysical property library implementing equations of state and transport models for pure and pseudo-pure fluids. CoolProp relies on formulations consistent with those used in REFPROP, a widely recognized software for fluid property prediction developed under NIST [27]. Unlike REFPROP, which was originally developed for more restricted computational environments, CoolProp can be seamlessly integrated into modern programming platforms such as Python and MATLAB, providing greater flexibility for numerical modeling [28].

Given a couple of allowed thermodynamical inputs (pressure, temperature, density, entropy, enthalpy, and others), any other thermodynamical or transport property can be computed. Moreover, it can easily identify phases and critical conditions.

Although its execution is fast, the number of queries required by the whole RL10 program is enormous. This led to the precomputation of many values and their insertion into property tables, which were then interpolated to retrieve the values to be used. The only drawback is the size of these precomputed tables and functions: as the tables get more accurate their size grows, and loading time increases accordingly. Anyway, it was deemed beneficial, since the loading time was far lower than the standard CoolProp calls.

3.3.2. Turbomachinery

The turbomachinery subsystem consists of the fuel pump, the oxidizer pump, and the turbine, all mounted on a common shaft and coupled through a quasi-steady mechanical power balance. Although these elements are physically and energetically interconnected, their modeling approaches are presented separately in the following sections for clarity.

Pumps The fuel pump consists of two centrifugal stages separated by an interstage duct, in which the interstage cool-down valve (FCV-1) is located. This valve is used to vent fuel during the engine start procedure. Due to the presence of FCV-1, each pump stage is modeled as a separate component. The LOX pump is a single-stage centrifugal impeller driven by the fuel turbine through a gear train. As a result, the LOX pump operates at a lower shaft speed than the fuel pump. Since the pumps share a centrifugal architecture, a single pump model has been developed and tailored to represent each stage. From a system-level perspective, the objective of these components is to provide the pressure rise required to sustain the propellant mass flow rates imposed by the expander-cycle architecture, while their operating point is determined by the balance between hydraulic demand and available mechanical power.

In the present work, the pump model is based on dimensionless performance maps for head and torque provided by Pratt & Whitney and documented in Binder [10], and follows the reference formulation proposed by Di Matteo [5]. Since the present work relies on the same manufacturer-provided performance maps as the reference model, the pump formulation at nominal operating conditions is necessarily equivalent. The originality of the present approach therefore lies not in redefining the component model, but in its integration, extension, and critical assessment within a quasi-steady system-level framework.

The pump performance curves for head and torque provided by P&W cover the full range of operating conditions required by the present analysis. This is achieved by combining available test data with generic performance trends, resulting in an extended map expressed as a function of the dimensionless parameter θ, defined as:

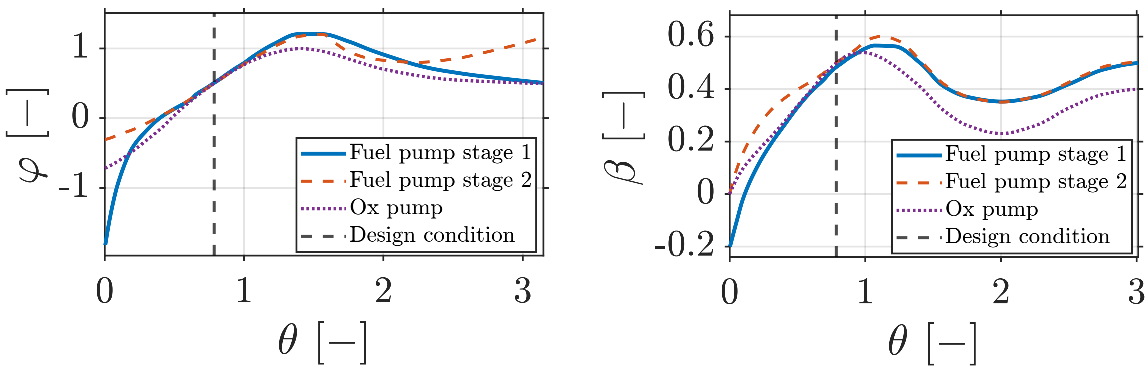
$$\theta = atan(\nu/n) \qquad \nu = \frac{Q}{Q_{nom}} \qquad n = \frac{\omega}{\omega_{nom}} \tag{3.1}$$

where ν and n are respectively the volumetric flow ratio and the speed ratio. The head TDH and torque τ are normalized using the values of torque, head and volumetric flow at the point of maximum pump efficiency taken from P&W in [10], and reported in Table A.2 in Appendix A. Moreover, starting from the normalized quantities, dimensionless head φ

and torque β are defined as follows:

$$\varphi = \frac{TDH/TDH_{nom}}{\nu^2 + n^2} \qquad \beta = \frac{\tau/\tau_{nom}}{\nu^2 + n^2} \tag{3.2}$$

This method has been introduced by the manufacturer to eliminate the singularities associated with zero values of flow rate or rotational speed, which is essential when dealing with off-nominal conditions. Note that since the maps come from an US producer, all the quantities need to be converted in Imperial Units. Additional maps provided by P&W were used to apply a corrective factor to the pump torque, as a function of the rotational speed ratio. Figure 3.2 shows the performance maps provided for the pumps.



(a) Extended Head Maps.

(b) Extended Torque Maps.

Figure 3.2: Pumps Performance Maps. Credits: [10]

These maps can be used to compute the outlet total pressure and the torque absorbed by the pump once the volumetric flow rate and the shaft rotational speed are known. To retrieve the nominal pump operating point, an iterative procedure is introduced, originally proposed by Di Matteo [5] and shown in Figure 3.3. The necessity of this procedure arises from the different definitions of total dynamic head adopted by the manufacturer and by the thermodynamic formulation derived from the pump enthalpy rise associated with the pressure increase across the pump:

$$TDH_{P\&W} = \frac{P^0_{out}}{\rho_{out}} - \frac{P^0_{in}}{\rho_{in}} \qquad TDH = \frac{h^0_{out} - h^0_{in}}{g} \tag{3.3}$$

At nominal conditions, the pump operating point is not determined by the downstream hydraulic demand, but by the intrinsic capability of the pump to deliver a physically consistent pressure rise under its design operating parameters. By enforcing the consistency between these two formulations through an iterative matching process, the nominal pump operating point can be identified, in particular the reference mass flow rate and the outlet thermodynamic state. This distinction becomes relevant when real-fluid thermodynamic properties are considered in the pump model. In the ideal incompressible limit, the two definitions of total dynamic head are equivalent, and the iterative matching procedure would not be required.

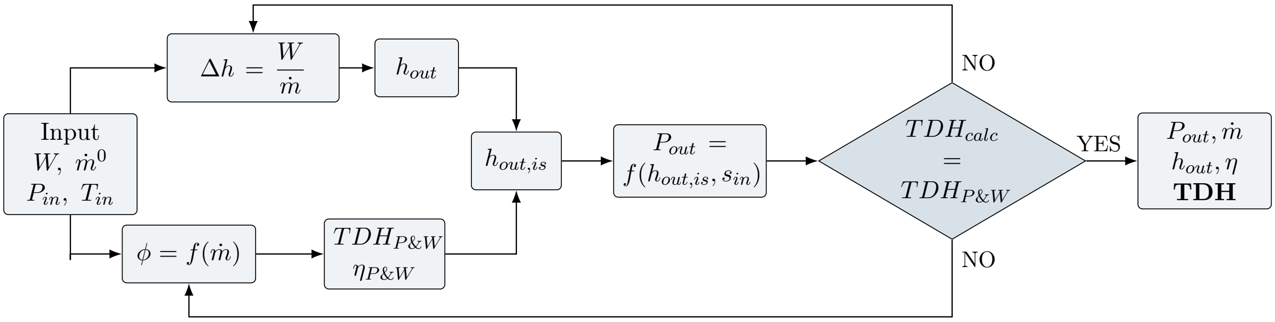


Figure 3.3: Iterative procedure for engine nominal parameters determination. Credits: [5]

Within this framework, to achieve consistency across the different formulations, a modified pump efficiency η is introduced. This quantity is directly derived from the real pump efficiency maps provided by Pratt & Whitney and reported in [10]. It does not represent a new physical efficiency of the pump, nor a tuning parameter introduced to enforce nominal matching conditions. Instead, it serves as a conversion factor between the hydraulic and thermodynamic definitions of total dynamic head, allowing the enthalpy-based formulation of pump work to remain consistent with the manufacturer's performance maps.

Since the efficiency maps are expressed as functions of the pump flow coefficient ϕ, the latter is evaluated using a correlation proposed by Binder, which relates it to the pump operating conditions and the pump geometry:

$$\phi = (\pi^2 D_{out,pump}^2\ b_{blade,pump}\ 0.004329)^{-1}\ \frac{Q}{\omega} \tag{3.4}$$

where $D_{out,pump}$ is the impeller exit diameter and $b_{blade,pump}$ is the impeller exit blade height. Again, it is important to note that, since the maps come from an US producer, all the quantities need to be converted to Imperial Units. Since this correlation is valid only within a limited range of flow coefficients corresponding to the available experimental data in [10] (see Figure A.2 in Appendix A), the use of the modified efficiency introduces an explicit validity limit for the pump model, particularly when extending the analysis toward off-nominal operating regimes. The resulting validity domain is reported in Figure A.3 in Appendix A.

A constant leakage flow is assumed in the fuel pumps to account for bearings cooling. Its value is taken from literature at nominal conditions [25] and applied throughout the operating range, as the leakage remains negligible compared to the main pump mass flow rate. In addition, minor corrective factors are introduced to compensate for inaccuracies arising from the digitization of the manufacturer performance maps. These corrections are identified under nominal operating conditions and are subsequently kept fixed, ensuring consistency of the pump model without affecting its off-design behavior.

Turbine The RL10 turbine is a two-stage axial-flow impulse turbine, driven by gaseous hydrogen and mechanically coupled to both the fuel and oxidizer pumps via the shaft. The turbine receives the fuel after its passage through the cooling jacket channels and extracts mechanical power from the flow to drive the turbopump assembly. In the present work, the turbine is not modeled as a detailed aerodynamic component but is represented by a quasi-steady formulation, in which its operating point is determined by the mechanical power balance with the coupled pumps.

Under this quasi-steady formulation, the turbine model relies on the following simplifying assumptions. First, the turbine is assumed to operate entirely in subsonic regime, without reaching choked flow conditions, even under off-nominal operation. This assumption is justified by the presence of the Venturi upstream, which represents the primary flow-limiting element in the fuel branch. Second, the working fluid within the turbine is not treated as an ideal gas. Instead, the real fluid thermodynamic properties are evaluated at the inlet and outlet states, and then a locally averaged specific heat capacity at constant pressure c_p is computed and assumed constant throughout the turbine expansion. This approach provides a more consistent representation of the enthalpy drop while avoiding the complexity of a full real-gas model. Additionally, no leakage is considered inside the turbine, and the fuel mass flow rate is therefore conserved across the component.

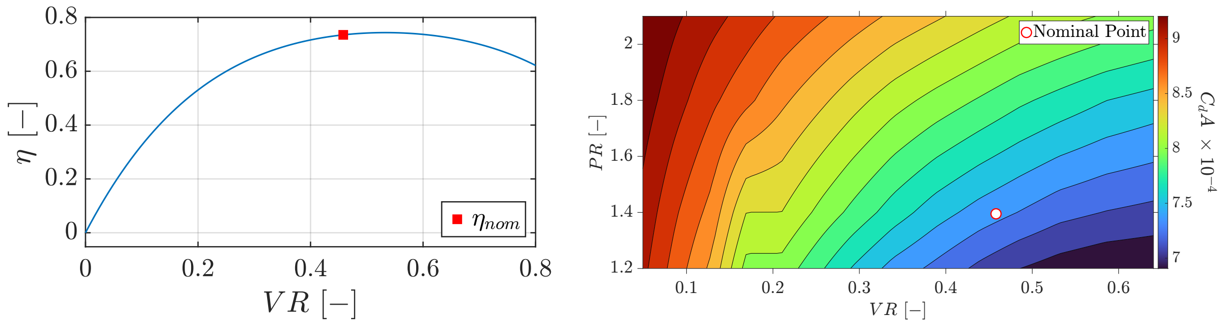
Under these assumptions, the turbine behavior is described using the performance maps provided by P&W, which constitute the core of the model implemented. The maps refer to the two stages combined. The characteristic parameters used to describe the turbine operating condition are the velocity ratio VR and the pressure ratio PR, defined as:

$$VR = \frac{U}{C_0} = \frac{\sqrt{2}\ D_{turb}\ \omega\ \frac{\pi}{60}}{\sqrt{2\ c_p\ J\ g\ \Delta h_{is}^0}} \qquad PR = \frac{P_{in}^0}{P_{out}^0} \tag{3.5}$$

where all the quantities are expressed in Imperial Units, D_{turb} is the turbine diameter, J is mechanical-to-thermal units conversion, g $[in/sec^2]$ is gravitational acceleration [1].

[1] the $\sqrt{2}$ factor is peculiar to the RL10 turbine maps [10] and would not normally be present for a single turbine stage
Conversions adopted by [10]: $J = 9337.92$; $g = 386.1\ in/s^2$

Figure 3.4a provides the turbine total-to-static efficiency η, which accounts for all losses occurring during the turbine expansion, as function of the velocity ratio. The second map shows the effective flow area A_{eff}, defined as the product of the geometric area A and the discharge coefficient C_D, as a function of the velocity ratio at different pressure ratios. The contour plot shown in Figure 3.4b is derived from the original Pratt & Whitney data reported in [10].



(a) Turbine efficiency η map

(b) Effective area A_{eff} map

Figure 3.4: Turbine performance maps. Credits: [10].

The effective area map also defines the admissible operating range of the turbine. As shown in Figure 3.4b, the pressure ratio range is limited to [1.2, 2.1]. This represents the operative range of the turbine, beyond which the turbine behavior is not defined. Hence, this limit will be used as a validity check in off-nominal operations.

Within this admissible range, the turbine operating point is determined through an internal iterative procedure. The model operates with known total inlet conditions and mass flow rate, provided by the upstream components, while the pressure ratio is not prescribed a priori and is therefore obtained through an iterative loop. For a given shaft rotational speed ω, imposed by the mechanical shaft balance, a trial value of the pressure ratio is assumed to evaluate the isentropic enthalpy drop across the turbine. The velocity ratio VR is then computed using Eq. 3.5, and the effective flow area is retrieved using the map shown in Figure 3.4b. The corresponding turbine mass flow rate is computed as:

$$\dot{m}_{turb} = A_{eff} \sqrt{P_{in}^0 \, \rho_{in}^0 \, \frac{2\,\gamma}{\gamma - 1} \, [PR^{-\frac{2}{\gamma}} - PR^{-\frac{\gamma+1}{\gamma}}]} \tag{3.6}$$

The resulting mass flow rate $\dot{m}_{turb}$ is compared to the mass flow rate imposed by the fuel-side system, and the pressure ratio is updated until convergence is reached.

Once the turbine pressure ratio consistent with the system-imposed operating conditions is found, the turbine efficiency, total enthalpy drop and torque delivered can be computed as:

$$\eta = f(VR, PR) \;\rightarrow\; \Delta h^0 = \eta\, \Delta h^0_{is} \;\rightarrow\; \tau = \frac{\dot{m}_{turb}\, \Delta h^0}{\omega} \tag{3.7}$$

where ω is the shaft rotational speed in $[rad/s]$.
The resulting torque must satisfy the mechanical power balance with the torque absorbed by the pumps. This coupling is addressed in the system-level formulation discussed in the following sections.

3.3.3. Cooling Jacket

The cooling jacket is the component that allows thermal energy to be extracted from the combustion gases and then used throughout the whole engine. The system is composed of 180 parallel axial channels disposed in a one-and-a-half pass configuration, meaning that the inlet is positioned along the divergent portion of the bell, the channels then continue towards the exit section of the nozzle, turn around and then come back towards the injector plate, where the outlet is located. The channel geometry varies along the engine longitudinal axis, while maintaining a rectangular cross-section with semicircular ends. The geometrical construction data are presented in Table A.1 and Figure A.4a in Appendix A [5, 10, 29].

The thermodynamic state of liquid hydrogen flowing in the cooling channels is always supercritical, both under nominal and off-nominal conditions, except for fast transients such as start-up and shut-down that are not in the scope of this analysis; this leads to smooth temperature and pressure variations with no discontinuities due to phase change [10]. This assumption is validated in Figure 5.2 in Section 5.1.

Regarding the modeling, first of all the thermophysical ideal model is developed: the coolant fluidic line is discretized, and at each node a steady state heat transfer analysis between the hot exhaust gases and the coolant is performed, determining the node temperatures and the next node coolant conditions.

In order to perform the necessary computations, some geometrical information (i.e. engine bell and channel geometry and roughness), initial conditions (i.e. coolant inlet temperature and pressure) and combustion gases information (i.e. temperature, pressure and transport properties along the axial coordinate) are needed as input. The whole cooling jacket model is based on the heat flux balance between the hot gas side and the cold coolant side, defined as:

$$\dot{q}_w = H(T_{aw} - T_{lb}) \tag{3.8}$$

In order to formulate the heat balance equation, the two extreme hot adiabatic wall temperature and cold liquid bulk temperature, and the global heat transfer coefficient are needed.

Both temperatures are known: the hot side temperature is obtained from the precomputed temperature distribution along the nozzle for assigned values of P_c and O/F, while the cold side temperature is either imposed as an input or computed from the previous node along the cooling channel. The overall heat transfer coefficient H is evaluated using the analogy of thermal resistances. Since the metal thickness is small compared to the nozzle radius (0.015 at most), heat conduction through the metal is approximated as one-dimensional through a flat wall, with convective heat transfer imposed at both surfaces:

$$H = \frac{1}{\frac{1}{h_g} + \frac{s}{k_w} + \frac{1}{h_l^*}} \tag{3.9}$$

where s is the local thickness of the metal between hot and cold sides, and k_w is its conductivity. The remaining coefficients h_g and h_l^* are the convective heat transfer coefficients of the two opposites metal sides.
The two sides are hereafter separately analyzed.

Cold Side

In order to compute the cold side convective heat transfer coefficient, the Nusselt number correlation can be used:

$$h_l = \frac{Nu\, k_l}{D_h} \tag{3.10}$$

where Nu is the Nusselt number, k_l is the coolant thermal conductivity evaluated at bulk temperature, as suggested by Pizzarelli [20], and D_h is the duct hydraulic diameter.

In turn, the Nusselt number is retrieved via the Gnielinski correlation [30]:

$$Nu = \frac{f_d/8(Re - 1000)Pr}{1 + 12.7\sqrt{f_d/8}(Pr^{2/3} - 1)} \tag{3.11}$$

where f_d is the Darcy friction coefficient, Re is the Reynolds number (computed from the known mass flux and hydraulic diameter, and the retrieved viscosity as $Re = \frac{GD_h}{\mu}$), Pr is the Prandtl number (intensive property retrieved beforehand).

The cold-side Darcy friction coefficient f_d can be finally computed using the Colebrook-White equation [31]:

$$\frac{1}{\sqrt{f_d}} = -2\log\left(\frac{2.51}{Re\sqrt{f_d}} + \frac{Ke}{3.71}\right) \tag{3.12}$$

where Ke is the dimensionless equivalent roughness (ε/D_h). Since the friction factor appears on both sides of the equations and inside a logarithm, its implicit solution is computationally expensive. For this reason, admitting a maximum relative error of approximately 2.8%, it is possible to really speed up the computation using the following explicit correlation [31]:

$$\frac{1}{\sqrt{f_d}} = -0.8686 \left[\ln\left(\frac{Ke}{3.71}\right) + \ln\left(1 + \frac{64.5262}{Ke\ Re}\right)\right] \tag{3.13}$$

Another peculiar phenomenon must be considered: the metal slivers separating each channel act as fins, enhancing the heat transfer. This is usually modeled in the literature [32] as:

$$m_{fin} = \sqrt{\frac{2h_l}{k_{wall}t}} \qquad \eta_{fin} = \sqrt{\frac{2k_{wall}}{h_l t}} \tanh\left(m_{fin} b\right) \qquad h_l^* = h_l \eta_{fin} \tag{3.14}$$

where t is the thickness of the metal between channels and b is the length of the metal slivers in radial direction.

The thermophysical properties of the coolant needed for the calculations (i.e. k_l, Pr, c_p) are either known or retrievable with CoolProp (Section 3.3.1). In order to obtain physically consistent results, it is essential to evaluate each property at the most appropriate reference temperature.

Density ρ, specific heat c_p and thermal conductivity k can be evaluated at bulk temperature $T_{l,b}$, Other physical or transport properties, however, must be evaluated at different characteristic temperatures. In particular, viscosity μ is evaluated at film temperature T_{film}, while density and viscosity in the friction pressure loss correlations (ρ_w and μ_w) are evaluated at cold wall temperature $T_{w,c}$ [20], as discussed in the following paragraphs.

Hot Side

As for the convective heat transfer in the hot side, the heat transfer coefficient has been computed using the Bartz equation [33]:

$$h_g = \frac{0.026}{D_t^{0.2}} \frac{\mu^{0.2} c_p}{Pr^{0.6}} \left(\frac{P_c}{c^*}\right)^{0.8} \varepsilon^{-0.9} \frac{1}{(\frac{1}{2}\frac{T_{wg}}{T_{ch}} b_{bartz} + \frac{1}{2})^{0.68} \cdot b_{bartz}^{0.12}} \tag{3.15}$$

where b_{bartz} is the ratio between total and static temperature $b_{bartz} = 1 + \frac{\gamma-1}{2} M^2$. The other thermophysical properties of the hot gases are computed using NASA CEA (Section 3.3.5). Since they are evaluated under free stream conditions (CEA is 1-D), they are first corrected to approximate boundary-layer conditions as $x_{bl} = x\left(\frac{T}{T_{aw}}\right)^{0.85}$. The exponent is tuned based on the Sutherland equation [34]. The properties evaluated through this correction are μ, c_p and λ. The adiabatic wall temperature is precomputed from CEA data as:

$$T_{aw} = T\left(1 + Pr^{0.33}\frac{\gamma - 1}{2}M^2\right) \tag{3.16}$$

Since the RL10A-3-3A engine is equipped with a one-and-a-half-pass cooling system, there are sections along the nozzle longitudinal axis that contain 180 channels, while others contain 360. As a result, the sections with 180 channels exhibit twice the effective flow area compared to the sections where the forward and return channels are stacked side by side. This effect is accounted for by halving the heat transfer coefficient in the double-stacked sections, so that half of the area is considered in the forward direction and the other half in the return path.

Algorithm

At each node, the heat flux balance between the hot and cold sides must be satisfied. However, in order to compute the thermophysical properties required to evaluate the convective heat transfer coefficients, the hot- and cold-side metal temperatures must be initially assumed.

Once the heat flux is determined, the metal wall temperatures are recomputed directly from the heat transfer equations. If the discrepancy with respect to the assumed values exceeds a prescribed tolerance, the updated metal temperatures are adopted as refined guesses and the iterative procedure is repeated. Otherwise, if convergence is achieved, the coolant temperature and pressure at the next node are computed, accounting for friction, acceleration, duct divergence, and curvature effects.

Since fluid properties, such as c_p, at the subsequent node depend on the local thermodynamic state, an iterative predictor–corrector approach is employed. Temperature and pressure are initially assumed and subsequently corrected until convergence is reached. The state variables are updated as follows:

$$T_{k+1} = T_k + \Delta T_{heat} + \Delta T_{acc} \tag{3.17}$$

$$P_{k+1} = P_k + \Delta P_{fric} + \Delta P_{acc} + \Delta P_{curv} \tag{3.18}$$

The temperature variations are derived from the heat transfer equation and from the total enthalpy conservation, and are computed as follows:

$$\Delta T_{heat} = \frac{\dot{Q}}{\dot{m}_f c_{p,mean}} \tag{3.19}$$

$$\Delta T_{acc} = -\frac{u_{k+1}^2 - u_k^2}{2c_{p,mean}} \tag{3.20}$$

As for the pressure terms, they are related to fluid friction along the conduit walls due to surface roughness, to variations in kinetic energy associated with fluid acceleration, and to additional pressure losses associated with conduit curvature.

The friction pressure loss is computed using the Darcy–Weisbach formulation, applied at the current node:

$$\Delta P_{fric} = \frac{1}{2}\rho u_k^2 f^* \frac{L}{D_h} p_{fact} \tag{3.21}$$

The term p_{fact}, defined as $p_{fact} = p/(\pi D_h)$, is introduced to account for conduit deviation from circular cross section, hence increasing its perimeter compared to the ones intended in the formulas.

The friction coefficient f^* is a refined parameter related to the friction coefficient found in the Colebrook-White equation (Eq. 3.12). It takes into account the temperature gradient between bulk liquid and cold-wall, and also the regime of the fluid in the boundary layer, as described by Pizzarelli [20]:

$$f^* = f_d \left(\frac{T_{l,b}}{T_{w,c}}\right)^m \qquad m = -0.6 + 5.6(Re_w^*)^{-0.38} \qquad Re_w^* = G\frac{D_h}{\mu_w}\frac{\rho_w}{\rho_{l,b}} \tag{3.22}$$

The acceleration pressure loss is computed exploiting the conservation of momentum [35]:

$$\Delta P_{acc} = \frac{G_{k+1}^2}{\rho_{k+1}} - \frac{G_k^2}{\rho_k} \tag{3.23}$$

Lastly, the most complex contribution is the computation of the pressure losses due to conduit curvature. The most complete and widely accepted formulation is that proposed by Idelchik [36] for smooth bends with deflection angles lower than 180°:

$$\Delta P_{curv} = \frac{1}{2}\rho u^2 (k_\Delta k_{Re} \xi_l + \xi_{fr}) \tag{3.24}$$

where k_Δ and k_{Re} are tabulated correction factors defined as: $k_\Delta = 1 + \varepsilon^2 \cdot 10^6$ and $k_{Re} = 1$. The parameters ξ_l and ξ_{fr} are defined as follows:

$$\xi_l = A_1 B_1 C_1 \qquad A_1 = 0.9\sin(\delta) \qquad B_1 = \frac{0.2}{\sqrt{R_c/D_h}} \qquad C_1 = 1 \tag{3.25}$$

$$\xi_{fr} = 0.0175\lambda\frac{R_c}{D_h}\delta° \qquad \lambda = 0.00035\frac{R_c}{D_h}\delta° \tag{3.26}$$

3.3.4. Valves and Feeding Lines

Valves and feeding lines are essential components of the RL10A-3-3A rocket engine, interconnecting the various subsystems, regulating propellant flow throughout the system, enabling start-up, shutdown, and throttling operations, and preventing catastrophic failures through safety valves.

For the scope of this thesis, both feeding lines and valves have been modeled as concentrated pressure drops, with their characteristic parameters derived from simulation data [10]. A more detailed analysis was considered infeasible due to the limited availability of geometric and mechanical data. A mixed modeling approach was also evaluated; however, the use of a coherent and homogeneous data source was prioritized in order to ensure consistent and comparable results across all lines.

The governing relation adopted is the concentrated pressure drop formulation:

$$\Delta P = \frac{1}{2}\rho u^2 \xi = \frac{\dot{m}^2}{2\rho A^2}\xi \tag{3.27}$$

Given the reference simulation data [10], the loss coefficient ξ is computed as:

$$\xi = \frac{2\Delta P_{ref}\rho_{ref}A^2}{\dot{m}_{ref}^2} \tag{3.28}$$

For valves and lines for which the aperture area is unknown (but constant in running condition), the A term in incorporated in a new parameter $\xi^* = \frac{\xi}{A^2}$. The resulting loss coefficients are reported in Table A.1 in Appendix A.

The actual pressure loss as system variables change can be easily evaluated, as the loss coefficient is assumed to remain constant. This assumption limits the validity of the model when the flow regime changes (e.g., variations in thermodynamic state or Mach number). However, since the RL10A-3-3A has limited throttle-ability, this approximation is considered reasonable within the operating range analyzed.

Almost all valves are inserted in series with the main propellant mass flow, with the only exception being the TCV, whose function is to regulate the amount of fuel entering the turbine and thus control the mechanical work extracted. For this reason, the TCV is placed in parallel with the turbine, allowing a fraction of the flow to bypass it. This is the only valve for which the pressure drop is not directly derived from the mass flow rate. Given an initial guess for the turbine pressure drop, and since both the turbine and the TCV share the same upstream and downstream nodes, the spilled mass flow $\dot{m}_{TCV}$ can be computed. Consequently, the turbine mass flow rate is obtained as the difference between the total inlet mass flow and the bypassed flow through the TCV. The corresponding pressure drop is then evaluated, and further iterations are performed if required.

Particular attention is devoted to the modeling of the Venturi channel. Its purpose is not to measure pressure drop, but to regulate mass flow through choking at the throat. Under nominal operating conditions, the Venturi is not choked, but operates sufficiently close to the choking condition to allow regulation of mass flow variations. This configuration introduces a permanent pressure loss due to irreversibility, which is commonly modeled in the literature as a fixed fraction of the throat-to-inlet static pressure [37]. This value is determined from nominal performance data and subsequently applied during runtime using the actual pressure levels. During each execution of the Venturi model, the critical pressure at the throat is first computed assuming isentropic expansion. Critical pressure and constant entropy uniquely defines the thermodynamic state, allowing the evaluation of density and velocity (with the Mach number equal to 1 at critical conditions). Combined with the geometric throat area, this determines the critical mass flow rate. If the inlet mass flow exceeds the computed critical value, the iteration of the entire fuel line is discarded, and the newly determined critical mass flow rate is imposed, restarting the computation. Otherwise, the aforementioned pressure loss is applied and the solution proceeds downstream.

3.3.5. Thrust Chamber and NASA CEA

The thrust chamber is modeled as a zero-dimensional, quasi-steady control volume in which propellant combustion occurs, and the resulting combustion products are expanded through the nozzle to determine overall engine performance. The zero-dimensional formulation adopted for the combustion chamber serves as the thermodynamic closure of the propulsion system, imposing the choked-flow condition at the nozzle throat and thereby linking the chamber pressure to the total mass flow rate. Chamber pressure and global thermochemical properties are evaluated at representative averaged conditions, without spatial resolution of the internal flow field.

A spatial discretization is instead introduced within the cooling jacket subsystem. In that case, local heat transfer and fluid property variations along the flow path must be accounted for to ensure consistency of the thermal balance. The discretization remains confined to the cooling jacket and does not imply a distributed treatment of the combustion chamber itself. The chamber provides global boundary conditions to the cooling model, while the cooling jacket resolves local heat exchange along the chamber wall.

The thermochemical state of the combustion products is determined using NASA Chemical Equilibrium with Applications (CEA). NASA CEA is a thermochemical equilibrium solver developed by NASA to evaluate the equilibrium composition and thermodynamic properties of reacting mixtures. For a specified set of reactants and imposed chamber pressure and oxidizer-to-fuel ratio OF, CEA determines the chemical equilibrium state by minimizing the total Gibbs free energy of the system, and provides the ideal thermodynamic and performance quantities, including chamber temperature, ideal characteristic velocity and thrust coefficient [38].

Given the objective of the present block, NASA CEA is employed using the `rocket` problem formulation, with `Infinite Chamber` assumption. A frozen-flow assumption is adopted for the nozzle expansion downstream of the throat, implying that the chemical composition is assumed to remain fixed during the expansion. This formulation represents the standard framework commonly adopted in thrust chamber modeling, allowing the evaluation of ideal thermodynamic and performance properties in a manner consistent with quasi-steady system-level analyses.

The direct use of NASA CEA within the system-level solver would significantly increase the computational effort. To avoid repeated calls to the CEA solver at each iteration, a set of pre-generated lookup tables, defined as function of chamber pressure P_c and oxidizer-to-fuel ratio OF, is introduced. During the simulation, the required thermochemical and performance properties are obtained by interpolating these tables, substantially improving the computational efficiency of the algorithm.

In the thrust chamber model, the combustion chamber and nozzle are treated as a coupled block, with the total mass flow rate and OF ratio imposed by the upstream propellant feed system. Chamber pressure is not fixed a priori, but is determined through an iterative cycle based on the choked flow condition at the throat. When choking condition is assumed, total mass flow rate in combustion chamber can be defined as:

$$\dot{m}_{chok,CC} = \frac{P_c \, A_{throat} \, C_{d,nozzle}}{c^*} \tag{3.29}$$

where A_{throat} is nozzle throat cross sectional area, $C_{d,nozzle}$ is the discharge coefficient, and c^* is the effective characteristic velocity. The effective characteristic velocity c^* is obtained as:

$$c^* = \eta_{c^*}(P_c, OF) \; c^*_{ideal} \tag{3.30}$$

where c^*_{ideal} is provided by NASA CEA, and η_{c^*} is a lumped efficiency parameter, provided by [10], which takes into account of all the inefficiencies and losses occurring in the combustion process.

Once the chamber pressure is consistent with the operating conditions imposed by the system, the performance parameters can be evaluated. The thrust coefficient is obtained from the ideal value supplied by NASA CEA, and then another lumped efficiency parameter, η_{c_T}, provided by [10], is applied. Then thrust T and specific impulse I_{sp} are evaluated:

$$c_T = \eta_{c_T}(P_c, OF)\ c_{T,ideal} \ \rightarrow\ T = c_T\ A_{throat}\ P_c \ \rightarrow\ I_{sp} = \frac{T}{\dot{m}\ g} \tag{3.31}$$

In conclusion, the combustion chamber provides the thermodynamic closure of the propulsion system, linking propellant feed conditions to overall engine performance.

3.4. Solution Algorithm and Numerical Strategy

In the present section, the solution algorithm adopted for the quasi-steady simulation of the engine is presented, along with the numerical and iterative strategies employed to ensure the convergence of the coupled system.
The objective of the algorithm proposed in this section is the evaluation of the engine equilibrium point under nominal operating conditions, accounting for the strong coupling between the different subsystems. Since the integrated engine system can't be solved explicitly, an iterative solution framework is required. The iterative process is interpreted as a numerical procedure to enforce physical equilibrium conditions.

3.4.1. Structure and Iterative Logic of the Solution Algorithm

The flowchart presented in Figure 3.5 reports the logical procedure followed in the solution algorithm. Its objective is to highlight the sequence according to which the different subsystems are linked together, emphasizing the coupling mechanism and presenting the proposed iterative solution.

The solution algorithm is therefore organized around two nested iterative levels:

- **Inner loop:** enforces the local mechanical equilibrium of the turbomachinery shaft by satisfying the torque balance between the turbine, the pumps, and the mechanical losses, determining the shaft rotational speed ω. This condition represents a necessary requirement for a physically admissible operating point.
- **Outer loop:** ensures the global consistency of the engine system and the choked-flow condition at the nozzle throat by closing the mass flow balance at chamber

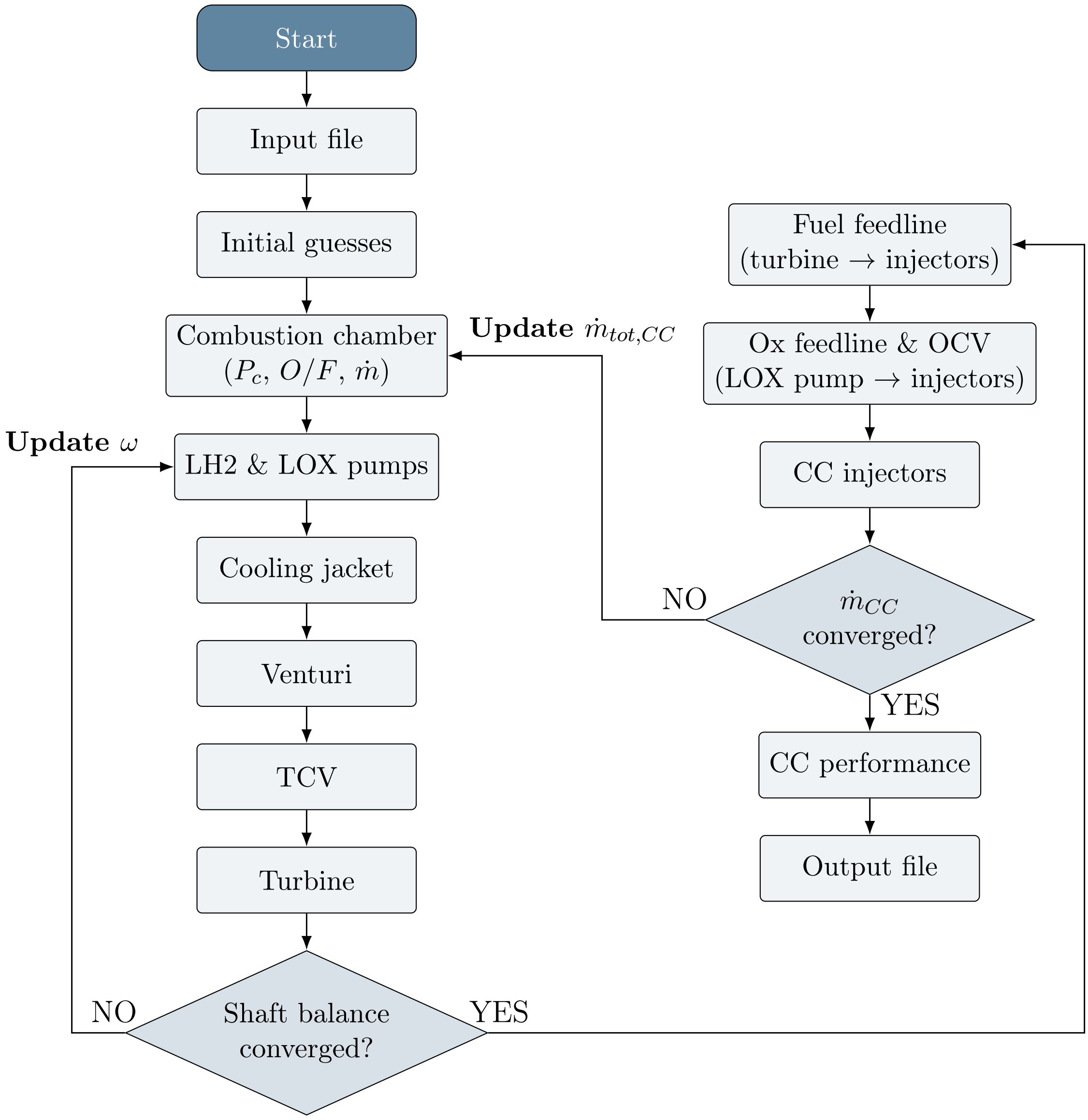


Figure 3.5: RL10A-3-3A simulation algorithm flowchart.

level.

The interaction between these two levels defines the overall solution strategy. The detailed sequence of operations is described in the following.

In principle, the outer-loop closure can be formulated either in terms of chamber pressure P_c or in terms of total mass flow rate entering the combustion chamber $\dot{m}_{tot,CC}$. A sensitivity analysis of the nominal formulation, reported in Appendix B, was carried out to assess the conditioning of the selected closure variable. The response of the cooling jacket and turbine subsystems to perturbations of P_c revealed a very weak dependency, as shown in Figure B.1. This behavior is directly related to the pump model adopted in the nominal

formulation, which evaluates pumps at their design operating conditions and constrains the delivered mass flow rate. Without an independent hydraulic closure loop on each propellant branch, chamber pressure cannot be a fully independent variable, but rather the result of the upstream hydraulic equilibrium imposed by the pump characteristics. Therefore, perturbations of P_c do not significantly modify the subsystem response, leading to a weakly sensitive outer residual. This limitation motivates the extended solution strategy presented in Chapter 4, where independent hydraulic branch closures restore the sensitivity of the outer equilibrium problem.
Conversely, perturbations of the total mass flow rate shows a stronger subsystem response, as shown in Figure B.2. For this reason, the nominal outer closure equation is formulated in terms of $\dot{m}_{tot,CC}$.

Once established the rationale behind the selection of the outer-loop variable, the structure of the overall solution algorithm can now be discussed. The solver starts by defining engine constants and geometrical parameters, and loading all the pre-generated lookup tables and performance maps required by each subsystem. Only TCV and OCV opening area ratio are user-imposed parameters.
The system variables iteratively updated in the solver are the shaft rotational speed ω, and the total mass flow rate entering the combustion chamber $\dot{m}_{tot,CC}$. Additional variables required by specific subsystems, such as the pump inlet mass flow rates $\dot{m}_{in,pump}$, chamber pressure P_c, OF ratio, and turbine pressure ratio PR, are initialized at this stage and updated within their respective blocks.

Once the initial guesses are defined, the combustion chamber block enforces the choked-flow condition (Eq. 3.29) to determine the chamber pressure P_c consistent with the current value of $\dot{m}_{tot,CC}$. This condition is satisfied through an internal iterative procedure within the combustion chamber model. The resulting chamber pressure represents a key coupling point in the algorithm, since the cooling jacket model requires the updated thermodynamic state of the combustion products.

Within the feed systems, the mass flow rates through the pumps, and consequently through the feed lines, are not imposed externally but are internally determined by the pump models through the iterative procedure described in Figure 3.3. Once the pump operating conditions are determined, the solution proceeds along the fuel branch. The resulting fuel mass flow rate is propagated through the feed line and the cooling jacket, where liquid hydrogen undergoes phase change and gains thermal energy. The fuel then flows through the Venturi, which acts as a flow regulator and may impose a local choking condition. It therefore not only influences the overall hydraulic equilibrium, but also defines the admissible operating domain in the fuel branch. Downstream of the Venturi, along the feeding line connecting it with the turbine, there is the TCV. This valve regulates the turbine inlet mass flow rate by spilling part of the flow through a bypass line. By changing its opening ratio, the fraction of fuel effectively expanding in the turbine is modified, thereby shifting the turbine operating point. The turbine then extracts mechanical power from the heated fuel and closes the mechanical loop of the system, contributing to the shaft power balance that determines the rotational speed.

Due to the presence of the turbine bypass regulated by the TCV, a local iterative procedure is implemented to ensure thermodynamic consistency between turbine mass flow $\dot{m}_{turb}$ and the mass flow spilled by the valve $\dot{m}_{TCV}$. Starting from a guess of the turbine outlet total pressure, $\dot{m}_{TCV}$ is computed from the corresponding valve pressure drop and the associated turbine mass flow is determined. The turbine block is then solved and the outlet pressure is updated until consistency is achieved. This procedure prevents an incompatible thermodynamic state downstream of the bypass.

At this point, mechanical consistency between the driving turbine and the pumps is enforced through the shaft torque balance equation:

$$\underbrace{I\,\frac{d\omega}{dt}}_{\text{quasi-steady assumption}} = 0 = \tau_{turbine} - (\sum \tau_{pumps} + \tau_{losses}) \tag{3.32}$$

The shaft rotational speed ω is iteratively updated until the mechanical equilibrium condition is satisfied.

Once mechanical equilibrium is achieved, the pressure losses along both fuel and oxidizer feed-lines are evaluated until just after the injectors. Here the total mass flow rate resulting from the system solution is compared with the current value of $\dot{m}_{tot,CC}$. If the resulting residual is lower than the tolerance imposed, the thrust chamber performance is retrieved and the simulation is concluded. Otherwise, the initial guess of mass flow rate entering the combustion chamber $\dot{m}_{tot,CC}$ is updated and the solution procedure is repeated until convergence is obtained.

3.4.2. Convergence Criteria and Numerical Stability

The convergence criteria adopted in this work rely exclusively on the physical closure conditions introduced in the previous section. They are defined on the shaft rotational speed and on the total mass flow rate for the two main iterative levels, and on additional specific quantities for the local iterative procedures within individual blocks. However, all convergence checks follow the same philosophy: only relative residuals are evaluated, scaled with respect to the corresponding reference values. This approach ensures consistency across different operating conditions and prevents the convergence assessment from being biased by the absolute magnitude of the quantities involved. The tolerances are selected according to the level of accuracy required by each variable. A conservative approach is adopted in selecting the magnitude of the tolerances, to prioritize numerical robustness over convergence speed. In Tables A.3 and A.4 (Appendix A), the iterative structure of the solver and the corresponding convergence tolerances and under-relaxation factors are summarized.

The numerical solution of both system-level and local iterative loops relies on fixed-point iterations combined with relaxation factors. These factors are selected on a case-by-case basis to achieve a balanced compromise between solution accuracy and convergence efficiency, with the explicit objective of preventing non-physical oscillations or overshoot phenomena during the iterative process.

In the shaft torque balance, turbomachinery inertia is introduced as a scaling quantity for the relaxation of the rotational speed update in order to provide a physical meaning to the numerical treatment. Within the present quasi-steady framework, inertia does not represent a dynamic effect, but serves exclusively as a numerical stabilization tool.

Additional checks on the physical admissibility of the solution are implemented within the solver. When such conditions are violated, the solution process is interrupted and a new set of initial guesses is generated. The same approach is adopted when the maximum number of iterations is reached, indicating that the solver is unable to converge to a consistent equilibrium point under the current conditions. A detailed description of these physical admissibility checks is provided in Chapter 4, where all the corresponding conditions are explicitly listed and summarized in Table 4.1.

3.5. Components Validation

This section presents a component-level validation performed for the turbomachinery and cooling jacket subsystems. The analysis is carried on fixing the shaft rotational speed at Binder nominal value [10] ($\omega = 31494\ rpm$) and the inlet conditions for each component. The model predictions are then compared with the corresponding nominal performance values provided by [10] and shown in Figure A.1 in Appendix A. The resulting relative errors are reported in Table 3.2, Table 3.3 and Table 3.4.

The objective of this analysis is not to provide a full experimental validation of the individual components, but rather to assess the consistency and correctness of the implemented models when operated at nominal conditions.

	Fuel Pump			Oxidizer Pump		
Quantity	**Binder**	**Model**	**Err. [%]**	**Binder**	**Model**	**Err. [%]**
$\dot{m}_{\text{out}}$ $[kg/s]$	2.7587	2.7736	0.54	13.948	13.999	0.36
P_{out} $[bar]$	73.58	73.72	0.19	42.819	43.016	0.46
T_{out} $[K]$	32.02	32.31	0.90	99.222	99.207	0.02
$\tau_{1^{st}stage}$ $[Nm]$	72.93	73.46	0.72	–	–	–
$\tau_{2^{nd}stage}$ $[Nm]$	79.17	80.23	1.33	–	–	–
τ_{tot} $[Nm]$	152.10	153.68	1.04	59.78	60.19	0.69

Table 3.2: Nominal validation of fuel and oxidizer pumps blocks against Binder reference data [10].

The validation of the fuel and oxidizer pumps at nominal conditions reported in Table 3.2 shows a very good agreement with Binder reference data [10], with deviations below 2%. The residuals reported are mainly associated to digitization and interpolation uncertainties of the original performance maps, as well as the simplified correction factors adopted. Nevertheless, the results confirm the physical consistency of the model adopted.

Quantity	**Binder**	**Model**	**Err. [%]**
PR $[-]$	1.402	1.399	-0.24
$\dot{m}_{\text{turb}}$ $[kg/s]$	2.747	2.747	0.00
VR $[-]$	0.4581	0.4524	-1.24
η $[-]$	0.7353	0.7340	-0.18
ΔH $[BTU/lb]$	92.02	94.15	2.32
τ $[Nm]$	178.27	182.41	2.32
P $[kW]$	587.9	601.6	2.32
$P_{t,\text{out}}$ $[bar]$	39.72	40.11	0.98
$T_{t,\text{out}}$ $[K]$	200.61	200.00	-0.31

Table 3.3: Nominal validation of the turbine block against Binder reference data [10].

The validation of the turbine block at nominal conditions reported in Table 3.3 shows again a good overall agreement with Binder reference data ([10]), even if the errors are slightly higher than 2%. These discrepancies are mainly attributed to uncertainties in digitizing and interpolating effective area map. Moreover, part of the mismatch highlighted in Table 3.3 can be attributed to the evaluation of the inlet total density, which is computed using the ideal gas relation: $\rho^0_{in} = \frac{P^0_{in}}{R_{H_2}\ T^0_{in}}$ and directly affect the turbine mass flow as showed in Eq. 3.6. This approximation, which is consistent with a quasi-steady formulation adopted, may introduce differences in turbine inlet thermodynamic state and consequently influence the predicted turbine overall operating point. Despite these approximations, errors remain below 3%, confirming the consistency of the model.

Quantity	**Binder**	**Model**		**Err. [%]**
Q_{tot}	7714 BTU/s	7613 BTU/s	8.031 MW	-1.3
ΔT	326.3 R	322.2 R	179.0 K	-1.2
ΔP	195.7 psi	192.3 psi	1.326 MPa	-1.8

Table 3.4: Validation of the cooling jacket block against Binder reference data [10].

The validation of the cooling jacket block reported in Table 3.4 shows good adherence to the binder simulation data. The biggest error is found in the pressure drop, a parameter that is found to have a very high sensitivity to input geometrical data, in particular roughness and bell profile (curvature as a consequence): these data have limited availability and their accuracy might not be perfect [29].

Even though the number of parameters to be taken into account to check for convergence is limited, the strict interconnection between those three properties leads to incorrect setup yielding unfeasible physical conditions or unacceptable errors.

4 | Throttling and Off-Nominal Engine Modeling

Extending the analysis to off-nominal operating conditions requires a change in the problem formulation. Under off-nominal conditions, the quasi-steady approach no longer defines a single closed problem, but rather an extended search for possible engine equilibrium configurations, each potentially corresponding to a distinct operating point. This extension is parametrized through the modulation of the Thrust Control Valve (TCV) and Oxidizer Control Valve (OCV) opening ratios, which introduce additional degrees of freedom into the system. As a consequence, the solution algorithm adopted for the nominal operating condition must be extended for the off-nominal analysis.

This extended search requires well-defined and clear criteria to assess the validity of the identified equilibrium points. Not all the configurations that converge numerically can be regarded as physical admissible points. For instance, a converged solution may correspond to a condition in which the required turbine pressure ratio exceeds the admissible operating range of the turbomachinery, or where local choking conditions in the Venturi or saturation limits in the cooling jacket are violated. In such cases, mathematical convergence does not imply physical admissibility. Moreover, certain valve configurations do not admit any acceptable quasi-steady equilibrium, as the corresponding solutions violate one or more physical or modeling consistency conditions, which will be explicitly defined in the following sections. Consequently, a dedicated set of diagnostic warnings and failure criteria need to be defined, in order to characterize the obtained solutions, identify the non-admissible configurations, and assess the physical consistency of the model results.

4.1. Extension of the Quasi-Steady Solution Algorithm

The nominal algorithm is specifically formulated to determine the nominal operating point of the engine. Within this framework, the operating conditions of the turbopump

are implicitly fixed at their design point, and the mass flow rates through the two propellant branches are determined by the corresponding nominal pump performance. The downstream hydraulic load does not actively enter into the determination of the pump operating point, since the branch mass flow rates are not treated as independent unknowns but are implicitly selected by the nominal closure of the system. This aspect is crucial to understand the necessity of an extended algorithm.
As a result, the pumps effectively impose the branch mass flow rates, while variations in valve opening ratios introduce only secondary perturbations to the hydraulic network. Under these conditions, valve modulation has a limited influence on the determination of the system equilibrium point, strongly restricting the model throttling authority.

Moving away from nominal conditions, this implicit closure is no longer valid. The branch mass flow rates become unknown quantities, and no reference operating point exists toward which the system can naturally converge. The original quasi-steady formulation therefore no longer defines a closed problem, and a revised solution strategy is required to explicitly reconstruct system closure under off-nominal conditions. To achieve this, additional degrees of freedom must be introduced into the formulation, allowing valve modulation to significantly influence the branch flow equilibrium. Most importantly, the pumps must respond to the downstream hydraulic load rather than remain constrained to their nominal design operating conditions. It is important to note that this limitation is not related to the engine's physical behavior, but rather to the closure assumptions used in the adopted formulation.

4.1.1. Role of Valve Modulation as Additional Degrees of Freedom

Under off-nominal conditions, the modulation of the Thrust Control Valve (TCV) and the Oxidizer Control Valve (OCV) represents the lever through which the system is shifted from the nominal operating point. Valve opening ratios do not directly alter the overall mass flow rate in the propellant branches, nor impose a specific operating point. Their effect on the system is indirect, through the modification of the hydraulic response of the feed system. In fact, by modulating the valve opening ratio, the pressure losses along the feed lines are modified, thereby changing the hydraulic load perceived by the pumps and, consequently, their operating point, even though the component models themselves remain the same.

In the specific case of the RL10A-3-3A, the two valves influence different aspects of the system behavior. Modulation of the Thrust Control Valve does not change the total fuel

mass flow rate in the branch, but rather adjusts the fraction of flow expanded in the turbine and, consequently, the power available at the shaft. The bypassed flow is then re-injected downstream of the turbine. As a result, the shaft power and the overall thrust level are adjusted without directly imposing the branch mass flow rate.
In contrast, modulation of the Oxidizer Control Valve alters pressure losses along the oxidizer line and, consequently, the oxidizer mass flow rate. In this way, the OF ratio and the total propellant mass flow entering the combustion chamber are altered themselves. However, due to the strong coupling between pumps, cooling jacket, turbine and combustion chamber, the impact of valve modulation is inherently non-linear.
Therefore, within the off-nominal framework valve opening ratios act as an additional degree of freedom of the system, enabling the existence of different equilibrium points from nominal one. Depending on the selected valve configuration, the obtained equilibrium point could be physically admissible or not.

In the present work no valves control law is implemented. Valve opening ratios are treated as user-imposed parameters, manually selected to explore the off-nominal behavior of the engine. Valve modulation is therefore not interpreted as a control action, but as a means to investigate the structure and limits of the quasi-steady equilibrium space. The introduction of an active control law would modify the treatment of valve modulation. In the real RL10A-3-3A engine, for instance, the TCV is regulated through chamber pressure feedback. In such a case, valve openings become dependent variables linked to selected system outputs, introducing a feedback mechanism that alters the closure of the quasi-steady problem. However, this modification is outside the scope of this work, as it would require a dynamic or control-oriented framework.

4.1.2. Extended Solution Algorithm and Equilibrium Admissibility

According to all the points discussed until now, an extended version of the nominal algorithm, previously presented in Figure 3.5, is proposed. This updated procedure accounts for the influence of valve modulation on the global system closure, in order to identify equilibrium points that satisfy the relevant consistency conditions.
The extended algorithm is organized in a hierarchical structure of nested loops, each enforcing a specific system closure, such as hydraulic compatibility, mechanical equilibrium of the shaft, and combustion chamber consistency. The overall structure of the extended solution algorithm can be summarized as follows:

- **Outer loop on chamber pressure** P_c: enforces global combustion chamber consistency by matching P_c with the choked-flow condition at the nozzle throat.

- **Shaft balance loop on rotational speed** ω: enforces mechanical equilibrium between turbine power and pump power.
- **Fuel branch hydraulic loop**: enforces hydraulic consistency of the fuel feed system by iteratively adjusting the branch mass flow rate.
- **Oxidizer branch hydraulic loop**: enforces hydraulic consistency of the oxidizer feed system by iteratively adjusting the branch mass flow rate.

The logic of global outer structure of the algorithm remains unchanged with respect to the nominal formulation, while the closure variable is now the chamber pressure P_c. The extension concerns the hydraulic closure of the propellant branches, with the introduction of two additional iterative loops, one for each propellant branch. As a result, the branch mass flow rates become independent unknowns, allowing the pumps to respond to the downstream hydraulic load rather than remain constrained to their nominal design conditions. The new algorithm structure is shown in Figure 4.1 and Figure 4.2.

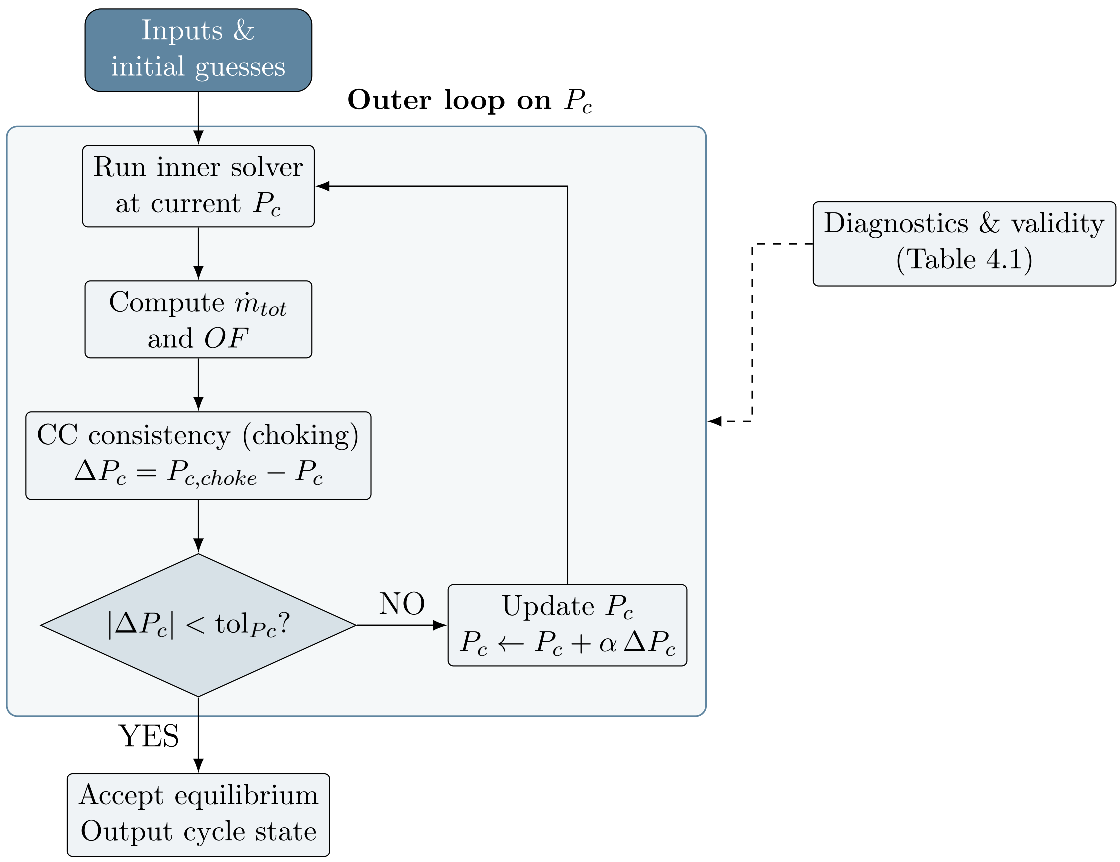


Figure 4.1: Outer-loop structure of the extended quasi-steady solution algorithm.

For a given engine configuration and valve opening ratios, the extended algorithm is initialized through a set of guessed global variables. At the highest level, as shown in Figure 4.1, an outer iteration is defined on the chamber pressure P_c, enforcing global consistency through the choked-flow condition at the nozzle throat. Chamber pressure P_c is initially guessed and iterated until consistency is achieved with the value required to ensure the choked-flow condition at the nozzle throat, evaluated using the total mass flow rate $\dot{m}_{tot}$ of the system. For a given value of P_c, the inner loops are executed to determine the corresponding $\dot{m}_{tot}$ resulting from the coupled hydraulic and shaft mechanical equilibrium of the system.

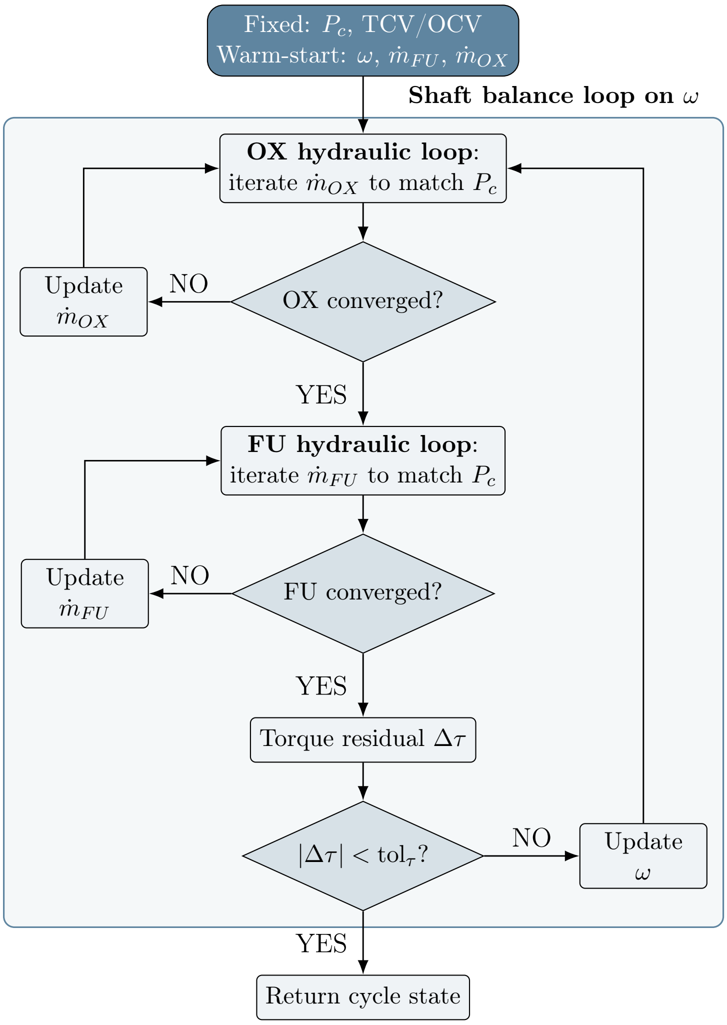


Figure 4.2: Inner-loop structure for the coupled hydraulic–mechanical equilibrium.

The inner loop, shown in Figure 4.2, is composed of three nested loops, which enforce the coupled mechanical-hydraulic equilibrium of the system. The internal organization of the subsystems follows the same logical sequence of single-block calls of the nominal algorithm, while the closure equations are enforced at different hierarchical levels.

For a given P_c, a trial value of ω is first assumed. Then the hydraulic equilibrium of each propellant branch is enforced by two local loops, one for the oxidizer and one for the fuel. The inlet mass flow rates $\dot{m}_{in}$ of the branches are treated as independent unknowns and iteratively updated until hydraulic consistency is achieved. For each branch, a guessed inlet mass flow rate is propagated along the entire feed system. The resulting flow conditions must be consistent with the fixed chamber pressure P_c imposed by the outer loop. Only when hydraulic consistency is satisfied for both branches the corresponding mass flow rates and thermodynamic states are provided to the shaft balance loop.

At this stage, the mechanical equilibrium is evaluated through Eq. 3.32. The shaft rotational speed ω is iteratively updated until the mechanical shaft balance is satisfied. When a coupled mechanical-hydraulic solution is obtained for the given P_c, the inner loop returns the corresponding propellant line states, the shaft rotational speed and the total mass flow rate. In Tables A.3 and A.4 (Appendix A), the iterative structure of the solver and the corresponding convergence tolerances and under-relaxation factors are summarized.

The extended quasi-steady algorithm described thus allows the computation of engine equilibrium operating points under off-nominal conditions. However, algorithmic convergence alone is not sufficient to ensure their physical admissibility. A dedicated set of diagnostic checks, of both numerical and physical nature, is therefore required to validate the obtained solutions.

4.2. Model Behavior at the Limits of Applicability

Under off-nominal conditions, the model may fail to converge, but it may also converge towards solutions that are physically inconsistent or lie close to the validity limits of one or more subsystem models. For this reason, a dedicated analysis of the model behavior in proximity to such critical regimes is required, in order to validate the solutions provided by the extended algorithm and to assess their physical admissibility.

These critical regimes define the limits of applicability of the proposed modeling framework. Such limits do not correspond only to numerical instabilities or lack of convergence, but also include violations of the physical assumptions adopted or of the validity limits of individual subsystem models. This distinction is essential for the correct interpretation of the off-nominal results obtained in this work. From a practical standpoint, these limits define the boundaries that the model solution must satisfy to be considered physically admissible.

4.2.1. Identification of Warning and Failure Conditions

A set of diagnostic checks is introduced within the numerical algorithm. These checks are used to distinguish physically admissible solutions from inadmissible ones and to characterize the type of criticality encountered. The diagnostic conditions are embedded directly in the solution procedure and are evaluated during the iterative process, allowing the feasibility of each candidate operating point to be assessed as the solution evolves. The resulting diagnostic framework is structured around two classes of conditions: warning conditions, which do not interrupt the solver procedure, and failure conditions, which indicate the absence of an admissible solution. The complete set of warning and failure conditions adopted in this work is summarized in Table 4.1, together with their scope, triggering criteria, and associated physical or numerical implications.

To clarify the distinction between warning and failure conditions, a representative example for each class is discussed. A typical warning condition is the occurrence of choked flow at the Venturi throat. In this case, the mass flow rate computed upstream exceeds the maximum flow rate that the Venturi can physically handle. Since the corresponding critical mass flow rate is known, the algorithm proceeds by limiting the effective mass flow rate accordingly, allowing the solution procedure to converge.

As a result, an equilibrium point can still be identified, even though it violates the Venturi operating constraint and the mass conservation in the fluid network. Such a solution is therefore flagged as a warning: the numerical equilibrium exists, but the corresponding operating point is not physically admissible within the current configuration due to the mass flow rate difference across the Venturi, hence a new mass flow rate update is required.

On the other hand, failure conditions correspond to situations that cannot be ignored by the solver and for which no physically meaningful solution can exist. For instance, the occurrence of non-physical thermodynamic states, such as negative total pressure within the circuit, makes any further continuation of the solution procedure meaningless. In these cases, the simulation is terminated and the corresponding operating point is classified as a failure.

Table 4.1: Diagnostic warning and failure criteria adopted for the assessment of off-nominal quasi-steady solutions. Unless otherwise specified, diagnostics are defined at propellant-line level and applied symmetrically to both fuel and oxidizer branches.

Flag name	Type	Scope	Trigger	Meaning / implication
Numerical and algorithmic diagnostics				
Max iter reached	FAIL	Global	Max iteration count exceeded	No numerical convergence; no equilibrium identified
TCV loop non convergence	FAIL	Turbine–TCV coupling	Internal coupling loop fails to converge	Coupling breakdown; operating point excluded
Shaft balance non convergence	FAIL	Shaft balance	Mechanical equilibrium not satisfied	No consistent shaft speed; no admissible equilibrium
Runtime exception	FAIL	Global	Runtime exception / numerical breakdown	Solution discarded
Physical inconsistency – propellant line (fuel and oxidizer)				
Static conversion failure	FAIL	Propellant line	Total–static conversion yields $P \leq 0$, $\rho \leq 0$, or NaN/Inf	Violation of basic thermodynamic consistency
Negative pressure drop	FAIL	Propellant line	$\Delta P < 0$ where losses are expected	Unphysical hydraulic behaviour
Non-physical Thermodynamic state	FAIL	Propellant line	$T \leq 0$, $\rho \leq 0$, or NaN/Inf	Non-physical state; point excluded
Chamber pressure below minimum	FAIL	Propellant line	$P_{\mathrm{cc,line}} < P_{\min}$ (e.g. 0.1 bar)	Insufficient pressure at chamber inlet; point excluded
Turbomachinery validity and physical limits				
Velocity ratio out of range	FAIL	Turbine	Velocity ratio outside validity range	Turbine model not applicable at this operating point

Continued on next page

Flag name	**Type**	**Scope**	**Trigger**	**Meaning / implication**
Flow Coefficient out of range	FAIL	Turbine	Flow coefficient outside validity range	Turbine model not applicable at this operating point
Pressure ratio saturated	WARN	Turbine	PR at map boundary	Near turbomachinery limit
Pressure ratio below minimum	WARN	Turbine	PR below minimum map value	Extrapolated regime; reduced accuracy
Negative shaft power	FAIL	Shaft	Net shaft power < 0	Mechanical inconsistency
Hydraulic choking and flow limiting				
Venturi choked	WARN	Venturi	Effective choking detected	Flow limitation; proximity to critical regime
Flow reversal	FAIL	Propellant line	$\dot{m} < 0$	Physically inadmissible flow reversal
Oxidizer-specific diagnostics				
OCV below P_{sat}	FAIL	OCV / oxidizer line	$P \leq P_{\text{sat}}(T)$ across OCV	Two-phase onset; invalid under single phase assumption
Model validity and quasi-steady limits				
Property extrapolation	WARN	Thermo-dynamics	Properties evaluated outside tabulated range	Reduced accuracy of thermodynamic evaluation

4.2.2. Interpretation of Warning and Failure Regimes

As previously introduced, warning and failure criteria are evaluated simultaneously during the solution procedure. The key difference lies in their impact on the admissibility of the operating point and on the continuation of the solver.

When a warning flag is triggered, the current operating point is marked accordingly, but the solution procedure is allowed to proceed. Warning conditions indicate proximity to the limits of applicability of the modeling framework, without preventing the existence of a numerical equilibrium. Due to the fixed-point nature of the solver, such conditions may arise during intermediate iterations and disappear as convergence is approached.

The opposite holds for failure conditions. When these occur, they completely invalidate the current solution attempt, as they indicate either physically inadmissible states, or numerical breakdowns that interrupt the simulation. To prevent this behavior, failure flags are introduced to stop the solver, mark the operating point as inadmissible, and restart the solution process with a new initial guess.

In conclusion, warning conditions delineate the boundary of the physically admissible solution space, thereby guiding the exploration of off-nominal equilibria, whereas failure conditions act as hard constraints that protect the solver from non-physical states and numerical instability. This classification provides the basis for the systematic mapping of the off-nominal operating domain presented in the following chapter, where the admissible and critical regions of the engine equilibrium space are explicitly identified.

5 | Results

This chapter presents the results obtained with the quasi-steady system-level model developed in this work. The analysis is structured to progressively assess the validity, limitations, and potential extensions of the proposed modeling framework.

First, the nominal operating point is evaluated and compared against Binder reference data [10]. This case represents the primary validation of the model, both in terms of overall system behavior and of consistency of the individual subsystem models. In fact, the identification of the nominal solution requires the simultaneous satisfaction of hydraulic, mechanical, and thermodynamic closure conditions at system level.

Starting from the validated nominal configuration, the analysis is then extended to off-nominal operating conditions through valve modulation. This investigation highlights the onset of critical regimes, numerical breakdowns, and physical limitations inherent to the quasi-steady formulation.

Finally, after identifying the limiting conditions and the main system bottlenecks, an extension analysis is performed by modifying selected components in order to improve the engine throttling capabilities. This last part aims at assessing whether the proposed modeling framework can be effectively extended and used as a basis for future developments.

5.1. Nominal Operating Condition

The nominal operating point represents the reference condition of the present work and constitutes the cornerstone for the validation of the quasi-steady modeling framework. At nominal conditions, all engine subsystems are strongly coupled, and the successful identification of a consistent nominal solution therefore validates both the individual subsystem models and the system-level coupling strategy presented in Chapter 3. Moreover, this result also confirms the physical assumptions underlying the present work, in particular the validity of the quasi-steady hypothesis at the nominal operating point. Since manufacturer data are incomplete, model predictions at nominal conditions are compared against the steady-state cycle performance data reported by Binder [10]. The results obtained in his work are presented in Figure A.1 in Appendix A. In Table 5.1 a comprehensive summary of the nominal results is reported.

Quantity	Description	Model	Binder	Error
Overall Engine Performance				
P_c [bar]	Chamber pressure	32.648	32.770	-0.40 %
O/F [-]	Mixture ratio	5.007	5.055	-0.95 %
$\dot{m}_{tot}$ [kg/s]	Total chamber mass flow	16.664	16.710	-0.26 %
F [kN]	Engine thrust	72.623	73.002	-0.52 %
I_{sp} [s]	Specific impulse	444.26	445.600	-0.30 %
RTL [-]	Rate thrust level	99.20	100.000	-0.80 %
Cooling Jacket				
Q_{CJ} [MW]	Heat pickup	8.000	8.140	-1.71 %
ΔP_{CJ} [bar]	Pressure drop	13.470	13.490	-0.16 %
Turbine				
$T_{t,in}$ [K]	Inlet total temperature	210.08	213.444	-1.58 %
PR [-]	Pressure ratio	1.3930	1.402	-0.64 %
VR [-]	Velocity ratio	0.4555	0.4581	-0.01 %
$\dot{m}_t$ [kg/s]	Turbine mass flow	2.74	2.747	-0.39 %
τ_t [Nm]	Turbine torque	178.30	178.300	+0.02 %
LOX Pump				
$P^{ox}_{t,out}$ [bar]	Discharge total pressure	42.532	42.810	-0.65 %
$\dot{m}_{ox}$ [kg/s]	Mass flow	13.890	13.950	-0.45 %
ω_{ox} [rpm]	Shaft rotational speed	12518	12598	-0.63 %
τ_{ox} [Nm]	Torque	59.381	59.800	-0.67 %
LH2 Pumps				
$P^{fu}_{t,out}$ [bar]	Discharge total pressure	72.739	73.560	-1.14 %
$\dot{m}_{fu,1}$ [kg/s]	1st stage mass flow	2.793	2.795	-0.04 %
$\dot{m}_{fu,2}$ [kg/s]	2nd stage mass flow	2.774	2.774	-0.04 %
ω_{fu} [rpm]	Shaft rotational speed	31295	31494	-0.63 %
$\tau_{fu,1}$ [Nm]	1st stage torque	72.808	72.960	-0.17 %
$\tau_{fu,2}$ [Nm]	2nd stage torque	79.511	79.200	+0.43 %

Table 5.1: Nominal operating point comparison against reference data [10].

Table 5.1 shows a very good overall agreement between the present quasi-steady model and the Binder reference data at nominal conditions. All the main performance parameters and subsystem quantities are reproduced with relative deviations generally below 1%, confirming the consistency of the modeling assumptions and the robustness of the system-level coupling at the nominal operating point. This comparison provides a quantitative assessment of the accuracy of the proposed model and establishes a consistent reference baseline for the subsequent off-nominal analysis.

The following figures are reported to assess the validity of the adopted algorithm and of the modeling assumptions. Figure 5.1 shows the shaft torque residual as a function of the rotational speed ω. At nominal conditions, a unique zero-crossing is observed, confirming the consistency of the shaft balance closure implemented in the algorithm. The zero-crossing point corresponds to the shaft rotational speed at which the power produced by the turbine exactly matches the power absorbed by the pumps and dissipated by mechanical losses. The presence of a unique zero-crossing point for the selected configuration indicates that the shaft balance loop converges toward a unique equilibrium value of ω, ensuring a well-defined mechanical closure of the system at nominal operating conditions.

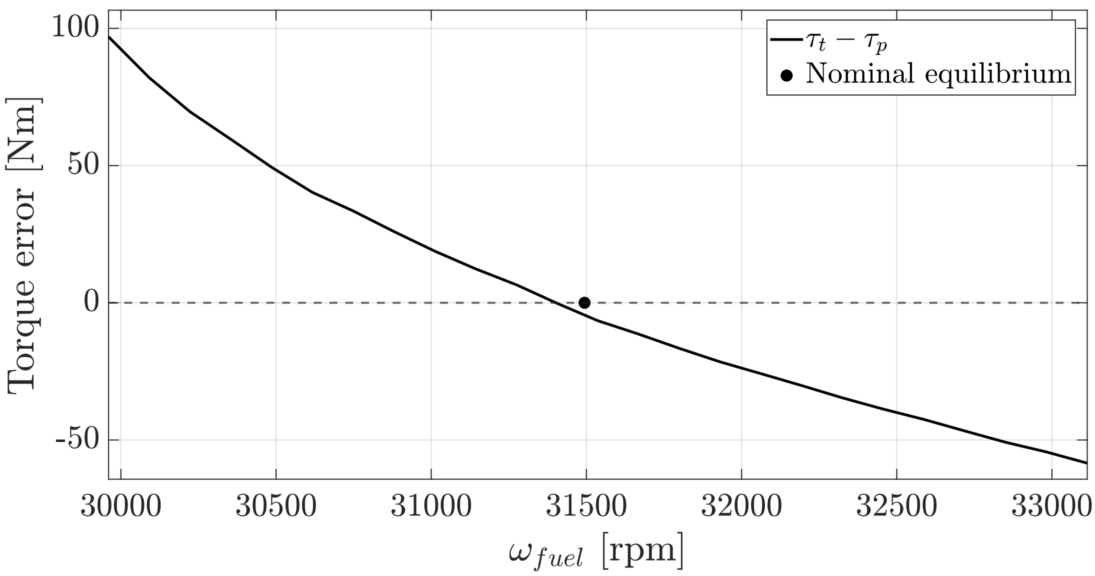


Figure 5.1: Shaft torque residual versus fuel pump rotational speed.

Figure 5.2 shows the cooling jacket pressure–temperature phase diagram for different inlet total pressures. The hydrogen saturation curve, together with the critical pressure and temperature, is reported as a reference to assess the thermodynamic regime of the coolant along the cooling path. As can be observed, even under a -35% reduction of inlet pressure, the coolant phase trajectory remains entirely above the critical limit. This result confirms that hydrogen operates in supercritical conditions throughout the cooling jacket, supporting the validity of the assumptions adopted in the cooling jacket modeling at nominal conditions.

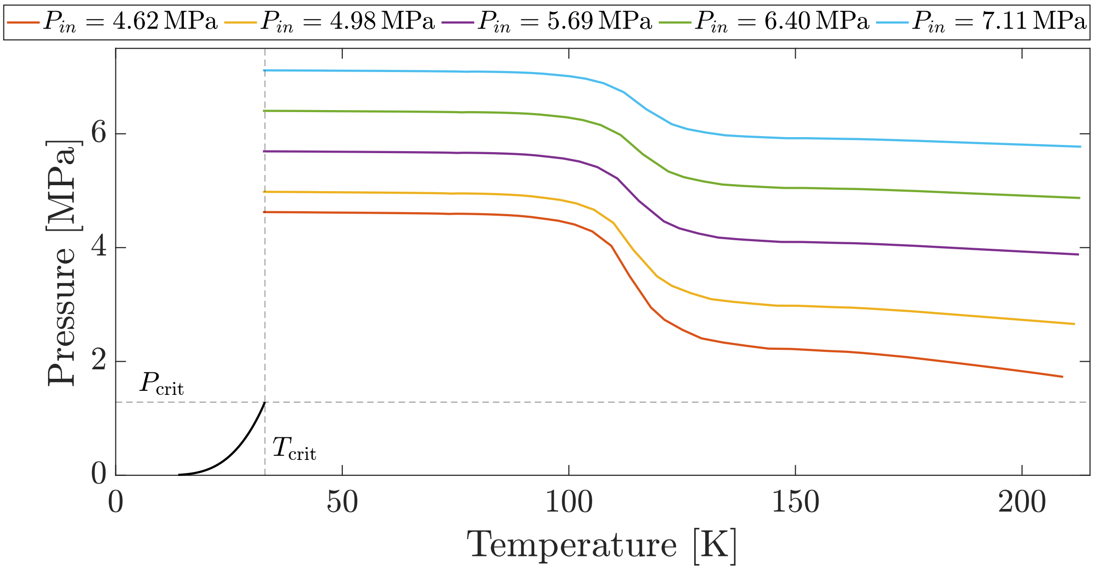


Figure 5.2: Cooling jacket pressure–temperature phase diagram.

5.2. Off-Nominal Operating Conditions

Starting from the validated nominal operating point presented in the previous section, the analysis is extended to off-nominal operating conditions for a systematic exploration of the quasi-steady equilibrium space. This time the engine is sampled as a series of independent steady states, and off-nominal operation conditions are obtained by modifying the valve opening ratios, namely the Turbine Control Valve (TCV) and the Oxidizer Control Valve (OCV).

Throttling is not treated as an imposed control action, but rather as a derived system property. It represents the capability of the coupled subsystems (turbomachinery, hydraulic network, cooling jacket, and combustion chamber) to satisfy the global closure conditions at different mass flow rates. Consequently, the identification of a throttled state coincides with the existence of a physically admissible solution for a given valve configuration.

Finally, the Rated Thrust Level (RTL), defined as the ratio between current thrust and nominal thrust as a percentage, is handled as a post-processing metric rather than a target variable. Each identified equilibrium point yields a specific RTL as an output, which is subsequently used to classify the identified operating points and define the boundaries of the admissible engine envelope.

The investigation of off-nominal conditions serves a dual purpose. On the one hand, it allows evaluating the robustness of the modeling framework when the engine departs from its reference operating point. On the other hand, it provides insight into the physical and numerical limitations of the quasi-steady approach, highlighting the conditions under which the system-level closure may become critical or fail to converge.

As the operating point moves away from nominal conditions, the strong coupling among turbomachinery, feed system, cooling jacket, and combustion chamber becomes increasingly relevant. Small variations in valve opening ratios can induce significant changes in mass flow distribution, pressure levels, and shaft equilibrium, potentially leading to choking phenomena, loss of hydraulic compatibility, or breakdown of the numerical solution procedure.

The nominal values for the valve openings are 73.8% for the OCV and 9% for the TCV[5, 39]. The extreme values for the OCV valve opening are 62.5% and 100%[39], while the extreme values for TCV are uncertain, thus a similar range between 0% and 40% has been selected. Tables 5.2 and 5.3 report the changes in system variables that occur while operating separately the TCV and OCV valves, compared with nominal conditions. Each table investigates the effect of a 10% increase and decrease in the respective valve opening ratio. This value was chosen in order to visualize the trend each system variable has in the specified valve operating direction, and to be able to compute the derivative as a finite difference.

As a reminder, the nominal conditions reported in the following tables are obtained using the off-nominal algorithm with nominal valve opening ratios. Since the off-nominal solution procedure differs from the nominal one, slight discrepancies may arise due to the different closure strategies adopted. Anyway, the differences are small (within 2%) and the values are globally aligned with the nominal results. The main aim of the off-nominal investigation is to identify global trends related to different valve opening ratios and infeasible operating conditions.

Quantity	Nominal	Closing valve	Change	Opening valve	Change
$TCV\%$ [%]	9	8.1	-10%	9.9	+10%
Overall Engine Performance					
P_c [bar]	33.096	33.210	+0.35%	33.010	-0.26%
O/F [-]	5.066	5.063	-0.06%	5.067	+0.02%
$\dot{m}_{tot}$ [kg/s]	16.923	16.979	+0.33%	16.869	-0.32%
F [kN]	73.657	73.909	+0.34%	73.467	-0.26%
I_{sp} [s]	443.677	443.726	+0.01%	443.957	+0.06%
RTL [-]	100.614	100.959	+0.34%	100.355	-0.26%
Cooling Jacket					
Q_{CJ} [MW]	8.106	8.127	+0.26%	8.088	-0.22%
ΔP_{CJ} [bar]	13.414	13.464	+0.37%	13.370	-0.33%
Turbine					
$T_{t,in}$ [K]	211.686	211.504	-0.09%	211.853	+0.08%
PR [-]	1.406	1.408	+0.08%	1.405	-0.09%
VR [-]	0.454	0.455	+0.15%	0.454	-0.11%
$\dot{m}_t$ [kg/s]	2.751	2.765	+0.52%	2.737	-0.48%
τ_t [Nm]	182.856	183.747	+0.49%	181.954	-0.49%
LOX Pump					
$P_{t,out}^{ox}$ [bar]	43.681	43.865	+0.42%	43.526	-0.35%
$\dot{m}_{ox}$ [kg/s]	14.133	14.179	+0.32%	14.088	-0.32%
ω_{ox} [rpm]	12704	12735	+0.24%	12678	-0.21%
τ_{ox} [Nm]	61.254	61.578	+0.53%	60.958	-0.48%
LH2 Pumps					
$P_{t,out}^{fu}$ [bar]	74.362	74.674	+0.42%	74.091	-0.37%
$\dot{m}_{fu,1}$ [kg/s]	2.829	2.839	+0.38%	2.819	-0.34%
$\dot{m}_{fu,2}$ [kg/s]	2.809	2.820	+0.38%	2.800	-0.34%
ω_{fu} [rpm]	31759	31839	+0.25%	31692	-0.21%
$\tau_{fu,1}$ [Nm]	74.587	75.007	+0.56%	74.219	-0.49%
$\tau_{fu,2}$ [Nm]	81.412	81.875	+0.57%	81.007	-0.50%

Table 5.2: Engine state parameters at varying levels of TCV aperture

Quantity	Nominal	Closing valve	Change	Opening valve	Change
$OCV\%$ [%]	73.84	66.46	-10%	81.22	+10%
Overall Engine Performance					
P_c [bar]	33.122	32.573	-1.66%	33.522	+1.21%
O/F [-]	5.070	4.893	-3.49%	5.207	+2.70%
$\dot{m}_{tot}$ [kg/s]	16.983	16.562	-2.48%	17.237	+1.50%
F [kN]	73.717	72.382	-1.81%	74.693	+1.32%
I_{sp} [s]	442.485	445.500	+0.68%	441.720	-0.17%
RTL [-]	100.697	98.873	-1.81%	102.030	+1.32%
Cooling Jacket					
Q_{CJ} [MW]	8.129	7.926	-2.49%	8.254	+1.54%
ΔP_{CJ} [bar]	13.466	13.418	-0.36%	13.426	-0.30%
Turbine					
$T_{t,in}$ [K]	211.713	206.430	-2.49%	215.850	+1.95%
PR [-]	1.408	1.412	+0.34%	1.402	-0.38%
VR [-]	0.454	0.456	+0.34%	0.453	-0.31%
$\dot{m}_t$ [kg/s]	2.758	2.771	+0.45%	2.738	-0.74%
τ_t [Nm]	183.527	182.640	-0.48%	183.235	-0.16%
LOX Pump					
$P^{ox}_{t,out}$ [bar]	43.782	43.987	+0.47%	43.467	-0.72%
$\dot{m}_{ox}$ [kg/s]	14.185	13.752	-3.05%	14.460	+1.94%
ω_{ox} [rpm]	12725	12687	-0.30%	12726	+0.01%
τ_{ox} [Nm]	61.527	60.196	-2.16%	62.232	+1.15%
LH2 Pumps					
$P^{fu}_{t,out}$ [bar]	74.568	73.949	-0.83%	74.759	+0.26%
$\dot{m}_{fu,1}$ [kg/s]	2.837	2.849	+0.44%	2.816	-0.73%
$\dot{m}_{fu,2}$ [kg/s]	2.817	2.830	+0.45%	2.796	-0.74%
ω_{fu} [rpm]	31815	31738	-0.24%	31813	-0.01%
$\tau_{fu,1}$ [Nm]	74.883	74.760	-0.16%	74.569	-0.42%
$\tau_{fu,2}$ [Nm]	81.739	81.603	-0.17%	81.392	-0.42%

Table 5.3: Engine state parameters at varying levels of OCV aperture

The results reported in Table 5.2 and Table 5.3 are consistent with the expected system response to variations in valve opening ratios. The TCV effectively regulates the turbine bypass flow. A reduction in the TCV effective area increases the fraction of fuel flowing through the turbine, increasing turbine power and consequently the pump discharge pressures. This results in increased chamber pressure and thrust. On the contrary, an increase in the TCV area enhances the bypass flow, reduces turbine work extraction, and leads to a decrease in chamber pressure and thrust. The OCV directly affects both thrust and mixture ratio. A reduction in the OCV effective area increases hydraulic losses in the oxidizer line, leading to a decrease in oxidizer mass flow. Since both propellants must reach the combustion chamber at compatible pressure levels, the fuel line responds by adjusting its mass flow accordingly. The combined reduction in chamber pressure and total mass flow results in a significant decrease in thrust as the OCV is progressively closed.

While the previous analyses focused on the system response in the vicinity of the nominal operating point, the following section investigates the global equilibrium structure of the engine across the entire admissible valve opening domain. The objective is to characterize how chamber pressure, mixture ratio, total mass flow rate, and thrust level evolve as the control valves are varied, and to identify the boundaries of physically feasible operating conditions. The results of this global sweep are presented in Figures 5.3 through 5.6.

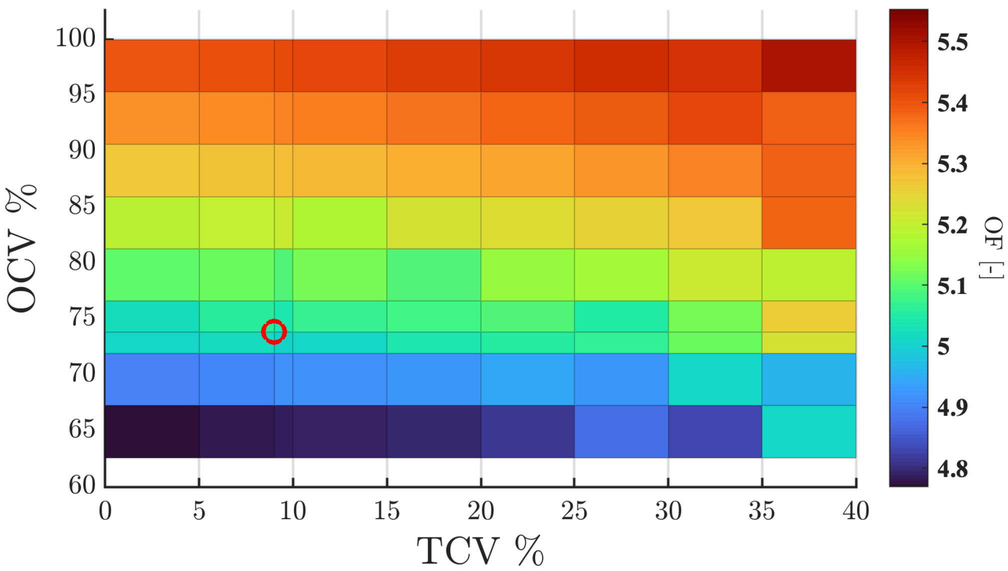


Figure 5.3: OF ratio in off-nominal conditions, as function TCV and OCV opening percentage. The red marker identifies the nominal operating condition.

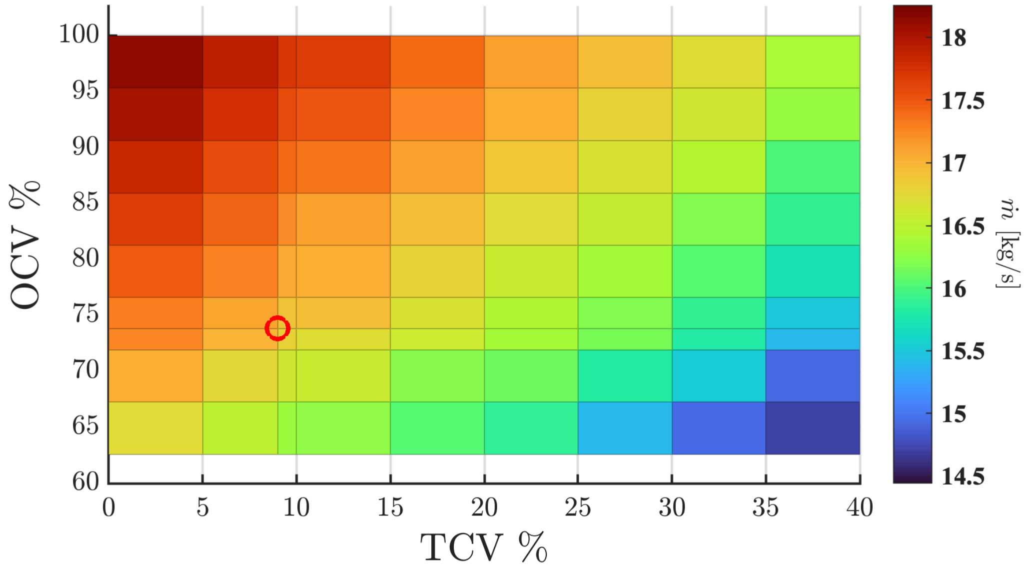


Figure 5.4: Total mass flow in off-nominal conditions, as function of TCV and OCV opening percentage. The red marker identifies the nominal operating condition.

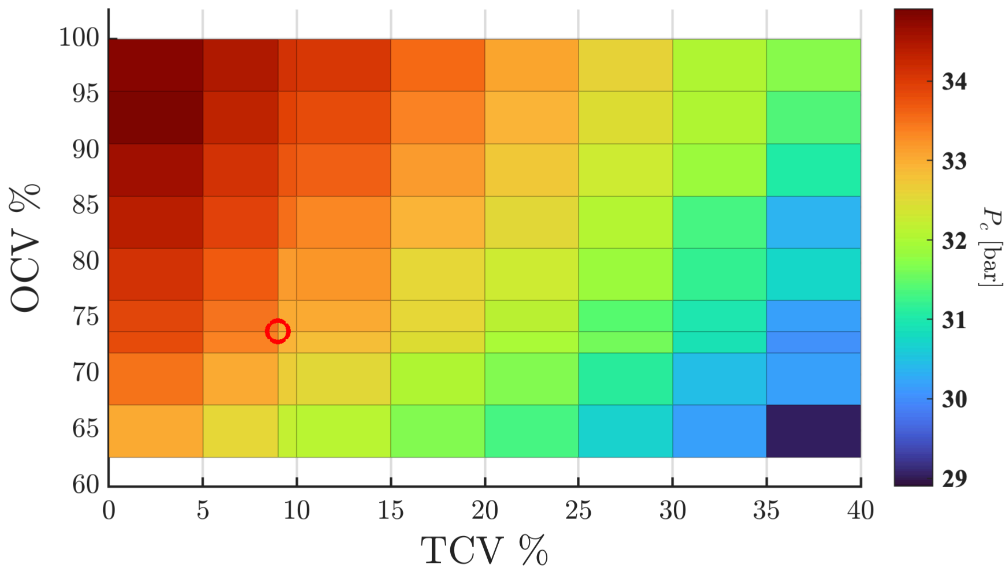


Figure 5.5: Chamber pressure in off-nominal conditions, as function of TCV and OCV opening percentage. The red marker identifies the nominal operating condition.

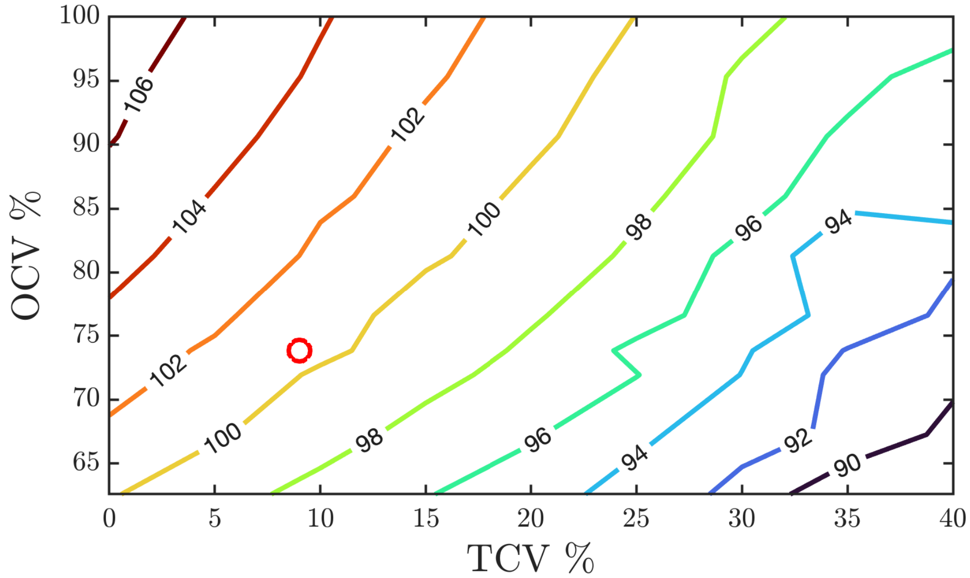


Figure 5.6: RTL in off-nominal conditions, as function of TCV and OCV opening percentage. The red marker identifies the nominal operating condition.

The global sweep highlights the distinct control roles of the TCV and OCV within the coupled engine architecture. It is evident that TCV is effective in modulating engine thrust via total mass flow and chamber pressure. Across the full TCV range analyzed, and for every value of OCV, the resulting variation in RTL is around 10%, as shown in Figure 5.6. This behavior is consistent with the physical role of the TCV in regulating the turbine bypass flow and therefore the available shaft power, which directly influences total mass flow rate and chamber pressure. On the contrary, TCV impact on the OF ratio is limited, as shown in Figure 5.3. A full TCV sweep produces OF ratio variations lower than 0.2, confirming that the TCV does not provide effective mixture control authority.

The OCV, instead, is capable of simultaneously changing both OF ratio and engine thrust level. Figure 5.3 shows that a full OCV sweep leads to an OF ratio variation of 0.5-0.6, and a corresponding RTL variation of 6-7% (Figure 5.6. This behavior comes from the direct modification of oxidizer line hydraulic losses caused by the OCV, which modifies the mass flow balance between the propellant branches and consequently impacts both chamber pressure and OF ratio.

The combined behavior of the valves outlined in this analysis demonstrates that engine performance parameters as thrust, chamber pressure, mass flow rate, and OF, cannot be independently controlled through a single valve. The achievement of well-defined performances requires precise and coordinated actuation of the two valves.

From a feasibility standpoint, the entire explored valve opening domain results in physically admissible equilibrium configurations. No violations of turbine power balance, Venturi choking constraints, or cooling jacket phase limitations are observed within the investigated range.

Despite the large sweep analyzed, the minimum achievable thrust level remains close to 90% of the nominal value. This limited throttling capability is consistent with the original design philosophy of the RL10A-3-3A, which prioritizes reliable in-space ignition and operational stability rather than deep throttling performance.

Some of the insights that were gathered relate to the numerical algorithm implemented. Since the corrections applied to the various system variables were related directly to the error via a coefficient (linear relation with a tuned constant coefficient), but the components often find themselves in non-feasible conditions, the corrections applied are not always the best fit for the specific system inputs. Other times, given the imperfect guesses, a root does not exist in the closing equation, hence the least error solution must be taken. All these imperfections related to the input data lead to very noisy relations between the variable and the associated error in the outer loop, finally failing to reach convergence. After a rough first scan of the solution space, assuming a planar relation between TCV-OCV-guess, each iteration can then be made to start with an educated guess: this leads to much cleaner results, with almost all points reaching convergence, as shown in Figure 5.7.

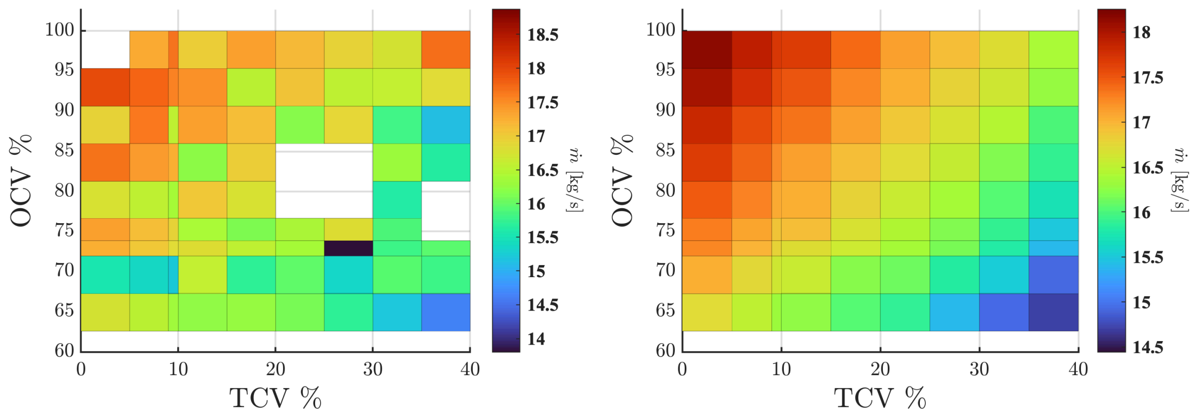


Figure 5.7: Comparison in off-nominal algorithm between uniform initial guess and custom educated guesses

The outcome of this analysis provides the necessary basis for the extension study presented in the following section, where selected system parameters are modified to investigate possible strategies for extending the operability range of the engine.

5.2.1. Critical Regimes Assessment

In this section, a targeted analysis of the mechanisms underlying the limitations of the engine throttling capability is presented. The objective is to identify and interpret the critical operating regimes that prevent the establishment of admissible quasi-steady equilibrium points below a given thrust level.

The analysis is carried out separately for the two flow paths and the results are then combined in a global status map. By exploiting the diagnostic flags introduced in Section 4.2.1, the critical regimes can be systematically tracked and classified. The ranges of shaft speed and mass flow rate are extended to explore the widest possible equilibrium space. Throughout this section, $\dot{m}_{req}$ denotes the mass flow rate imposed to the system, while $\dot{m}_{eff}$ represents the mass flow rate that the system is actually able to handle.

For a given shaft speed, once a physically inconsistent condition is encountered (e.g. negative pressure in the cooling jacket or violation of hydraulic compatibility), the analysis is not continued toward higher mass flow rates. At fixed rotational speed, increasing the required mass flow does not introduce additional degrees of freedom capable of restoring physical consistency, and therefore cannot lead the system back to a feasible equilibrium point. From a numerical point of view, once a major physical failure is reached, the solution exhibits an oscillatory behavior between acceptable, warning, and failure states, which are interpreted as numerical instability rather than physically meaningful solutions.

First, the analysis is performed at fixed shaft rotational speed, close to the nominal value. The outcomes are reported in Figure 5.8.

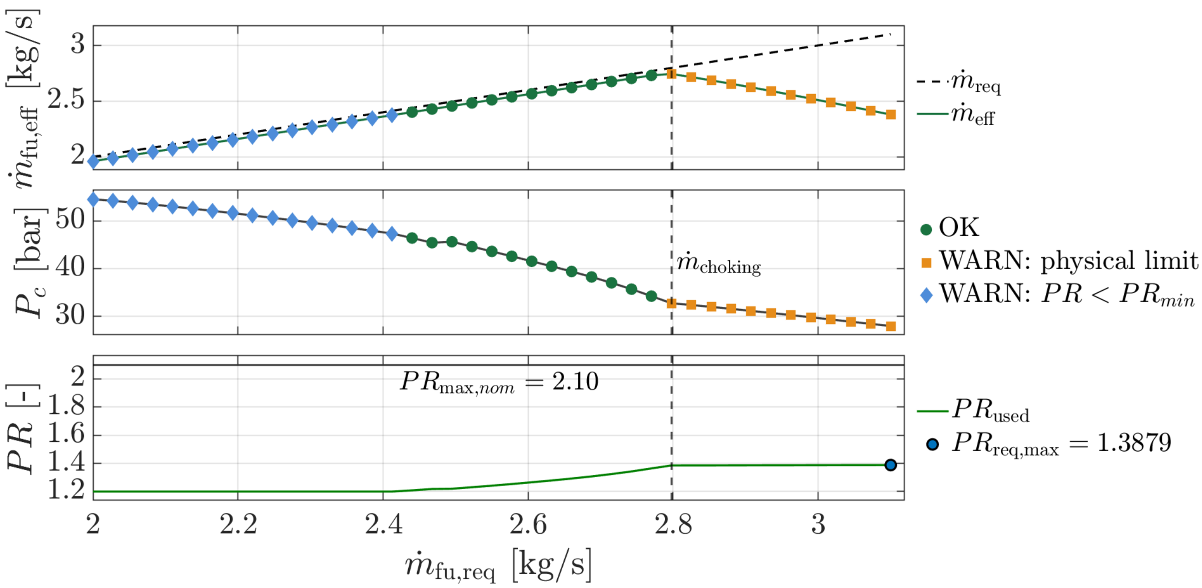


Figure 5.8: System response at $\omega = 31500$ rpm.

At fixed shaft rotational speed, as the required mass flow rate increases, the system initially adapts to the imposed demand. As can be noticed in the upper plot of Figure 5.8, the effective mass flow rate follows the imposed value up to a certain point, beyond which the two curves diverge. This separation point corresponds to the onset of choking in the Venturi, leading to a decoupling between the required and effective mass flow rates.

As the mass flow rate increases, the system response is characterized by a progressive reduction of the chamber pressure P_c, as shown in the second plot. This behavior arises because the pressure rise provided by the pumps becomes insufficient to counterbalance the increasing pressure losses along the flow paths.

Moving to the turbine pressure ratio evolution, the pressure ratio is clamped at the minimum admissible value ($PR_{\min} = 1.2$) in the initial operating region, corresponding to conditions for which the system would physically require lower pressure ratios than those covered by the turbine operating range and are therefore flagged as warning states.
Conversely, over the entire analyzed domain the turbine pressure ratio remains significantly limited, reaching a maximum value of approximately 1.39. This behavior is a consequence of the Venturi reaching choked-flow conditions, which naturally saturate the mass flow rate admitted to the turbine and prevent the system from demanding operating conditions leading to high turbine pressure ratios.

Based on the above considerations, three distinct operating regions can be identified in the plots. Acceptable equilibrium points are marked in green, Venturi-choked conditions in yellow, and operating points associated with sub-minimum turbine pressure ratios in blue ($PR < PR_{\min}$). No failure conditions have been detected in this fixed-speed analysis.

This fixed-speed analysis indicates that the loss of equilibrium can originate from different conditions, depending on the operating domain. In particular, in the high mass flow rate region Venturi choking becomes the dominant limiting mechanism, while in the opposite region turbine pressure ratio constraints are the primary limitation.

A global analysis is therefore performed by sweeping both the required mass flow rate and the shaft rotational speed, allowing the limiting mechanisms to be identified over the entire throttling domain rather than along a single operating line.

The outcomes of the global analysis are presented in Figure 5.9, Figure 5.10 and Figure 5.11. In the first figure, the Venturi ratio is reported over the entire domain. Venturi ratio is defined as the ratio between the effective mass flow rate and the required one ($\dot{m}_{eff}/\dot{m}_{req}$), and provides the measure of Venturi capability to manage the imposed mass flow rate. Values equal to unity indicate that the Venturi allows the required mass

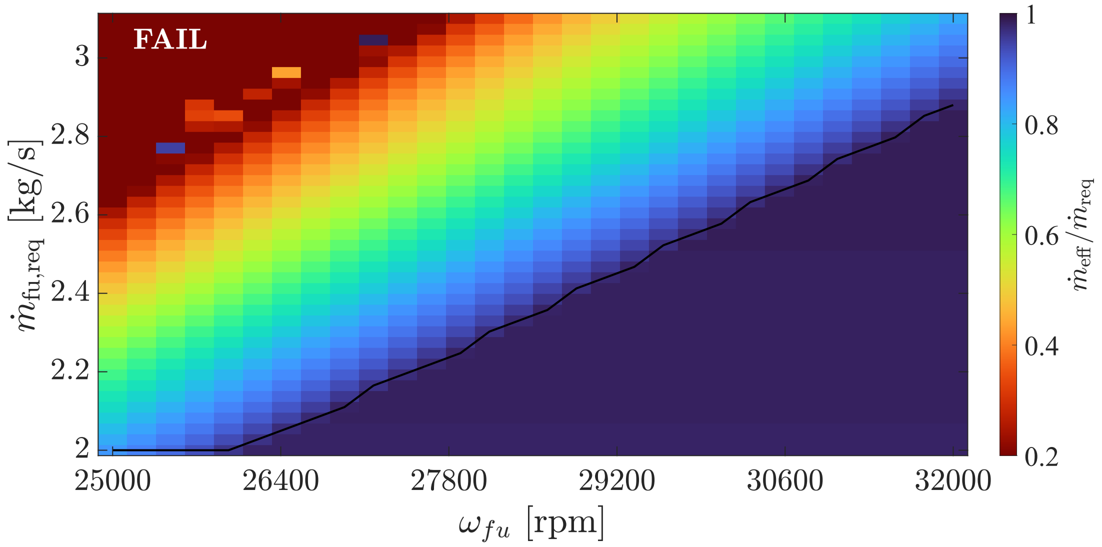


Figure 5.9: Venturi Ratio Map

flow rate to pass, while values lower than one correspond to choked-flow conditions, for which the effective mass flow rate is reduced with respect to the imposed demand.

The Venturi ratio map in Figure 5.9 shows a clear continuity with the fixed-speed analysis previously discussed. Venturi choked-flow conditions occur predominantly at high required mass flow rates, where the Venturi represents the primary limiting mechanism for system adaptability under throttling conditions. As the shaft speed increases, higher required mass flow rates can be sustained before Venturi choking.

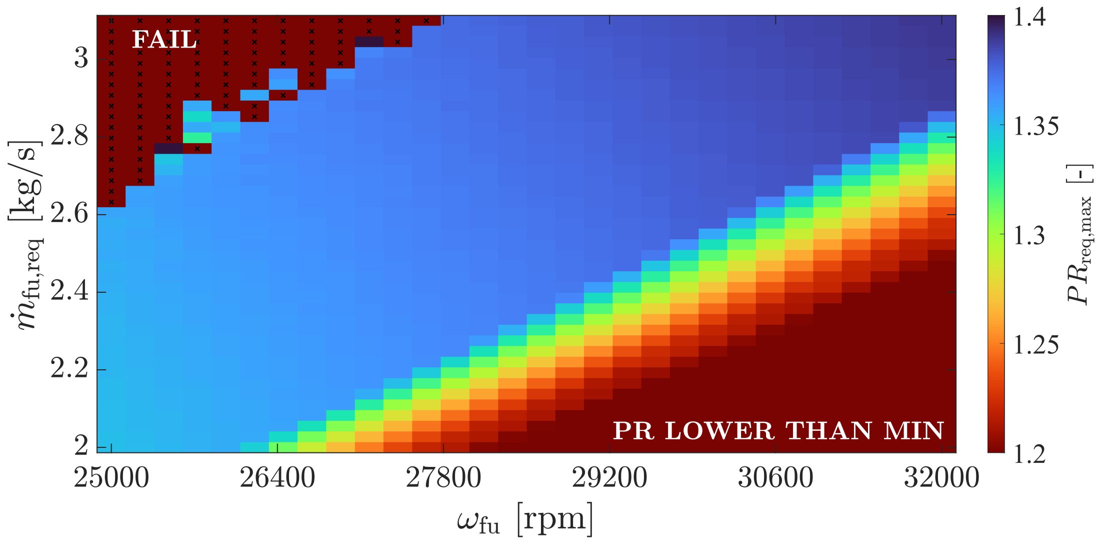


Figure 5.10: Turbine Pressure Ratio Map

Even turbine pressure ratio in Figure 5.10 shows a similar behavior with the fixed-speed analysis. At high value of shaft speed, and correspondingly low required mass flow rates, the pressure ratio required by the system falls below the minimum threshold admissible by the turbine performance maps.

Failure conditions are mainly detected in the upper-left portion of the operating domain, corresponding to low rotational speeds combined with high required mass flow rates. This behavior is physically consistent, as under these conditions the available pump head and turbine power are insufficient to sustain the imposed mass flow demand, leading to the loss of a physically admissible equilibrium. In particular, nonphysical thermodynamic states rise at the inlet of the cooling jacket, ultimately causing the solver to fail.

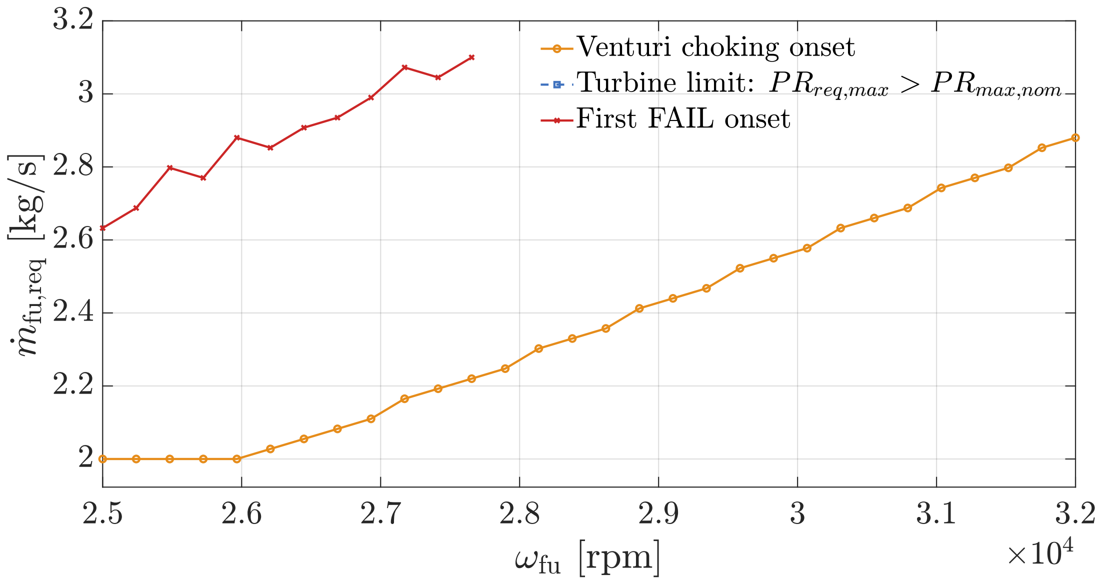


Figure 5.11: Onset Frontiers.

The onset frontiers plot (see Figure 5.11) summarizes the limiting boundaries identified in the previous maps and clearly shows that the turbine never approaches its upper pressure ratio limit. This is consistent with the behavior previously discussed, where Venturi choking conditions naturally saturate the mass flow rate admitted to the turbine, thereby preventing the system from reaching operating conditions associated with high turbine pressure ratios.

In general, the Venturi ratio map and the turbine pressure ratio map reveal two distinct and complementary regions of limited operability. Venturi-related limitations dominate at low rotational speeds and high mass flow rates, while turbine pressure ratio constraints become relevant in the opposite region of the operating domain. The failure conditions are instead mainly concentrated in the upper-left region. This complementary behavior is consistent with the trends observed in the fixed-speed analysis and provides a clear indication of the limiting mechanisms governing engine operation.

Moving to the oxidizer branch, the same analysis is performed to identify the critical regimes arising in this flow-path. The common variable between the fuel and oxidizer analysis is ω_{fu}, as the oxidizer shaft speed is directly linked to the fuel one through the gearbox. Over the analyzed range, the oxidizer branch does not show the onset of any critical regimes. This behavior is consistent with the simpler architecture of the oxidizer feed line with respect to the fuel one, which involves fewer coupled components and reduced sources of hydraulic and thermodynamic constraints. As a result, the oxidizer branch does not represent a limiting factor for the system operability within the explored domain. The only critical aspect that requires attention remains the Oxidizer Control Valve, for which flashing conditions must be avoided.

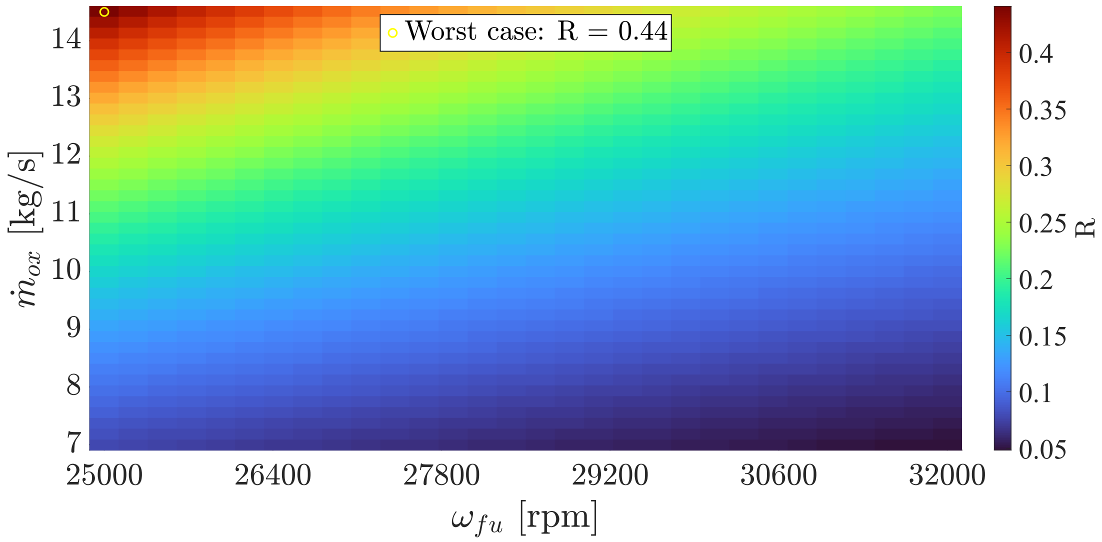


Figure 5.12: OCV saturation margin map.

To assess this, the saturation proximity index is introduced and tracked over the explored domain:

$$R = \frac{\Delta P_{OCV}}{P_{in,OCV} - P_{sat}(T_{in})} \tag{5.1}$$

The values $R \geq 1$ indicate a potential onset of saturation (FAIL), while $R \geq 0.8$ is adopted as a conservative warning threshold.

In the explored domain, the worst-case value is $R_{max} \approx 0.44$, well below the critical threshold $R = 1$, as shown in Figure 5.12; therefore, saturation-driven limitation is not expected in the LOX line and a dedicated LOX status map would be redundant. In conclusion, the overall status map over the entire analyzed domain is reported in Figure 5.13. The resulting map is related only to the fuel branch, as from the oxidizer analysis no critical regimes have been detected.

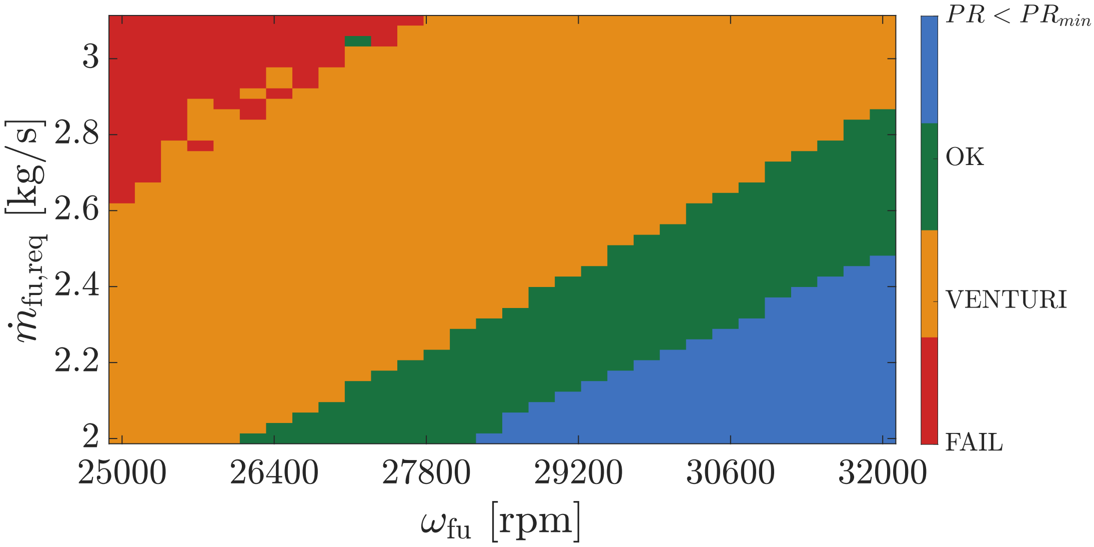


Figure 5.13: Engine Status Map

The resulting status map closely reproduces the trends identified in the previous maps, clearly highlighting that the primary limiting factor for engine throttling is Venturi choking, followed by the structural limits of the turbine. Non-physical regimes are instead confined to a well-defined portion of the operating domain.

An extension of the analysis toward lower mass flow rates is expected to enlarge the failure region, as progressively lower shaft rotational speeds prevent the pumps from providing adequate fluid pressurization.

5.3. Extension Analysis

This section investigates the possibility of acting on the limiting mechanisms previously identified to improve engine throttle-ability. As outlined before, the main limiting mechanisms are the onset of choked-flow condition at the Venturi throat, and the requirement for the turbine pressure ratio to remain above the minimum threshold.

Focusing on the latter limitation, the minimum pressure ratio currently adopted is set to 1.2, as declared by the manufacturer. This value already corresponds to a borderline operating condition, as it implies a small pressure drop across the turbine, and consequently, a minimum level of mechanical work extracted by the turbine from the fluid. Further reductions of pressure ratio would drive the turbine into an operating regime in which the extracted power is insufficient to sustain the pumps. In practice, turbine operating conditions below this threshold would correspond to an almost non-expanding flow through the turbine. Therefore, it is not possible to further relax this limitation not because of the modeling framework adopted, but due to physical and design constraints of the turbomachinery . The analysis must instead focus on the manipulation of the Venturi characteristics.

Given the operating principle of the Venturi, the only parameter effectively available for this type of analysis is the throat area. By progressively increasing the throat area by a constant scaling parameter k_A, the onset of choking within the Venturi can be delayed, as shown in Figure 5.14.

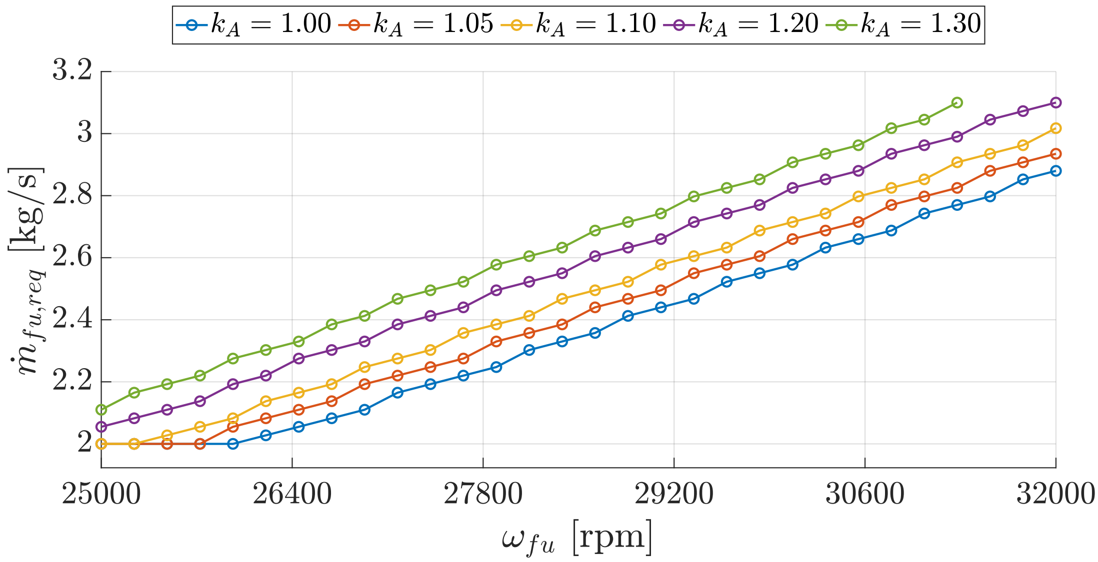
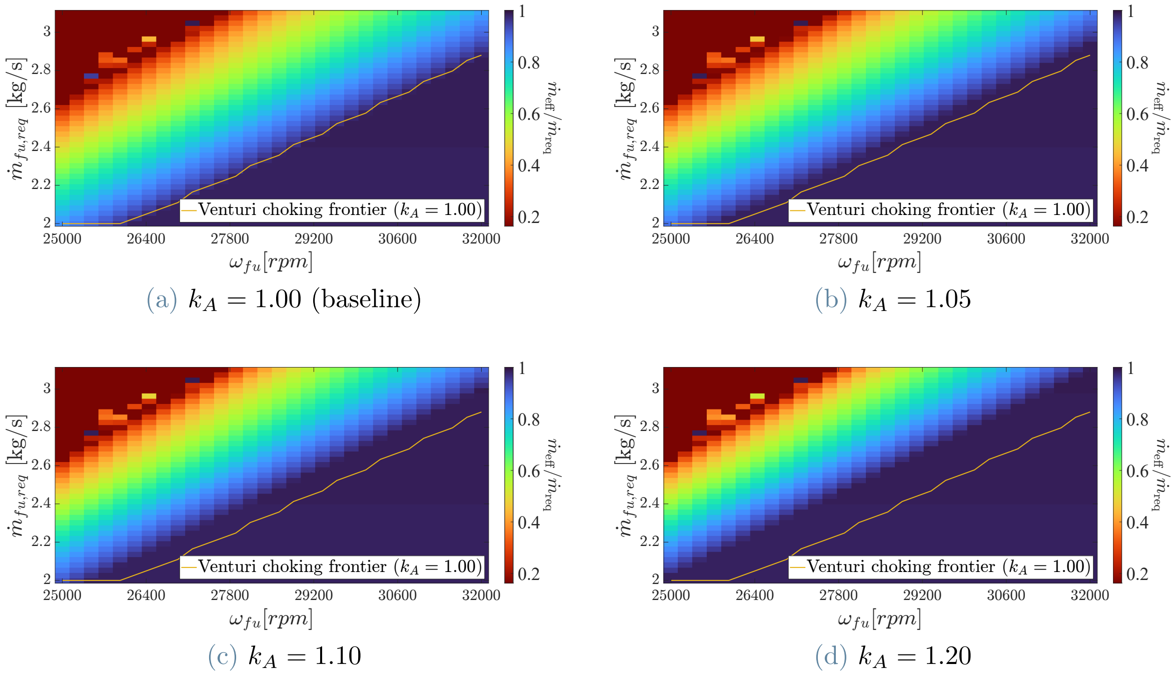


Figure 5.14: Venturi choking frontiers for different throat areas

In Figure 5.15, the Venturi ratio maps corresponding to four increasing values of the scaling parameter k_A are reported. Each subfigure represents the off-nominal equilibrium domain obtained for a different effective Venturi throat area, while the choking frontier identified for the nominal configuration ($k_A = 1.00$) is shown in all cases as a reference.

As can be observed, increasing the Venturi throat area causes the Venturi choking frontier to shift upwards, without eliminating it, as expected. As a result, the admissible solution region progressively enlarges, potentially allowing a larger number of engine equilibrium operating points. Concurrently, the enlargement of the Venturi throat area produces an increment of the mass flow rate entering the turbine, which in turns implies that an higher turbine pressure ratio is required. Therefore, it is crucial to monitor also the turbine behavior, to verify whether it arises as new limiting condition for engine operability.



(a) $k_A = 1.00$ (baseline)

(b) $k_A = 1.05$

(c) $k_A = 1.10$

(d) $k_A = 1.20$

Figure 5.15: Ratio between effective and requested fuel mass flow rate $\dot{m}_{\text{eff}}/\dot{m}_{\text{req}}$ for increasing Venturi throat area.

While for the first four values of k_A the situation does not differ from the critical regimes analyzed in Section 5.2.1, as shown in Figure B.3 in Appendix B, a critical condition occurs at the maximum Venturi throat area enlargement considered. The corresponding onset frontiers for this limiting case are reported in Figure 5.16.

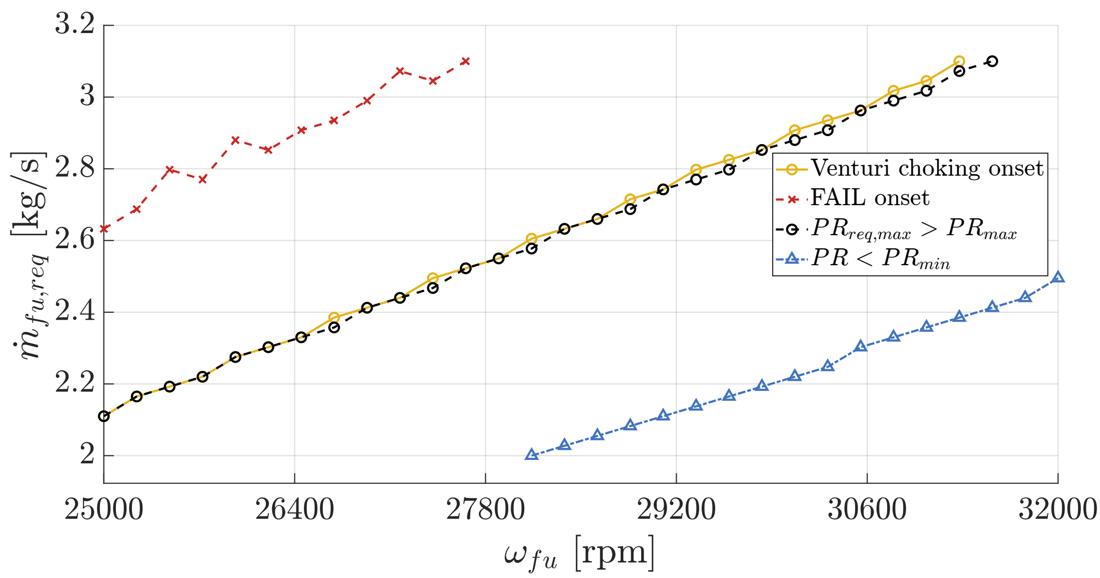


Figure 5.16: Onset Frontiers for $k_A = 1.3$

The key point to emphasize is the presence of the $PR_{\text{req,max}} > PR_{\text{max}}$ frontier which is observed only for the largest k_A case. This behavior highlights the coupling between the Venturi throat area enlargement and turbine operating regime, as the mitigation of Venturi choking progressively shifts the limiting mechanism toward the turbine. This is stressed even more in Figure 5.17, where the maximum required pressure ratio is reported as function of the shaft speed for each value of k_A.

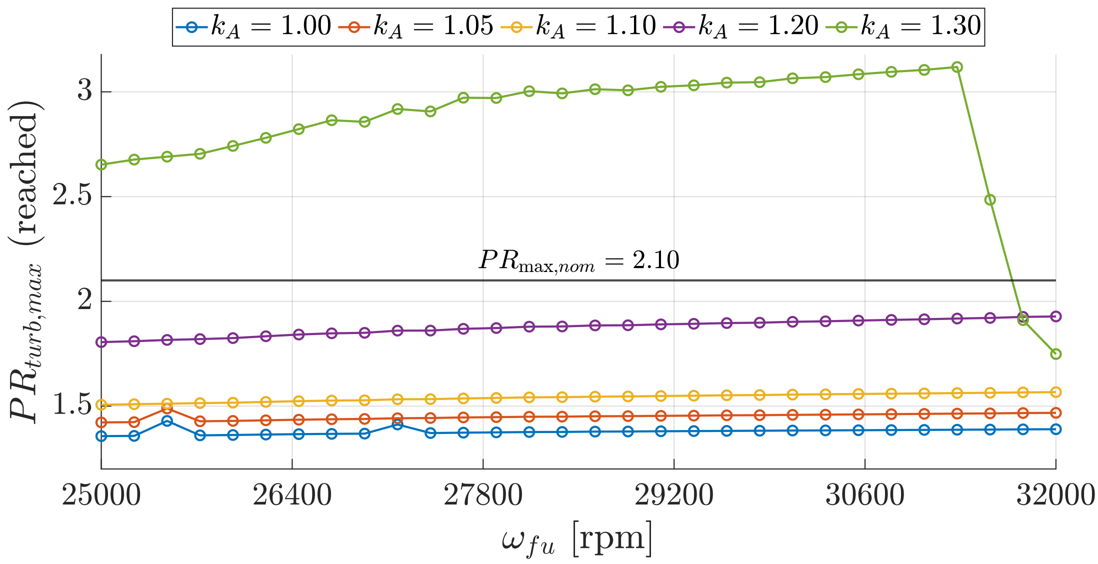


Figure 5.17: Maximum required turbine PR for different throat areas

Figure 5.17 identifies the practical limit in the enlargement of the Venturi throat area. Larger values of the scaling parameter k_A would not further extend engine throttle-ability, as they would simply shift the dominant limiting mechanism from Venturi choking to turbine operation. As the Venturi throat is enlarged, an higher mass flow rate enters the turbine, increasing the hydraulic load and the required pressure ratio. For a given inlet thermodynamic state, the turbine admits a maximum extractable power, and an associated maximum pressure ratio achievable. This limit is commonly defined as turbine limit load condition [40], and occurs when the last stage rotor of the turbine blading reaches choked conditions and the flow expansion capability is saturated.

Consequently, the mitigation of Venturi choking is achieved at the expense of increasingly demanding turbine pressure ratios, ultimately leading to the violation of the admissible turbine operating range. As observed in the figure, for the largest considered scaling factor ($k_A = 1.3$), the required pressure ratio exceeds the admissible threshold over a significant portion of the operating domain, indicating that the turbine cannot sustain such conditions.

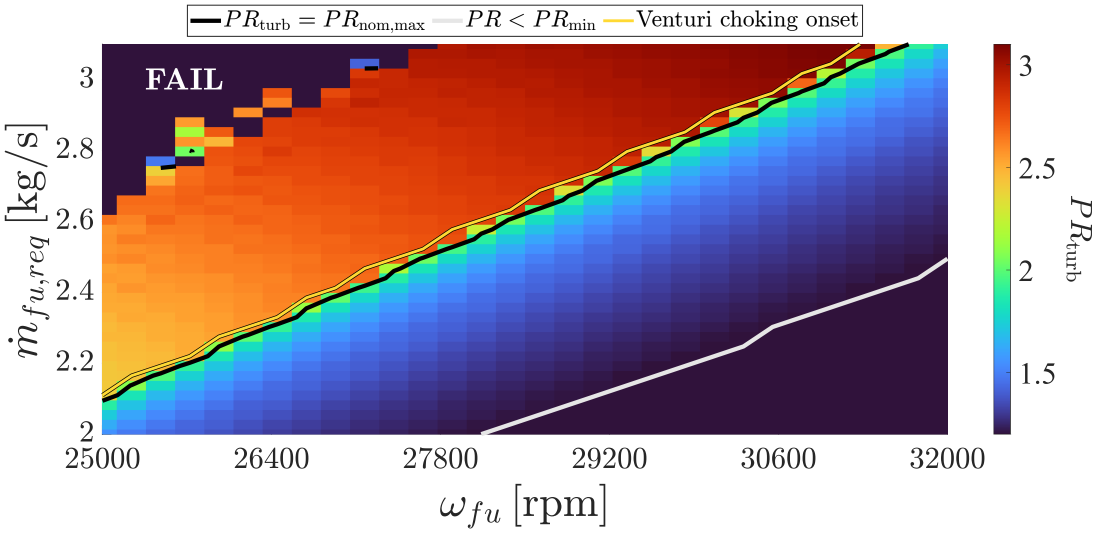


Figure 5.18: Turbine pressure ratio map with $k_A = 1.3$

In Figure 5.18, the distribution of the turbine pressure ratio PR obtained for this configuration is reported. As can be observed, the benefit associated with an increase in the Venturi throat area is progressively counterbalanced by the limited operating range of the turbine. For $k_A = 1.3$ a limit condition is reached, as the admissible operating region, included between the turbine pressure-ratio lower limit (white line) and upper limit (black line), does not exhibit a significant further enlargement with respect to the previous cases (see Figure 5.14). Further increments of k_A would mainly translate into more critical turbine operating conditions, rather than into a meaningful extension of the set of reachable equilibrium points.

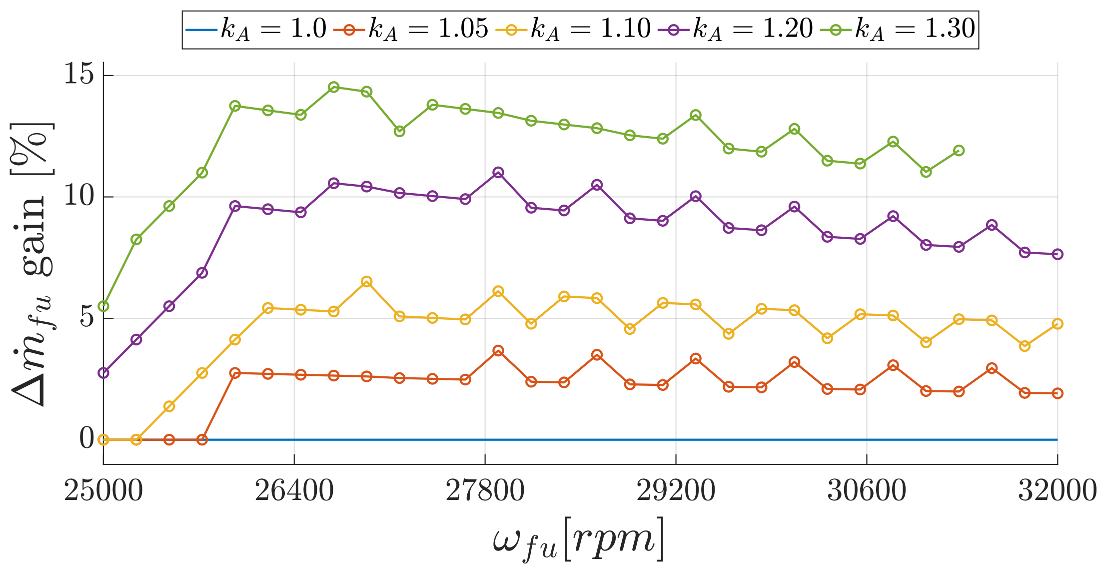


Figure 5.19: Gain in reachable fuel flow at fixed shaft speed (relative to k_A=1.00).

In conclusion, the effective gain in terms of admissible fuel mass flow rate is summarized in Figure 5.19. As can be observed, increasing by 20% the Venturi throat area can lead to a rise of almost 10% of the fuel mass flow rate the system can manage over the entire shaft speed range analyzed.

In this section, it is shown that enlarging the Venturi throat area makes it possible to extend the domain of admissible equilibrium solutions. In particular, an increase of approximately 10% in the fuel mass flow rate circulating in the system can be achieved with respect to the nominal configuration.

However, this enlargement is strictly governed by turbine operating constraints, which emerge as the primary limiting mechanism of the architecture. Once the Venturi's regulating effect is relaxed, the required turbine pressure ratio increases to manage the higher mass flow. While for scaling factors up to $k_A = 1.2$ the required pressure ratio remain within the manufacturer's declared operating range, configurations exceeding this threshold demand pressure ratios beyond admissible limits. Consequently, these high-flow configurations cannot be considered admissible equilibrium solutions for the system. Furthermore, the introduction of higher mass flow rates into the turbine may triggers complex local phenomena that necessitate a high-fidelity turbomachinery analysis. Specifically, the increased hydraulic load may induce off-design aerodynamic losses, such as flow separation and shock wave formation, which compromise turbine internal efficiency. Structurally, the increased momentum and pressure gradients could lead to aero-elastic instabilities or exceed the mechanical and thermal limits of the turbine blades and bearings. Regarding the fluid network, higher flow velocities may potentially lead to non-equilibrium transients like water hammer effect, related to the increased fluid momentum, or an incomplete gasification during the regenerative cooling. As the present study feeding lines are modeled through a lumped quasi-steady formulation, these phenomena cannot be captured. Therefore, numerical convergence alone does not guarantee the structural or operational feasibility of the physical components under the modified flow conditions.

In conclusion, the extension analysis presented here should be interpreted as a preliminary exploration of a potential system lever to enhance throttle-ability, rather than as a finalized design recommendation. Its purpose is to identify the possible margins of improvement in engine throttle-ability, and to provide a preliminary quantification of the achievable gain. Nevertheless, any practical implementation would require a detailed analysis, and probably a redesign and validation, of the relevant components, particularly the turbine.

6 | Critical Review

The results presented in the previous chapters demonstrate the capability of the proposed quasi-steady system model to reproduce the engine's nominal operating conditions and to analyze its throttling capabilities. However, the achievement of solver convergence at multiple equilibrium points under different operating conditions is not sufficient by itself to assess the validity of the present formulation. A critical review of the work done, the underlying assumptions, and the modeling choices adopted throughout the model implementation is therefore essential to clarify the actual usefulness of such a framework for engine development and system-level diagnostic analyses.

This chapter, therefore, aims to critically analyze the quasi-steady modeling framework developed in this work, highlighting not only its capabilities but also its structural limitations and the criticalities identified. Finally, the implications of these limitations are discussed from the perspective of extending the present framework towards transient simulations and alternative engine architectures.

6.1. Assessment of the Quasi-Steady Modeling Approach

In the present work, the quasi-steady assumption is adopted as a system-level modeling hypothesis rather than as a numerical simplification to approximate a transient simulation. The engine is represented as a set of coupled subsystems, interconnected according to the actual engine architecture. The engine operating conditions are determined by simultaneously enforcing a set of closure equations, including hydraulic compatibility in the propellant lines, turbopump shaft power balance, and combustion chamber consistency. For a given set of boundary conditions and control parameters, the model computes isolated equilibrium points, without introducing time as an independent variable and without describing the temporal evolution between different operating states. As a consequence of this formulation, the model's objective is defined a priori: the identification of engine operating equilibrium points at or close to steady regimes, and the assessment of how variations in control elements, such as valve settings in throttling scenarios, modify the system's global equilibrium. The framework is not intended to reproduce fast transient regimes, such as engine start-up or shutdown, nor to describe the trajectory connecting different equilibrium states.

Given this formulation, the primary strength of the present work lies in the ability to obtain a global operating point that emerges from the simultaneous compatibility of all coupled subsystems. Due to the strong system coupling inherent to the engine architecture, each subsystem adjusts its operating condition within the overall balance and does not necessarily operate at its design point. Isolated component-level analyses are therefore insufficient to describe the engine behavior. In this context, the successful simulation of the nominal operating point indicates the internal consistency of the system with respect to the adopted modeling assumptions and architectural choices, rather than validating the standalone performance of individual components. The nominal solution is obtained without global tuning parameters or imposed system-level constraints, and therefore represents a non-trivial consistency check of the overall formulation.

Beyond the nominal simulation, the implemented quasi-steady framework enables exploration of engine equilibrium points across a wide operating domain by modulating control parameters, such as valve aperture ratios. The model evaluates how equilibrium operating points change with boundary conditions and whether a physically consistent solution exists. Critical regimes are identified directly during the solution process and interpreted as indicators of physical inconsistency at the system level, rather than solely as numerical failures of the solver. By detecting, classifying, and handling critical regimes throughout the simulation, the admissible operating domain can be explored without interrupting the overall analysis.

Within this perspective, the quasi-steady approach should be interpreted primarily as a diagnostic system-level modeling framework. It systematically interrogates the engine's structure by enforcing equilibrium closure during controlled variations in control parameters, thereby allowing the identification of feasible operating regions. The structural limitations inherent to this formulation are discussed in the following section.

6.2. Limitations and Identified Criticalities

After having outlined the quasi-steady framework and its underlying assumptions, this section examines the principal limitations associated with the adopted modeling approach. These limitations can be divided into two categories: those inherently related to the quasi-steady hypothesis, and those arising from specific modeling choices adopted in the present implementation.

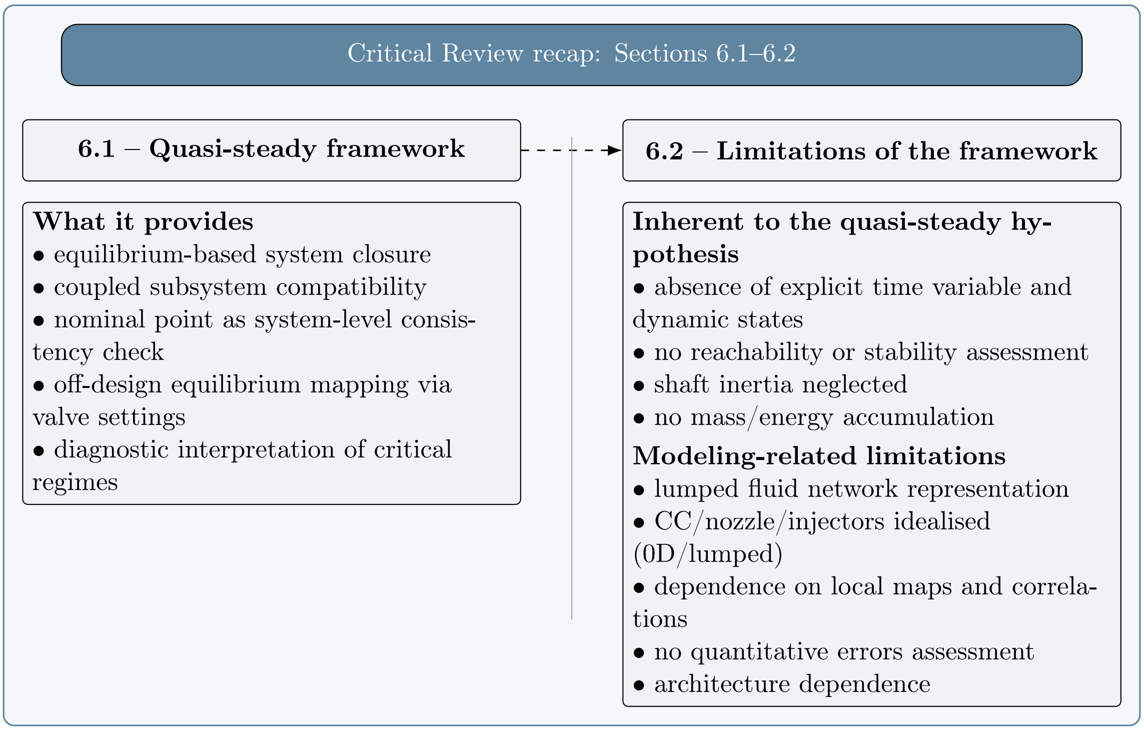


Figure 6.1: Schematic recap of the quasi-steady framework (Section 6.1) and the main limitation classes discussed in Section 6.2.

In Figure 6.1, a schematic recap of the quasi-steady framework discussed before and the main limitations identified are reported. Each point of the schematic recap will be discussed in the following pages.

6.2.1. Limitations Inherent to the Quasi-Steady Assumption

The most important limitation inherent to the quasi-steady formulation is the absence of an explicit time scale. Time is not introduced as an independent variable, and no dynamic states are defined within the system. As a consequence, the solutions obtained, i.e. the engine operating equilibrium points, are independent of each other and represent standalone configurations that the engine can theoretically achieve for a given set of boundary conditions and valve settings.

Within this framework, the model cannot describe the trajectory connecting different operating points, nor assess whether a given equilibrium is dynamically reachable during engine operation. Stability properties cannot be determined, since no information is available on the temporal evolution of the system variables. The quasi-steady formulation, therefore, restricts the analysis to the identification of internally consistent equilibrium states, without addressing their dynamic feasibility.

The absence of temporal states also affects the role of certain physical balances. In the integral formulation of the shaft power balance (see Eq. 3.32), the time derivative of the shaft rotational speed and the associated inertia term are neglected. The shaft balance is thus enforced purely as an algebraic closure condition, ensuring mechanical compatibility of the equilibrium point, without influencing its selection through dynamic effects. Similarly, no mass or energy accumulation is introduced in any subsystem, including the fluid network model. The feeding lines do not include control volumes or dynamic states capable of storing mass, momentum, or energy. Fluid inertia, compressibility effects, and thermal storage within the pipelines are not represented. The network, therefore, does not possess intrinsic temporal memory: variations in pressure, mass flow rate, or temperature are instantaneously propagated through algebraic compatibility relations, without delay or dynamic filtering. In this sense, the fluid lines cannot serve as dynamical elements of the system, since no internal state variables exist to govern their time evolution. Moreover, the adopted representation does not introduce distributed thermal inertia or dynamic thermal feedback within the feeding lines. Since no energy storage or time evolution is considered in these components, variations in the thermal state of the fluid do not generate delayed interactions with the rest of the system. The system is therefore treated as memoryless, and all subsystem interactions are resolved instantaneously at the algebraic level.

From a practical viewpoint, the solution process relies on an iterative enforcement of algebraic closure equations until convergence is achieved. The update of the system variables is therefore not driven by the engine dynamics, but by numerical consistency conditions. As a consequence, the framework is suitable for diagnostic analyses of well-defined engine configurations, but not for describing the temporal evolution of the engine. This clearly defines the domain of applicability of the approach and directly reflects the quasi-steady assumption adopted in the present work.

6.2.2. Modeling-Related Limitations

In addition to the structural consequences of the quasi-steady hypothesis, additional limitations arise from the specific modeling choices adopted to represent individual subsystems.

Lumped Spatial Representation of the Feeding Lines The propellant feeding lines are represented through a lumped-parameter approach, without spatial discretization.

The thermodynamic state is evaluated only at the inlet and outlet of each line, while the evolution of the fluid along the pipelines is not resolved. Accordingly, the feeding lines act as algebraic compatibility constraints linking pressures, temperatures, and mass flow rates at the network nodes through concentrated pressure-loss relations. Spatially distributed thermo-fluid effects and local phenomena within the pipelines are therefore not represented within the present framework. Within the operating conditions investigated in the present work and for moderate throttling levels, this approximation yields physically acceptable results and preserves the global consistency of the solution. However, its validity cannot be extended to regimes where distributed effects or strong local nonlinearities become dominant.

Zero-Dimensional Combustion and Injector Modeling Within the quasi-steady architecture, the combustion chamber and the nozzle are modeled using a zero-dimensional approach, assuming combustion as a sequence of equilibrium states computed through NASA CEA and corrected by tabulated efficiency factors. This representation provides a consistent thermodynamic closure at the system level, but does not resolve the internal spatial structure of the combustion process. Similarly, the chamber injectors are represented through a lumped-parameter approach and treated as concentrated pressure losses upstream of the combustion process. Feed–combustion interactions are therefore reduced to algebraic compatibility conditions rather than being resolved through distributed or detailed internal modeling. In addition, no dedicated heat-transfer model for the injector assembly is included. Thermal interactions between the propellants and the injector structure are not explicitly represented, and possible temperature variations occurring in the injector passages are neglected. As a consequence, the thermodynamic state of the propellants at the chamber inlet is assumed to be unaffected by injector-level heat exchange phenomena, although such effects may influence the initial conditions of the combustion process.

Dependence on Local Models Within the adopted quasi-steady framework, each subsystem is represented through local models that rely on performance maps, empirical correlations, and tabulated coefficients, which are typically engine-specific and must be available a priori. As a consequence, the global engine behavior is strongly dependent on the accuracy of the local models used to represent individual subsystems, which directly affect the system-level solution. These models are generally valid within limited operating domains and are derived from experimental data or calibrated correlations. Their application outside the corresponding validity ranges may introduce uncertainties that are not explicitly quantified within the present framework. The equilibrium states identified by the algorithm are therefore internally consistent with the imposed closures, but their quantitative accuracy remains intrinsically linked to the validity of the adopted local models.

Validity Boundaries The proposed model does not provide an internal indicator capable of quantifying its own accuracy. The FAIL and warning conditions introduced in Section 4.2.1 do not provide a quantitative evaluation of the error that led to a FAIL condition, but rather highlight the occurrence of a non-physical condition or a numerical breakdown. As a result, the introduced FAIL conditions:

(i) effectively identify the limits of the framework and of the modeling choices adopted;

(ii) enable the distinction between admissible and unfeasible solutions;

(iii) do not provide a quantitative measurement of the error committed.

Furthermore, the framework has been developed for a specific expander-cycle architecture. Several compatibility relations and coupling strategies implicitly reflect this configuration. While individual subsystem models may be reused in other contexts, the direct application of the overall framework to different engine architectures would require a critical revision of the imposed structural assumptions and perhaps a rearrangement of the solution process logic. This limitation will be discussed further in Section 6.4.

Overall, the structural limitations of the present work have been outlined. The absence of explicit dynamics, the lumped representation of the fluid network, the zero-dimensional modeling of the combustion subsystem, and the reliance on local maps define the boundaries within which the identified equilibrium states can be interpreted. Within these assumptions, the framework provides internally consistent solutions. However, its validity remains confined to the adopted modeling choices and to the reference engine architecture. If the aim is to describe transient phenomena, such as engine start-up or shutdown, the framework must be extended to formulations capable of explicitly representing temporal evolution.

6.3. Extension to Transient Simulation

Building upon the limitations discussed in Section 6.2, a conceptual extension of the proposed framework towards a time-driven formulation is addressed in this section. The objective here is not to introduce a fully developed transient model, but rather to outline how the existing quasi-steady architecture can be conceptually reorganized to account for explicit time dependence and system dynamics. In this perspective, the focus is placed on identifying which elements of the current framework can be directly reused, and which components require a reformulation due to their role in introducing memory and temporal evolution.

6.3.1. Framework Re-organization

The transition from a quasi-steady to a time-driven formulation does not consist in replacing all the existing subsystem models, but in reorganizing their structural role within the overall framework. The introduction of explicit time dependence fundamentally modifies the nature of the problem, shifting it from a global algebraic compatibility search to the integration of a set of coupled dynamic states.

Table 6.1: Conceptual transition from the quasi-steady framework to a time-driven formulation

Subsystem	Role in QS framework	Reused in time-driven	Reformulation
Fluid Network	Algebraic compatibility constraint based on pressure losses	Network topology, nodal structure, loss correlations	Introduction of fluid mass, momentum and energy conservation equations with explicit time dependence
Shaft and rotating assembly	Mechanical closure enforcing instantaneous power balance	Turbomachinery models and torque evaluation	Inertia; time-dependent shaft dynamics
Turbo-pumps	Instantaneous evaluation based on performance maps	Maps, efficiency models, choking logic	None (baseline dynamic formulation)
Valves	Lumped pressure losses enforcing flow compatibility	Flow coefficients and effective areas	None in the baseline dynamic formulation
Injectors	Lumped representation of injector pressure losses	Global injector loss characterization	More detailed injector model
Cooling jacket	Spatially discretized thermal model under steady assumptions	Spatial discretization and heat transfer correlations	Thermal capacitance; time-dependent energy balance
Combustion chamber and nozzle	Zero-dimensional thermodynamic closure	Global thermodynamic closure; performance efficiencies	Axial spatial discretization for energy balance consistency
System-level coupling	Simultaneous enforcement of global compatibility	Subsystem interaction logic and closure structure	Reorganization around explicit time variable

In order to identify which elements of the current architecture can be preserved and which require reformulation, the subsystems are reclassified according to a physical criterion: the presence of quantities capable of storing mass, momentum, or energy. Whenever such storage mechanisms are involved, the corresponding subsystem inherently acquires temporal memory, and its behavior can no longer be fully described through purely algebraic relations. Conversely, subsystems that do not involve accumulation can be evaluated instantaneously as functions of the current system state, without introducing explicit time integration. The distinction therefore reflects an intrinsic structural property of the subsystem physics, rather than a modeling preference.

Table 6.1 summarizes this reclassification, highlighting which elements of the quasi-steady architecture can be directly reused and which require reformulation. The table is not intended to serve as a reference for the implementation of the time-driven model, but rather as a conceptual scheme of the structural modifications required when explicit system dynamics is introduced.

Among the subsystems requiring explicit reformulation, the most significant modifications concern the fluid network. In the quasi-steady framework, the hydraulic circuit is treated as a global algebraic compatibility constraint, where pressures and mass flow rates are determined instantaneously through pressure-loss relations, without any internal storage or inertia effects. When explicit time dependence is introduced, this representation is no longer sufficient, and the fluid network must be restructured as a sequence of capacitive elements (like volumes and heat exchangers) and momentum elements (like junctions and valves). The capacitive elements introduce mass and energy accumulation through the integration of the conservation equations. For each control volume, the evolution of density ρ and specific internal energy u is governed by the mass and energy balances:

$$\frac{d(\rho V)}{dt} = \sum \dot{m}_{\text{in}} - \sum \dot{m}_{\text{out}} \tag{6.1}$$

$$\frac{d(\rho V u)}{dt} = \sum \dot{m}_{\text{in}} h_{\text{in}} - \sum \dot{m}_{\text{out}} h_{\text{out}} + \dot{Q} \tag{6.2}$$

The control volumes are typically assumed fixed, so that variations of V are neglected and no external work is present. Through these equations, state variables associated with mass and energy storage are introduced in the system, thereby providing each control volume with temporal memory. Pressure is then obtained through an appropriate constitutive relation, linking density and internal energy via the equation of state.
The energy balance includes the heat transfer term $\dot{Q}$, which accounts for thermal interactions with the surroundings. While in the quasi-steady formulation the heat exchange is accounted for only within the cooling jacket, here distributed thermal interactions must be consistently included in each control volume, since they directly influence the time evolution of the fluid state.

In addition to mass and energy accumulation within the control volumes, momentum conservation must be explicitly introduced in the fluid network. This requirement arises from the need to represent fluid inertia in the connecting elements. The momentum elements are therefore responsible for describing the temporal evolution of mass flow rates between adjacent capacitive elements. In this formulation, the mass flow rate $\dot{m}_{ij}$ is no longer determined instantaneously from the pressure difference, but becomes a dynamic variable whose evolution is governed by a lumped momentum balance. Such a formulation is obtained by integrating the momentum conservation equation over a representative control length connecting two adjacent nodes. A simplified integral form of the momentum conservation equation can be expressed as:

$$\frac{\partial \dot{m}_{ij}}{\partial t} = \frac{A}{L}\left(\Delta P_{ij} - \frac{R\dot{m}_{ij}^2}{\rho}\right) \tag{6.3}$$

where i and j denote two adjacent nodes in the network, A and L represent an effective cross-sectional area and characteristic length, and R accounts for distributed loss effects [41]. In contrast with the quasi-steady formulation, pressure differences no longer enforce instantaneous flow compatibility, but generate acceleration or deceleration of the fluid through inertial effects.
As a consequence of this reformulation, pressures and mass flow rates are no longer imposed through a global algebraic closure, but emerge from the time evolution of the distributed control volumes and their interconnections. Therefore the operating condition of the hydraulic circuit is dynamically achieved through the integration of the system states.

Moving forward to the shaft and rotating assembly, in the quasi-steady formulation the mechanical equilibrium at the turbopump shaft is enforced as an algebraic closure condition. When passing to a time-driven formulation, the temporal evolution of the shaft rotational speed is driven by the torque difference between turbine and pumps in combination with the rotational inertia. The complete mechanical balance becomes:

$$I\frac{d\omega}{dt} = \tau_{turb} - \sum \tau_{pumps} \tag{6.4}$$

where I represents the equivalent rotational inertia of the assembly. In this configuration, the shaft speed ω becomes a state variable of the system, and discrepancies between turbine and pump torques generate acceleration or deceleration of the rotating assembly, rather than being instantaneously compensated through algebraic enforcement.

The next subsystem to consider is the cooling jacket, whose extension towards a time-driven formulation is conceptually more straightforward. In the quasi-steady framework, a spatial discretization of the coolant path is already implemented, and for each node the thermal interaction between the coolant and the hot gases is evaluated under steady-flow assumptions, without any explicit temporal evolution of either the coolant or the wall temperature. When explicit time dependence is introduced, thermal inertia must be included for both the coolant and the surrounding solid wall. The local fluid temperature and wall temperature therefore become dynamic state variables, evolving according to the corresponding energy balances. The temperature field is no longer determined solely by the instantaneous operating conditions, but results from the accumulation of energy within the coolant and the structural material. In this formulation, fluid energy storage and wall thermal capacity introduce additional memory effects into the system. These thermal transients may influence the propellant thermodynamic state upstream of the turbine and, consequently, affect the global engine response.

With regard to the combustion chamber subsystem, the introduction of an explicit time-dependent formulation is not required for the present conceptual extension, following the same approach proposed by Ruth et al. [12]. The combustion chamber is therefore still evaluated quasi-steadily at each time step, without introducing additional dynamic state variables associated with the gas phase. This modeling choice is supported by a physical rationale. The dominant transient phenomena discussed above, namely hydraulic inertia, shaft dynamics, and wall thermal capacity, have characteristic time scales that are significantly larger than those associated with the combustion process itself. Therefore, the combustion chamber can be evaluated instantaneously, while the slower thermo-fluid and mechanical dynamics govern the overall engine evolution. This assumption is consistent with the approach adopted by Binder [10], where the chamber performance is described through a spatially discretized model and thermodynamic properties are evaluated via NASA CEA under instantaneous equilibrium assumptions. As a matter of facts, introducing internal chamber dynamics would require defining a compressible gas control volume and associated closure relations for mass, momentum, and energy. Such an extension represents a higher level of model refinement, not required as a first step for the transition towards a time-driven architecture. It can therefore be incorporated in future developments without modifying the structural organization of the dynamic framework.

The remaining subsystems do not require a structural reformulation in order to account for engine dynamics. The models implemented within the quasi-steady framework for turbopump, valves, Venturi, and injectors can be re-used within the extended time-driven formulation. These components do not introduce additional state variables in the baseline dynamic architecture, since their internal physics is not associated with mass, momentum, or energy storage at the system level. In the time-driven configuration they are evaluated instantaneously at each time step, consistently with the current system state, without introducing explicit time integration within their internal models. What changes is not their mathematical formulation, but their functional role within the solution process. In the time-driven formulation, they provide constitutive relations that close the system locally at each time step, while the global operating condition emerges from the integration of the dynamic states.

6.3.2. Reformulation of the Solution Algorithm

After having discussed in the previous section the modifications required to reformulate the individual subsystems in a time-driven perspective, the next step is the transition of the overall solution process from a quasi-steady to a time-driven formulation. In the quasi-steady approach, the engine operating point is obtained by solving a root-finding problem, where a set of nonlinear algebraic constraints is enforced iteratively until convergence is reached. In the time-driven formulation, the problem is instead treated as an initial value problem. The evolution of the engine variables, as pressures, temperatures, mass flow rates, and shaft speed, is described through explicit time dependence, and the engine state at time t is updated through time integration of the governing equations.

In this framework, the engine state can be collected in a vector $x(t)$, which explicitly includes the variables associated with mass, momentum and energy storage within the system. Then the evolution of the state vector is governed by the conservation equations associated with each subsystem. In the present architecture, the state vector and its dynamic formulation can be expressed as:

$$x(t) = \begin{bmatrix} \rho_i \\ u_i \\ \dot{m}_{ij} \\ \omega \\ T_{\mathrm{wall},k} \end{bmatrix} \qquad \begin{cases} \dot{x} = f(x, z, t), \\ 0 = g(x, z, t), \end{cases} \tag{6.5}$$

where the first two groups in the state vector correspond to the thermodynamic states of the fluid control volumes, namely density and specific internal energy, $\dot{m}_{ij}$ represents the mass flow rate between two adjacent nodes, ω is the shaft rotational speed, and $T_{wall,k}$ denotes the thermal states of the cooling jacket. In Seymour et al. [41] is suggested that, for numerical reasons, pressure p and specific enthalpy h may be more convenient choices as state variables for the mass and energy conservation equations, only if density

ρ and specific internal energy u can be consistently retrieved from them. However, in the present conceptual reformulation, density and specific internal energy are maintained for clarity and consistency with the conservation equations previously introduced.
In the dynamic formulation adopted $z(t)$ collects the algebraic variables arising from instantaneous component relations, such as turbopump operating points, injector pressure drops, valve flow coefficients and choking conditions. These quantities do not introduce additional state variables, but are evaluated consistently with the current value of $x(t)$ at each time step. This problem statement definition is consistent with transient propulsion system modeling strategies based on control volumes and shaft dynamics, as reported in the literature [5, 6, 19].

The resulting system is defined by a set of coupled differential–algebraic equations, in which accumulation effects are captured through the state vector $x(t)$, while component-level compatibility is enforced through the algebraic constraints. This conceptual transition is summarized in Figure 6.2, where the outer quasi-steady loop on chamber pressure is replaced by a time-marching integration procedure.

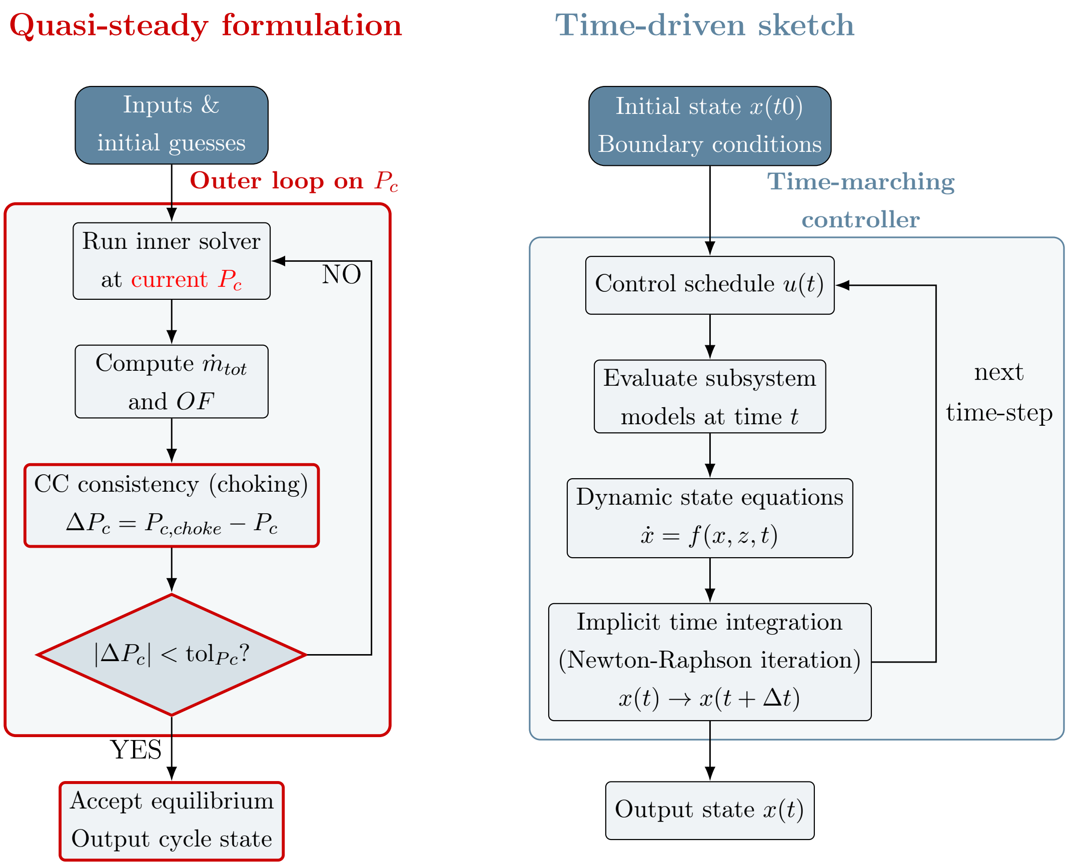


Figure 6.2: Conceptual transition from the quasi-steady outer loop to a time-driven formulation.

Figure 6.3 illustrates the modification of the logic previously associated with the inner loop, which is now reformulated as the evaluation of the subsystems at each time step. In the quasi-steady formulation, the operating point is determined through the iterative enforcement of the hydraulic and mechanical closure equations, where convergence of mass flow rates and shaft torque residuals defines the admissible equilibrium configuration. In the time-driven framework, at each time-step all subsystems are evaluated to compute the derivatives of the state vector $\dot{x}(t)$ along with the algebraic variables $z(t)$.

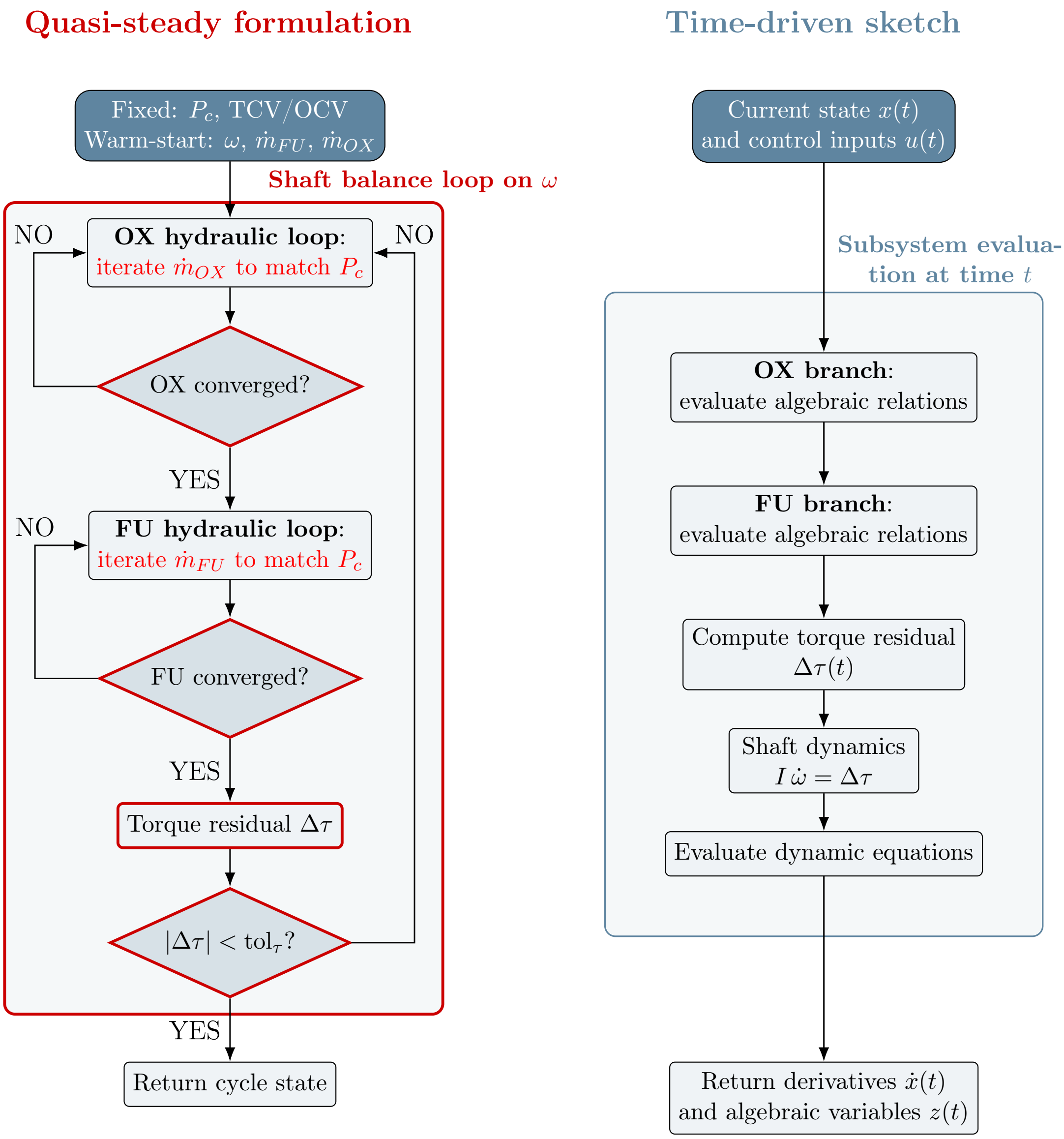


Figure 6.3: Conceptual transition from the quasi-steady inner loop (left) to a time-driven formulation (right).

The resulting set of differential-algebraic equations is then passed to the time-integration scheme, where it is advanced in time as described above. The equilibrium solutions previously identified in the quasi-steady analysis can now be interpreted as steady solutions of the dynamic system, corresponding to conditions for which $\dot{x} = 0$.

Due to the strong coupling between hydraulic, mechanical and thermal subsystems, the resulting differential–algebraic system may exhibit stiff behavior [13]. Typically, the dynamic phenomena occurring in the system, such as pressure variations in the fluid network, shaft inertia and thermal capacitance, evolve on different characteristic time scales. For this reason, in propulsion system simulations implicit time-integration strategies are generally adopted [19].

When implicit time-integration schemes are adopted, such as trapezoidal or Gear methods [16, 41], the state at the new time level appears implicitly in the discretized equations. For instance, in the classic trapezoidal time-integration scheme, the update of the state vector can be written symbolically as:

$$x_{n+1} = x_n + \frac{\Delta t}{2} \left[f(x_n, z_n) + f(x_{n+1}, z_{n+1}) \right] \tag{6.6}$$

At each time step, the computation of x_{n+1} and z_{n+1} therefore requires the solution of an iteration loop, because the derivative evaluated at the new time level depends on the unknown state x_{n+1} [41]. The state is iterated until the nonlinear residual resulting from the time discretization of the differential–algebraic system $R(x_{n+1}, z_{n+1})$ is driven to zero. This internal nonlinear problem is typically solved through a multivariable Newton–Raphson iteration, as adopted in transient propulsion solvers present in literature [16, 41]. The iterative loop introduced in this case is not associated with the enforcement of physical equilibrium, as in the quasi-steady formulation, but with the numerical resolution of the implicit time-integration scheme.

The transition from a quasi-steady to a time-driven formulation therefore does not modify the physical subsystem models, but fundamentally changes the mathematical structure of the problem and the logic of the solution process.

6.4. Extension to Different Engine Architecture

The discussion presented so far has focused on the extension of the quasi-steady framework towards a time-driven formulation. In this section, the architectural generalization of the model towards different engine cycles is discussed.

The framework developed in this work has been formulated for an expander-cycle architecture. As a consequence, the coupling strategy adopted, the closure equations and the solution process architecture are structured to represent the fluid network and the subsystem interactions of this configuration. A representative example is the linking between cooling jacket and turbine. In the present formulation the turbine is directly powered by the propellant heated in the cooling jacket. The shaft power balance is therefore intrinsically linked to the heat absorbed along the regenerative cooling path. This energetic coupling between cooling jacket and turbine is characteristic of expander-cycle engines and it is explicitly reflected in the subsystem interaction logic and closure relations.

Table 6.2: Structural reusability of subsystem models across different engine architectures

Subsystem	**Reusability across architectures**	**Remarks / Required modifications**
Pumps	Direct *(performance maps required)*	Independent of the global cycle architecture; no structural reformulation required.
Turbine	Direct *(performance maps required)*	The thermodynamic driving source may vary with the cycle architecture; the turbine model remains unchanged.
Valves and lumped losses	Direct	Local hydraulic relations depend on engine-specific geometry and flow coefficients.
Cooling jacket	Conditional	Reusable only when regenerative cooling is present. If absent, the subsystem is removed; if partially present, the model must be adapted accordingly.
Combustion chamber (0D)	Direct	Extendable also for gas generator subsystems without altering its mathematical structure.
Fluid network	Architecture-dependent	Subsystem interconnections must be redefined according to the specific cycle configuration.
Global closure strategy	Architecture-dependent	Closure equations, such as the shaft power balance, depend on the selected engine architecture.

While the logical sequence of the blocks depends on the engine cycle selected, several subsystem models remain directly reusable across different architectures. Table 6.2 summarizes the reusability of each component of the modeling framework.

As an example of a possible architecture-driven re-organization of the implemented framework, the SSME case study presented by Naderi et al. in [1] is analyzed. In this case, we are dealing with a staged combustion cycle, which presents an extremely complex architecture. For each propellant branch a preburner is present, and the turbomachinery is distributed between high- and low-pressure turbopump assemblies. In general, with respect to an expander-cycle, the number of components that must be modeled increases. From a modeling perspective, the increased complexity does not affect the modeling approach at component level. Pumps, turbines, valves, combustion chambers and cooling jackets can still be described through the same quasi-steady framework implemented in Chapter 3. Even the types of closure equations considered, namely shaft power balance, hydraulic consistency of the feed system, and the global combustion consistency at chamber or preburner level, remain conceptually unchanged. However, the interconnection among subsystems and the organization of the global closure equations are completely different.

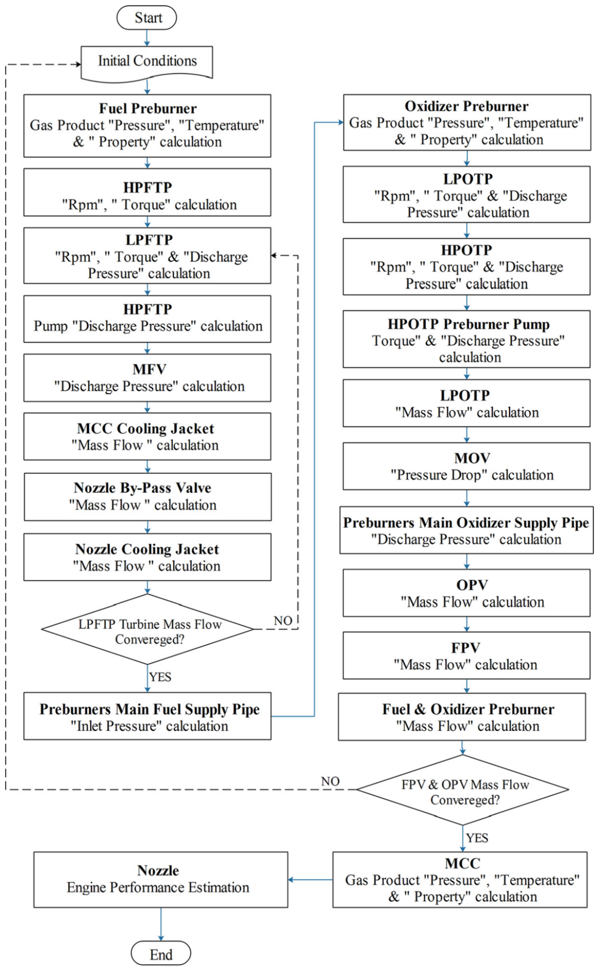


Figure 6.4: Nested loop algorithm implemented by [1] for SSME staged combustion cycle

The nested loop structure reported in Figure 6.4 of Naderi's work shows that the solver is organized in branch-specific iterative loops: the fuel line and the oxidizer line are solved sequentially, each one requiring local guesses on mass flow rates and shaft rotational speeds.

Power balance is enforced separately for the high-pressure and low-pressure turbomachinery assemblies, and then the global combustion chamber compatibility is checked only after convergence of the branch-level quantities. Therefore, the resulting algorithm is composed of multiple nested loops, in which local compatibility conditions are imposed first, while global consistency is imposed only at the outermost level.

In expander-cycle engines, the strong coupling introduced by regenerative cooling and the single drive shaft of the turbomachinery assembly requires a unified iterative structure, as shown in Figure 4.1 and in Figure 4.2. Shaft power balance and hydraulic compatibility must be enforced simultaneously rather than sequentially, while the global combustion chamber compatibility is verified in the outer loop.

Moreover, in the staged combustion configuration, the turbine is driven by combustion gases produced in dedicated pre-burners, rather than by the heat absorbed by the fuel in the cooling jacket as in an expander-cycle engine. The system is therefore defined as combustion-driven rather than heat-transfer-driven, and this results in a different coupling structure among the components, in terms of closure equations and modeling of the fluid network. In particular, the working fluid that powers the turbine is now a high-temperature combustion gas, characterized by different thermodynamic properties with respect to the cryogenic coolant of an expander-cycle, thus requiring a different thermodynamic modeling treatment. Since the combustion gases chemical composition is already an output of the NASA CEA program, and it also possesses a tool to compute the physical properties of any fluid, given its composition and a couple of thermodynamic parameters, it would be easy and beneficial to exploit this capability to implement both fluid mixing and physical properties retrieving.

In general, when moving to different and more complex architectures, the number of independent closure variables increases, and the quasi-steady problem acquires a larger algebraic dimensionality, while preserving the same underlying equilibrium-based formulation.

In conclusion, while component-level models and local balance equations can be substantially reused, the engine architecture determines the algebraic topology of the quasi-steady problem. The sequence of compatibility enforcement and the level at which energetic closure is imposed must be redefined according to the specific energy distribution of the cycle and to the architectural peculiarities that govern subsystem interaction.

7 | Conclusions and Future Developments

The present chapter provides the final considerations on the modeling framework developed in this work, summarizing its main contributions and results, and clarifying its intended scope of application and domain of validity. The model is conceived as a quasi-steady equilibrium framework for the evaluation of physically admissible engine configurations and their performance. The primary limitations identified throughout the analysis are discussed, and possible extensions and improvements are outlined as directions for future development.

7.1. Summary of Contributions

The present work develops and validates a fully coupled quasi-steady system-level model of a cryogenic expander-cycle engine. The engine is decomposed into a set of interconnected subsystems, each described by a dedicated physical model, and its operating equilibrium configurations emerge from the simultaneous satisfaction of shaft power balance, hydraulic compatibility and global thermodynamic consistency.

The nominal engine operating point is obtained as the converged outcome of the fully coupled equilibrium problem, rather than through parameter tuning of individual subsystems. The resulting solution shows excellent agreement with the NASA Binder reference model, with relative errors below 1% for most variables, in particular chamber pressure P_c, total mass flow rate $\dot{m}_{tot}$ and shaft rotational speed ω. The nominal simulation therefore not only validates the individual subsystem models, but above all the adopted coupling strategy and the underlying modeling assumptions.

The implemented framework is then extended to the analysis of engine off-nominal operating regimes. By introducing additional degrees of freedom through modulation of valve opening ratios, the model enables a systematic exploration of the admissible engine equilibrium configurations under throttling operations. The analysis shows that physically consistent equilibrium states can be individuated down to 90% of the nominal thrust T, and the corresponding variations of the main performance parameters are quantified.

In the proposed framework, the solutions obtained are critically assessed through a set of diagnostic checks embedded within the solver architecture. For each configuration analyzed, the solver is capable of detecting, classifying and handling non-physical conditions, such as negative pressures within the fluid network, or numerical breakdowns such as non-convergence of iterative loops. When these critical conditions arise, the solver flags the current configuration and proceeds with the simulation. In this way, the model identifies the region of physically admissible equilibrium configurations and provides insight into the reasons why other operating points are not feasible, highlighting the critical regimes that limit the engine throttling capability. Therefore, these diagnostic outcomes are treated as indicators of the physical boundaries of the proposed quasi-steady model, and not just as numerical artifacts.

Based on the identified critical regimes and on their interpretation, a possible extension of the admissible operating domain is associated with an enlargement of the Venturi throat area. This option is analyzed in terms of the maximum fuel mass flow rate achievable at fixed shaft rotational speed prior to Venturi choking onset, resulting in up to a 10% increase in the allowable fuel mass flow rate within the engine. The Venturi effective area therefore emerges as a system-level design parameter influencing the global equilibrium structure. However, the structural and physical implications of these modifications on the fluid network require detailed analysis at component level, particularly on the turbine and the feeding lines, as detailed in Chapter 5.3.

Finally, a conceptual reformulation of the proposed framework towards a time-driven formulation is presented, outlining how the quasi-steady equilibrium problem can be recast into a differential–algebraic structure suitable for transient simulations. The quasi-steady model is critically extended to alternative engine architectures, discussing how its modular structure can be reorganized for different engine cycles while preserving component-level formulations and redefining the architecture dependent closure strategy.

7.2. Scope and Limitations

The present modeling framework need to be intended as a diagnostic tool for the evaluation of the given configuration and its feasibility assessment. Once assigned the boundary conditions, namely propellants thermodynamic state inside the tanks, the valve settings, which are the only inputs required, and together with suitable initial guesses, the model determines whether the specified configuration is physically admissible and, if so, evaluates its corresponding performance. However, the scope of the work is not restricted to configuration diagnostics. The framework establishes a baseline set of quasi-steady component models representative of a liquid rocket engine, which can be partially reused and reorganized when considering alternative engine architectures.

The proposed framework is based on a quasi-steady equilibrium formulation, in which the engine operating condition is described as the solution of a coupled nonlinear algebraic system. Its validity is therefore restricted to regimes where accumulation effects remain negligible and the engine can be reasonably interpreted as a sequence of steady configurations. The model is not intended to reproduce transient phenomena, start-up or shut-down phases, nor dynamic instabilities. Operating conditions leading to hydraulic incompatibilities, choking phenomena or loss of subsystem consistency are identified as non-feasible configurations and mark the boundaries of validity of the quasi-steady description. Within these limits, this work provides a consistent equilibrium based framework for the evaluation of physically admissible operating points of cryogenic expander-cycle engines.

7.3. Future Developments

The future developments of the present work can be grouped into three main directions: refinement of the implemented models, extension of the quasi-steady framework to alternative engine architectures, and reformulation towards a time-driven approach for transient regime simulation.

First, several targeted improvements can be introduced at component level. More detailed models for the feeding lines, including spatial discretization, would allow a more accurate description of propellant evolution within the fluid network. Similarly, injector modeling can be refined to improve the representation of local flow and mixing effects. The lumped-parameter approach adopted in this work provides a consistent approximation under the operating conditions considered [10]. However, its applicability may be limited in regimes where distributed effects or strong local nonlinearities become dominant. In addition, the zero-dimensional combustion chamber model could be extended to a spatially discretized formulation in order to achieve a more detailed description of the combustion process.

In addition, the Venturi throat area has emerged as a system-level parameter influencing the admissible operating domain and throttling capability. A rigorous assessment of possible Venturi enlargement requires higher fidelity models of the feeding lines and of the turbomachinery, in order to verify the hydraulic and mechanical compatibility of the modified configuration. Such an analysis would allow the evaluation of the structural feasibility and performance trade-offs associated with Venturi redesign.

The next development direction identified concerns the extension of the quasi-steady framework to alternative engine architectures. The component level models and the local balance equations, which represents the foundations of the current approach, can be substantially reused. What requires a restructuring is the coupling strategy and the solution algorithm organization, which is strictly dependent on the architecture selected. The implementation of staged combustion or gas-generator cycles would therefore require the definition of additional closure variables and the redesign of the iterative solution process.

The most challenging future development concerns the transition to a time-driven formulation for transient regime simulation. The ability to reproduce start-up and shutdown phases would require the reformulation of the quasi-steady algebraic system into a differential–algebraic structure in which time is explicitly treated as an independent variable. This evolution would involve the introduction of state variables associated with engine dynamics and mass and energy accumulation, leading to increased model complexity and stricter numerical stability requirements. Nevertheless, the quasi-steady platform developed in this work provides a consistent structural foundation for such an extension.

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A | Reference Engine Data and Nominal Performance

A.1. Engine Geometry and Component Data

This Appendix aims at providing insights on the geometrical values used to model the RL10-3-3A engine, along with intermediate variables computed at subsystem level.

Table A.1: RL-10A-3-3A construction data [5].

Name	**Value**	**Units**
Fuel Turbopump		
1st stage impeller diameter	179.6	[mm]
1st stage exit blade height	5.8	[mm]
2nd stage impeller diameter	179.6	[mm]
2nd stage exit blade height	5.588	[mm]
Oxidizer Turbopump		
Impeller diameter	106.7	[mm]
Exit blade height	6.376	[mm]
Turbine		
Mean line diameter	149.86	[mm]
Mass moment of inertia	0.008767	[kg·m^2]
Ducts & Valves		
FINV flow area	0.0041	[m^2]
FCV-1 flow area	0.00038	[m^2]
FCV-2 flow area	0.00019	[m^2]
Pump discharge duct	0.0011	[m^2]
Venturi (inlet - throat)	0.0023 – 0.00067	[m^2]

Continued on next page

Name	Value	Units
TCV flow area[a]	1.01×10^{-5b}	[m^2]
Turbine discharge housing (inlet - exit)	0.013 – 0.003	[m^2]
Turbine discharge duct	0.003	[m^2]
FSOV flow area	0.0021	[m^2]
OINV flow area	0.0031	[m^2]
OCV flow area[a]	3.96×10^{-4b}	[m^2]
Cooling jacket[29]		
Number of channels	180	[-]
Channels roughness	2×10^{-6}	[m]
Metal conductivity	16.3	[W/mK]
Thrust chamber		
Chamber diameter	0.1303	[m]
Throat diameter	0.0627	[m]
Nozzle area ratio	61	[-]
Chamber/nozzle length	1.476	[m]
Number of injectors	216	[-]

[a] values at nominal full-thrust condition.
[b] this flow area includes the discharge coefficient for the orifice, which is unknown.

The construction data reported in Table A.1 are used in the present work as geometric and structural reference parameters for the implementation of the engine model. In particular, the turbomachinery characteristic dimensions (impeller diameters and blade heights) are employed to ensure geometric consistency in the definition of pump performance maps and scaling relations.

The flow areas associated with ducts, valves and Venturi sections constitute the geometric basis for the hydraulic network modeling. These quantities are used to compute pressure losses through appropriate discharge coefficients and loss correlations, allowing the determination of the corresponding resistance terms within the quasi-steady formulation. In this sense, the tabulated data do not directly define the operating point, but provide the geometric constraints required for a physically consistent evaluation of mass flow rates and pressure drops throughout the propellant branches.

A.2. Binder Nominal Simulation

Simulation performed in [10] for the RL10A-3-3A nominal operating condition, used as reference to validate the results obtained in this work in the same conditions. As this work is produced by NASA, all the quantities reported are in Imperial Units.

NASA Lewis Research Center RL10A-3-3A Engine Model Output

OVERALL ENGINE PERFORMANCE

Chamber Pressure (at injector face)	475.5	psia
Engine Thrust	16412	lbf
Engine Inlet O/F	4.993	
Combustion O/F	5.055	
Specific Impulse	445.6	sec
Fuel Consumption	6.161	lb/sec
Oxidizer Consumption	30.76	lb/sec

COMBUSTION CHAMBER / NOZZLE

Nozzle Stagnation Pressure	473.5	psia
Combustion Temperature	5816	R
c* Efficiency	0.9892	
c* (ideal)	93885	in/sec
Engine Exit Pressure (ambient)	0.01	psia
Fuel-LOX heat transfer at injector	40.36	BTU/sec

FUEL-SIDE ENGINE CONDITIONS

Location	static pressure (psia)	total pressure (psia)	total temp (deg R)	mass flow (lb/sec)	density (lb/cu.ft)	total enthalpy (BTU/lb)
Fuel Tank	27.68	27.68	38.60	....	4.337	-105.1
Fuel Pump Inlet	26.08	27.19	38.6	6.161	4.337	-105.1
Fuel Pump Interstage	532.2	534.8	47.65	6.118	4.285	-68.21
Fuel Pump Discharge	1058.9	1067.3	57.63	6.082	4.220	-27.84
Cooling Jacket Inlet	1018.3	1026.6	57.94	6.082	4.185	-27.84
Cooling Jacket Exit	816.8	830.9	384.2	6.082	0.3855	1240.5
Turbine Inlet	796.7	813.8	384.2	6.056	0.3767	1240.5
Turbine Hiusing Exit	562.2	576.2	361.1	6.082	0.2868	1149.2
Fuel Injector Plennum	537.9	537.9	359.4	6.082	0.2742	1142.4
Chamber	475.5					

OXIDIZER-SIDE ENGINE CONDITIONS

Location	static pressure (psia)	total pressure (psia)	total temp (deg R)	mass flow (lb/sec)	density (lb/cu.in)	total enthalpy (BTU/lb)
Oxygen Tank	42.35	42.35	174.7	....	69.07	66.42
Ox Pump Inlet	36.33	39.96	174.7	30.76	69.07	66.42
Ox Pump Discharge	601.5	621.1	178.6	30.75	69.02	68.84
Oxid. Injector Plennum	529.4	539.3	182.1	30.75	68.31	70.15
Chamber	475.5					

TURBOMACHINERY PERFORMANCE

	delta-Head (feet)	Volume Flow (gpm)	Speed (rpm)	Torque (lbf-in)	Efficiency
Fuel Pump					
1st stage	16858	637.8	31494	645.5	0.5854
2nd stage	17901	641.0	31494	700.7	0.5686
LOX Pump	1212	199.9	12598	529.1	0.6408

Turbine Pressure Ratio (T-S)	1.402	
Turbine Exit Pressure (static)	580.6	psia
Turbine Exit Temperature	360.9	R
Turbine delta-H	92.02	BTU/lb
Turbine Efficiency	0.7353	
Turbine Torque	1577.8	lbf-in
Veocity Ratio (U/Co)	0.4581	

COOLING JACKET PERFORMANCE

Cooling Jacket delta-P (total)	195.7	psid
Cooling Jacket delta-T	326.3	R
Total Heat Pick-up	7714.2	BTU/sec
Maximum Metal Temp	1500.8	R

VALVE PERFORMANCE

Thrust Control Open Area	.004836	sq.in.
Turbine Bypass Flow	.02634	lb/sec
Oxidizer Control Open Area	.5783	sq.in.
OCV primary flow	28.10	lb/sec
OCV bypass flow	2.659	lb/sec

Figure A.1: Binder nominal simulation for the RL10A-3-3A. Credits: [10]

A.3. Turbomachinery

In this section additional materials for a better understanding of turbomachinery model are reported.

In Table A.2, the nominal values provided by P&W are reported. Since they originate from a U.S. manufacturer, the data are given in Imperial units.

The nominal head, volumetric flow rate, and torque (the latter computed from the available performance data) are used as reference quantities for the definition of the dimensionless parameters employed in the pump performance maps.

Table A.2: Reference turbomachinery data adopted in the model (Credits: [10], based on PW engine data).

Fuel Turbine (combined stage)

Quantity	Value	Unit
Meanline diameter	5.90	in
Pressure ratio	1.39	–
Mass flow rate	5.89	lbm/s
Temperature drop	24.5	R
Shaft speed	31537	rpm
Mass moment of inertia	0.0776	lbf·in·s^2
Drag torque	20.0	lbf·in

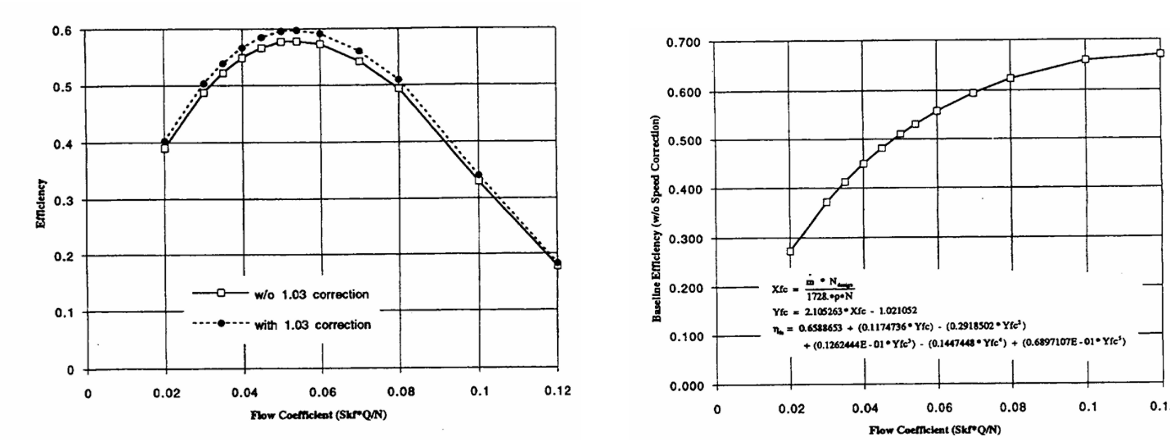
Fuel and LOX Pumps

Quantity	Fuel[a]	LOX	Unit
Impeller diameter	7.07	4.20	in
Exit blade height	0.230 0.220	0.251	in
Pump head	16969 17989	1212	ft
Mass flow rate	6.051 6.008	31.40	lbm/s
Temperature rise	9.23 10.2	3.85	R
Pump efficiency	0.5810 0.5619	0.6422	–
Shaft speed	31537	12615	rpm

[a] First and second stage values reported for the fuel pump.

Values refer to typical engine operating conditions as predicted by the model. Data reported by Binder based on Pratt & Whitney engine information.

In Figure A.2 the flow coefficient ϕ map is reported for both pumps. These maps outline the limits of validity of the correlation adopted in the model.



(a) Fuel pump

(b) Oxidizer Pump

Figure A.2: Flow coefficient ϕ map for both pumps. Credits: [10]

Based on the flow coefficient ϕ limits, a validity domain map has been produced varying fuel mass flow rate $\dot{m}_{fu}$ and shaft rotational speed ω. The purple region in Figure A.3 represents the points where the correlation adopted in Eq. 3.4 in Chapter 3.

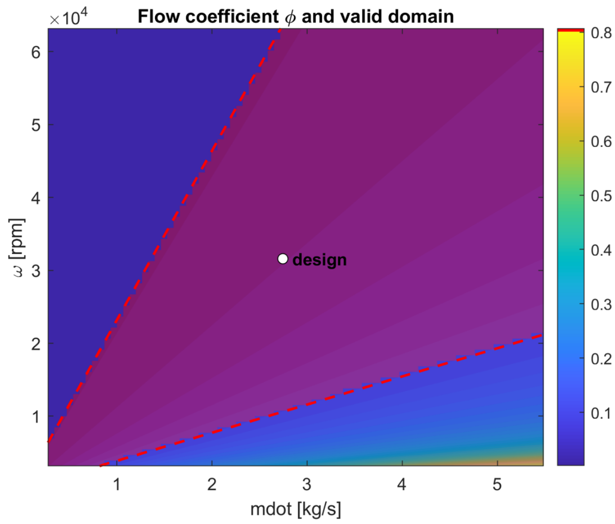
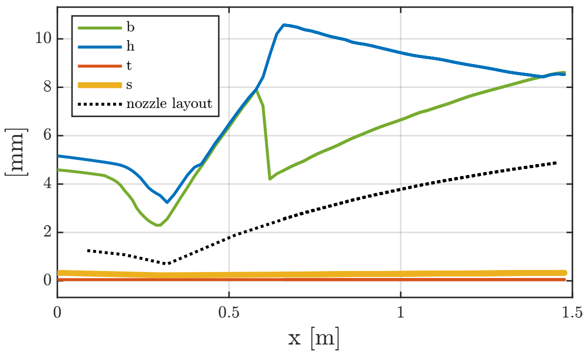


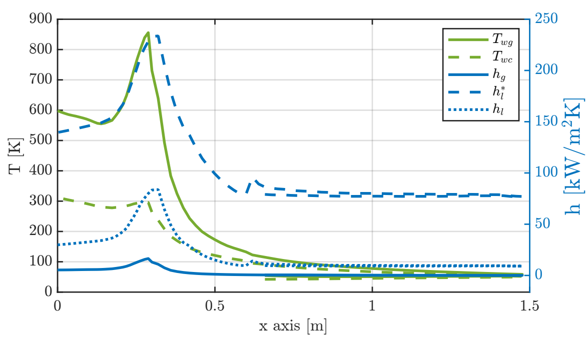
Figure A.3: Flow coefficient ϕvalidity domain.

A.4. Cooling Jacket

Figure A.4 shows the cooling channels geometry available beforehand [29], and the resulting distribution of heat transfer coefficients and temperature along the longitudinal axis.



(a) Geometrical data of channels[5]



(b) Characteristic temperatures and related heat transfer coefficients

Figure A.4: Cooling channel geometry and associated thermal parameters.

The difference between h_l and h_l^* is all due to the fin effect of the wall.

Figure A.5 shows a mode detailed view of the heat flux per unit area and the hot has side heat transfer coefficient. It is evident that in the throat section, due to gas compression and curvature effect the heating is much more accentuated.

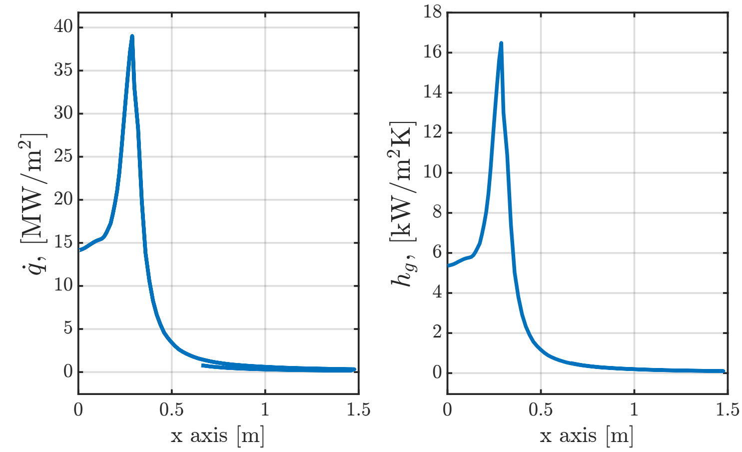
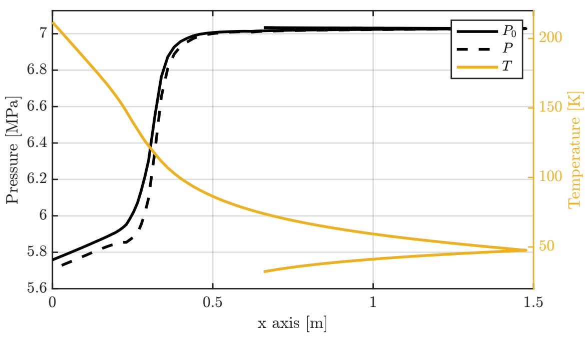


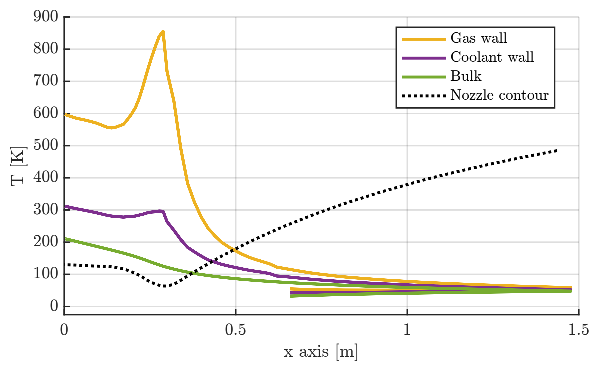
Figure A.5: Hot gas heat transfer coefficient and heat flux

The small "ripple" visible in all the figures from around 0.65m until the nozzle exit is due to the one-and-a-half-pass configuration. Since the representation is in engine axis coordinates, the ripple is the half-pass, while the full-length plot is the full pass.

Figure A.6 shows the coolant physical properties (temperature and pressure, both static and total) along its journey in the cooling jacket, along with all important temperatures.



(a) Progression of temperature and pressure along the cooling jacket



(b) Overview of characteristic temperature

Figure A.6: Thermal behavior along the cooling jacket.

A.5. Iterative Structure and Convergence Parameters of the Solver

In Table A.3 the closure condition associated with each loop of the nominal and off-nominal algorithm is defined.

Table A.3: Physical closure conditions of the quasi-steady solver under nominal and off-nominal regimes.

Loop	**Iterated variable**	**Closure condition**
	Nominal algorithm	
Outer loop – mass flow consistency	$\dot{m}_{tot,guess}$	Total mass-flow closure: consistency between $\dot{m}_{tot,guess}$ used to compute P_c through the choking loop and $\dot{m}_{tot,line}$ delivered by the coupled propellant branches
Choking loop	P_c	Choked-flow condition verification at throat for the current OF and $m_{tot,guess}$
Shaft loop	ω	Mechanical equilibrium at common shaft (turbine torque = pumps torque + drag torque)
Pump local procedure	$\dot{m}_{in,pumps}$	Pump suction mass-flow consistency
TCV local loop (bypass)	$\dot{m}_{TCV}$	Turbine bypass hydraulic consistency with imposed valve area and downstream conditions
Turbine loop	PR	Turbine operating-point compatibility at current ω and $\dot{m}$
	Off-nominal algorithm	
Outer loop – choking consistency	P_c	Chamber-pressure closure: consistency between the imposed P_c and the value obtained from the choking relation using the mass flow and mixture ratio resulting from the internal branch balances P_c^{choke}
Choking loop	P_c	Choked-flow condition verification at throat for the current OF and $\dot{m}_{tot}$

Continued on next page

Loop	**Iterated variable**	**Closure condition**
Shaft loop	ω	Mechanical equilibrium at common shaft (turbine torque = pumps torque + drag torque)
Ox branch hydraulic loop	$\dot{m}_{ox}$	Injector inlet pressure compatibility in oxidizer branch ($P_{c,ox} \to P_c$)
Fuel branch hydraulic loop	$\dot{m}_{fu}$	Injector inlet pressure compatibility in fuel branch ($P_{c,fu} \to P_c$)
TCV local loop (bypass)	$\dot{m}_{TCV}$	Turbine bypass hydraulic consistency with imposed valve area and downstream conditions
Turbine loop	PR	Turbine operating-point compatibility in off-design regime at current ω and $\dot{m}$

The closure conditions summarized in Table A.3 define the physical constraints that must be satisfied at convergence for the operating point. Each loop is therefore associated with a specific residual equation, whose vanishing ensures global mass balance, mechanical equilibrium, hydraulic consistency, turbine map compatibility and choking enforcement in the thrust chamber.

Once the physical meaning of each loop is established, the numerical implementation requires the definition of convergence thresholds and relaxation strategies. The iterative tolerances and under-relaxation coefficients adopted in both the solution procedure are reported in Table A.4. These parameters are selected to guarantee numerical stability of the nested structure while preserving convergence speed, especially in the strongly coupled shaft and turbine loops.

Table A.4: Iterative parameters of the quasi-steady solution procedure under nominal and off-nominal regimes.

Iterated variable	**Tolerance on ϵ_{rel}**	**Under-relaxation**
	Nominal algorithm	
$\dot{m}_{CC}$ (outer loop)	10^{-4}	–
ω (shaft loop)	10^{-3}	$\alpha_\omega = 2.4$ correction: $\frac{\alpha_\omega}{I}$
$\dot{m}_{TCV}$ (local loop)	10^{-4}	$\alpha_{TCV} = 0.70$
$\dot{m}_{in,pumps}$ (local)	10^{-4}	$\alpha_{pump} = 0.70$
PR (turbine loop)	10^{-3}	$\alpha_{PR} = 0.50$
P_c (choking loop)	10^{-3}	$\alpha_{P_c} = 0.40$
	Off-nominal algorithm	
P_c (outer loop)	10^{-3}	–
ω (shaft loop)	10^{-3}	$\alpha_\omega = 2.4$ correction: $\frac{\alpha_\omega}{I}$
$\dot{m}_{TCV}$ (local loop)	10^{-4}	$\alpha_{TCV} = 0.70$
$\dot{m}_{ox}$ (local)	10^{-4}	$\alpha_{pump} = 0.70$
$\dot{m}_{fu}$ (local)	10^{-4}	$\alpha_{pump} = 0.50$
PR (turbine loop)	10^{-3}	$\alpha_{PR} = 0.50$
P_c (choking loop)	10^{-3}	$\alpha_{P_c} = 0.40$

B | Additional Results

B.1. Sensitivity Analysis on Nominal Outer Loop

To assess the conditioning of the outer-loop closure variable in the nominal formulation, a sensitivity analysis was performed by perturbing the chamber pressure P_c and total mass flow rate entering the combustion chamber $\dot{m}_{tot,CC}$ around the converged operating point, while monitoring the response of the cooling jacket and turbine subsystems.

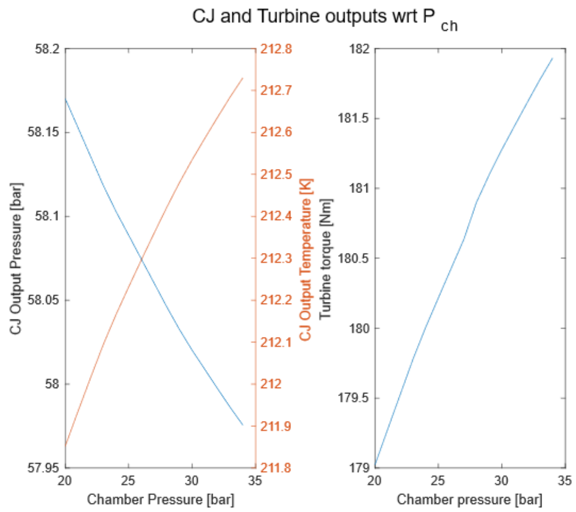


Figure B.1: Sensitivity of cooling jacket outlet pressure and temperature, and turbine torque, to perturbations of chamber pressure in the nominal formulation.

Figure B.1 shows a very limited variation of all monitored quantities. In particular, the cooling jacket outlet temperature and pressure exhibit only marginal deviations, and the turbine torque varies within a narrow range compared to its nominal value. This behavior indicates that the upstream hydraulic and thermal equilibrium is only weakly affected by perturbations of chamber pressure. In contrast with the previous case, perturbations of $\dot{m}_{tot,CC}$ induce significantly more pronounced variations in all monitored quantities, as shown in Figure B.2. The cooling jacket thermodynamic state changes more visibly, and the turbine torque exhibits a substantial variation compared to its nominal value.

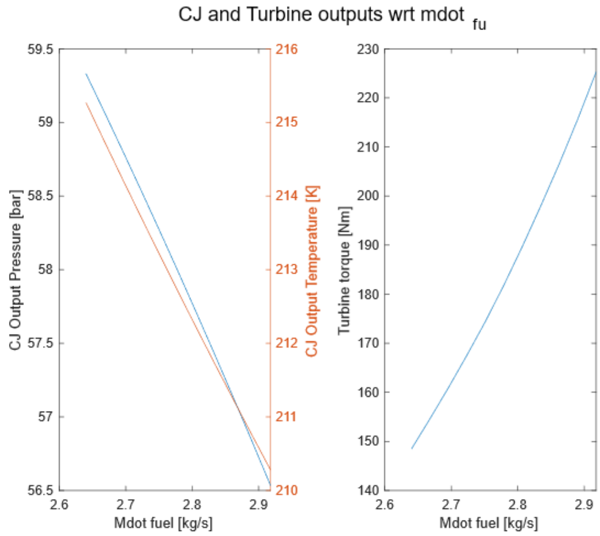
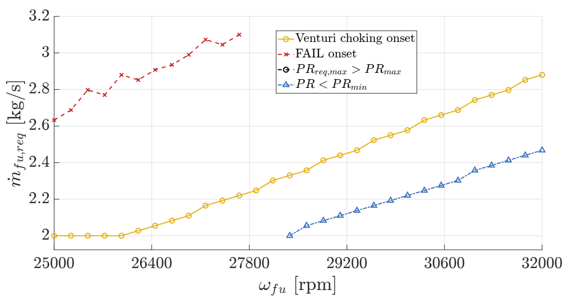
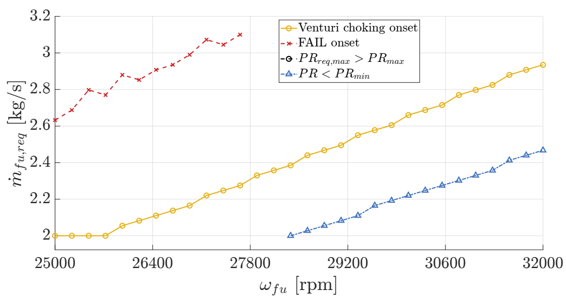


Figure B.2: Sensitivity of cooling jacket outlet pressure and temperature, and turbine torque, to perturbations of total mass flow rate in the nominal formulation.

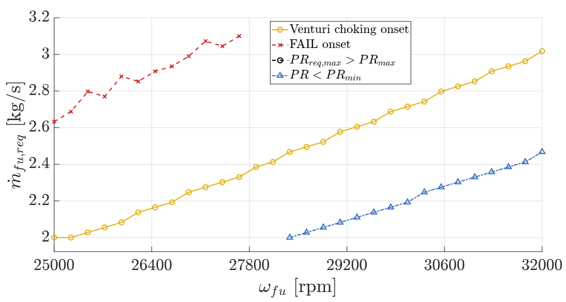
B.2. Extension Analysis



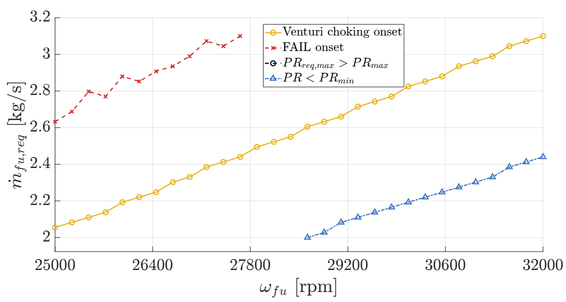
(a) $k_A = 1.00$ (baseline)



(b) $k_A = 1.05$



(c) $k_A = 1.10$



(d) $k_A = 1.20$

Figure B.3: Venturi choking frontiers in the $(\omega_{fu}, \dot{m}_{fu,\text{req}})$ plane for different values of the Venturi effective area parameter k_A.

In the context of the extension analysis presented in Chapter 5, Figure B.3 reports the onset frontiers for all analyzed k_A values, except for $k_A = 1.3$.

All configurations exhibit the same overall limiting structure, with the Venturi choking line being rigidly shifted as k_A increases. No additional limiting mechanisms emerge for $k_A \leq 1.20$, confirming that the enlargement of the Venturi throat primarily translates the choking frontier without altering the nature of the system-level constraints.

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List of Symbols

Variable	Description	SI unit
A_{throat}	nozzle throat area	m^2
b	fin length	m
$b_{blade,pump}$	pump impeller exit blade height	m
c^*	characteristic velocity	$\frac{m}{s}$
c_p	specific heat at constant pressure	$\frac{J}{kg\,K}$
c_T	thrust coefficient	-
D_h	hydraulic diameter	m
$D_{out,pump}$	pump impeller exit diameter	m
D_t	throat diameter	m
D_{turb}	turbine diameter	m
f_d	Darcy friction coefficient	-
g	gravitational acceleration	$\frac{m}{s^2}$
G	mass flux	$\frac{kg}{m^2 s}$
h	specific enthalpy	$\frac{J}{kg}$
h_g	hot-side convective heat transfer coefficient	$\frac{W}{m^2 K}$
h_l	cold-side convective heat transfer coefficient	$\frac{W}{m^2 K}$
h_l^*	corrected cold-side convective heat transfer coefficient	$\frac{W}{m^2 K}$
H	equivalent heat transfer coefficient	$\frac{W}{m^2 K}$
I_{sp}	specific impulse	s
k	thermal conductivity	$\frac{W}{mK}$
Ke	dimensionless equivalent roughness	-
$\dot{m}$	mass flow rate	$\frac{kg}{s}$
m_{fin}	fin parameter	-
M	Mach number	-
Nu	Nusselt number	-

Variable	**Description**	**SI unit**
OF	oxidizer-to-fuel ratio	-
P	static pressure	Pa
P_c	chamber pressure	Pa
Pr	Prandtl number	-
PR	turbine pressure ratio	-
q	heat flux	$\frac{W}{m^2}$
Q	volumetric flow rate	$\frac{m^3}{s}$
R_c	curvature radius	m
Re	Reynolds number	-
s	cooling jacket wall thickness in radial direction	m
t	fin thickness	m
T	temperature	K
TDH	total dynamic head	m
u	velocity	m/s
VR	turbine velocity ratio	-
γ	specific heat ratio	-
η	efficiency	-
η_{fin}	fin efficiency	-
μ	dynamic viscosity	$Pa\ s$
ρ	density	$\frac{kg}{m^3}$
τ	torque	Nm
ω	shaft rotational speed	rpm
^{0}	total quantity	-
$_{aw}$	wall adiabatic	-
$_{is}$	isentropic quantity	-
$_{lb}$	liquid at bulk	-

List of Acronyms

Acronym	**Description**
LRE	Liquid Rocket Engine
$P\&W$	Pratt & Whitney
LOX	Liquid Oxygen
$LH2$	Liquid Hydrogen
$SSME$	Space Shuttle Main Engine
TCV	Thrust Control Valve
OCV	Oxidizer Control Valve
RTL	Rate Thrust Level
CEA	Chemical Equilibrium with Applications

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